SAC 101

Fundamentals of Soil Science

Lecture-wise Theory Study Material

B.Sc. (Hons) Agriculture - First Year, First Semester
Tamil Nadu Agricultural University (TNAU)
2025 Syllabus

29 Lectures - Complete Coverage

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Preface and How to Use This Guide

This book is a **complete, lecture-wise theory study material** for the course **SAC 101 - Fundamentals of Soil Science**, taught in the **first year, first semester** of the **B.Sc. (Hons) Agriculture** programme under the **2025 syllabus of the Tamil Nadu Agricultural University (TNAU)**. It covers **all 29 lectures** of the syllabus, in the same order in which they are taught, with nothing skipped or merged.

How the Book is Organised

- **One chapter per lecture.** Each chapter is titled to match the syllabus line.
- **Headed sections** expand every sub-topic with clear, plain-English definitions.
- **Tables** summarise comparisons and standard ranges for quick revision.
- **Worked examples** show full step-by-step calculations.
- **Text diagrams (box art)** explain structures and processes that are usually shown as figures.
- **Exam focus** at the end of every lecture gives key points, short-answer questions with model answers, and long-answer questions with model answer outlines.
- **Indian and Tamil Nadu examples** appear throughout, including the soils of Tamil Nadu, the Cauvery delta and TNAU recommendations.
- The book ends with a **Quick Revision appendix**, a **Glossary** and a **list of References**.

How to Use This Book

1. **Read one lecture at a time**, just after (or before) the class on that topic.
2. **Study the tables first** - they compress a whole topic into a few lines.
3. **Work through the numerical examples** with a pen; soil science has several standard calculations that reappear in examinations.
4. **Attempt the questions in the Exam focus section** without looking at the model answers, then check.
5. **Use the Quick Revision appendix** in the last few days before the examination.

A Note on Values and Ranges

Soil properties vary with location, depth and management. Throughout this book, wherever a value is not fixed, the **standard published range** is given rather than a single false-precise figure. For example, particle density is given as 2.6-2.7 g/cm3 (average 2.65) and field capacity as about -33 kPa (1/3 bar). Where a standard average is widely used, it is quoted as such.

A Note on Diagrams

The diagrams are drawn as **text (box art)** so that they can be reproduced easily in notes. They show the essential structure or process; for photographic figures, refer to the prescribed textbook.

This guide is meant to **supplement**, not replace, classroom teaching and the prescribed textbook. Use it for revision, for structured answers, and for practice. All the best.

Lecture 1: Soil Definition - Soil as a Three Dimensional Natural Body; Pedological and Edaphological Concepts of Soil

1.1 What is Soil?

Soil is the thin, loose, unconsolidated layer of mineral and organic material that covers the surface of the Earth and supports the growth of land plants. It is a dynamic natural body that forms at the interface of the atmosphere, lithosphere, hydrosphere and biosphere.

The most widely accepted scientific definition is that of **Joffe (1936)**:

> "Soil is a natural body composed of mineral and organic constituents, having a definite genesis and a distinct nature of its own."

Soil Survey Staff / USDA defines soil as "the unconsolidated mineral or organic material on the immediate surface of the earth that serves as a natural medium for the growth of land plants."

For an agriculture student, remember this simple working definition:

- Soil is the **natural medium** for plant growth.
- It is **three dimensional** - it has length, breadth and depth (the third dimension, depth, is what makes soil a "body" rather than a surface film).
- It is **dynamic** - it changes with time, climate, organisms and management.

1.2 Soil as a Three Dimensional Natural Body

Soil is called a **three dimensional natural body** because:

- It occupies **space in three dimensions** - it extends over the landscape (length and breadth) and downward into the earth (depth).
- The third dimension, **depth**, is studied as the **soil profile** - a vertical section from the surface down to the parent material.
- A soil body has **definite boundaries** (top, sides and bottom) and **distinct horizons** arranged one below the other.
- It has a **definite genesis** (origin) and a **distinct nature of its own** - it is not merely powdered rock; it is a product of the interaction of climate, organisms, relief, parent material and time.

A single soil body is called a **pedon** (the smallest three-dimensional unit, usually about 1 m wide and 1-2 m deep). Many adjacent pedons that are similar form a **polypedon**, which is the basic unit used in soil classification.

1.3 Pedological Concept of Soil

The word **pedology** comes from the Greek *pedon* (soil) and *logos* (study). Pedology is the study of the **origin, nature, classification and genesis** of soil.

Under the **pedological concept**, soil is treated as:

- A **natural body** that has formed in place from parent material by the combined action of climate, organisms, relief and time.
- A body with a **definite profile** - distinct horizons (A, B, C) arranged vertically.
- An object of study for the **soil scientist / pedologist**, whose main interest is **how the soil formed and how to classify it**.

The pedologist studies the soil **as a whole**, as one would study a rock or a mineral - by examining the whole profile and its horizons.

Key point: The pedological concept is concerned with soil **formation and classification**, not directly with crop production.

1.4 Edaphological Concept of Soil

The word **edaphology** comes from the Greek *edaphos* (soil/ground) and *logos* (study). Edaphology is the study of soil **in relation to the growth of higher plants**, especially with respect to crop production.

Under the **edaphological concept**, soil is treated as:

- A **medium for plant growth** that supplies nutrients, water, air and mechanical support to roots.
- A body whose value is judged by **its productivity** - how well it supports crop growth.
- An object of study for the **agronomist**, whose main interest is **how to manage the soil for better yields**.

The edaphologist studies the soil **from the crop's point of view** - physical condition, fertility, moisture, aeration and pH.

Key point: The edaphological concept is concerned with soil **productivity and management**.

1.5 Comparison of Pedological and Edaphological Concepts

Feature	Pedological concept	Edaphological concept
Greek root	Pedon = soil (as a body)	Edaphos = soil (as ground for plants)
Main interest	Origin, genesis and classification of soil	Crop production and management

Feature	Pedological concept	Edaphological concept
How soil is viewed	A natural body with a profile	A medium for plant growth
Chief worker	Pedologist (soil scientist)	Edaphologist (agronomist)
Unit of study	Pedon / polypedon (whole profile)	Root zone / plough layer
Emphasis	How soil formed	How soil feeds the crop
Typical question	"Which soil order does this belong to?"	"Which fertilizer and how much?"
Example of interest	Horizon sequence A-B-C	Available N, P, K and pH

1.6 Fundamental Branches of Soil Science

Soil science is built on four fundamental branches, sometimes called the **four pillars**:

- **Soil physics** - physical properties such as texture, structure, density, water and air.
- **Soil chemistry** - chemical properties such as pH, salinity, colloids and ion exchange.
- **Soil biology** - living organisms in soil and organic matter decomposition.
- **Soil fertility and plant nutrition** - nutrients in soil and their supply to crops.

Added to these are **soil genesis and classification** (pedology), **soil survey**, **soil conservation** and **soil microbiology**.

1.7 Why Soil is Important in Agriculture (India and Tamil Nadu)

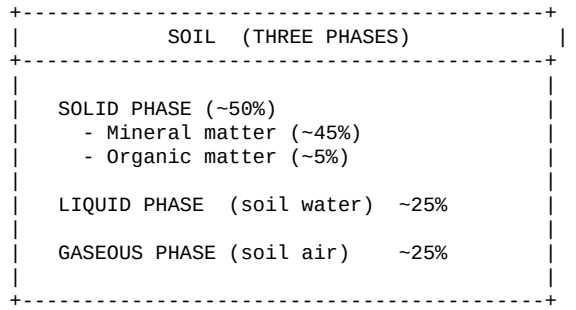
- Soil is the **base of food security** - about 95% of human food comes from the soil.
- India has about **140 million hectares of cultivated land** and Tamil Nadu has about **7 million hectares** under cultivation.
- Soil provides **anchorage, water, air and nutrients** to crops.
- In Tamil Nadu, major soil groups are **red soils, black soils, laterite soils, alluvial soils and coastal soils**, each with different management needs.
- Sound knowledge of soil helps in **choosing crops, irrigating correctly, applying the right fertilizer dose and conserving soil**.

1.8 Soil as a Three Phase System (Introduction)

Soil is made of **three phases**:

- **Solid phase** - mineral matter + organic matter.
- **Liquid phase** - soil water (soil solution).
- **Gaseous phase** - soil air.

The relative proportions of these three phases change with moisture and management. An ideal agricultural soil has roughly **50% solid, 25% water and 25% air** by volume. (This is developed in Lecture 2.)



1.9 Exam Focus

Key points

- Soil is a natural, three dimensional body with a definite genesis and distinct nature (Joffe, 1936).
- The three dimensions are length, breadth and **depth**; depth is studied as the soil profile.

- A pedon is the smallest 3-D soil unit; a polypedon is a group of similar pedons.
- **Pedology** = study of origin and classification of soil.
- **Edaphology** = study of soil as a medium for plant growth.
- Soil is a three phase system - solid, liquid and gas.
- Soil science has four pillars - physics, chemistry, biology and fertility.
- An ideal soil has about 50% solid, 25% water and 25% air by volume.

Short-answer questions

1. *Define soil.*

Model answer: Soil is a natural, three dimensional body composed of mineral and organic matter, having a definite genesis and a distinct nature of its own, which serves as a medium for the growth of land plants.

2. *Differentiate pedology and edaphology.*

Model answer: Pedology studies the origin, genesis and classification of soil as a natural body, while edaphology studies soil in relation to the growth of higher plants (crop production). The pedologist classifies soil; the edaphologist manages it for yield.

3. *Why is soil called a three dimensional natural body?*

Model answer: Because it occupies space in three dimensions - it extends over the landscape and downwards in depth. The third dimension (depth) forms the soil profile with distinct horizons, giving soil a definite boundary, genesis and nature of its own.

Long-answer questions

1. *Explain the pedological and edaphological concepts of soil with a comparison table.*

Outline: Introduction - define pedology and edaphology - explain pedological concept (natural body, profile, classification) - explain edaphological concept (medium for plant growth, productivity, management) - comparison table - conclusion that both are complementary views of the same soil.

2. *"Soil is a three dimensional natural body." Justify the statement with reference to the pedon and the soil profile.*

Outline: Define soil - meaning of three dimensional - pedon and polypedon - soil profile and horizons A, B, C - definite genesis and boundaries - three phase system - importance in agriculture - conclusion.

Lecture 2: Components of Soil - Soil as a Three Phase System; Composition of the Earth's Crust

2.1 Introduction

Soil is a **three phase system** made up of solid, liquid and gaseous phases. The solid phase forms the framework; the liquid and gaseous phases occupy the pore spaces between the solid particles. The proportion of each phase varies with the amount of water present.

2.2 The Three Phases of Soil

1. Solid phase

- Occupies about **50% of the soil volume** in a typical agricultural soil.
- Made of two parts: **mineral matter** (about 45%) and **organic matter** (about 5%).
- Mineral matter comes from the weathering of rocks and minerals; organic matter comes from plant and animal residues.
- Forms the **skeleton and matrix** of the soil and provides **adsorbing surfaces** (colloids) for nutrients.

2. Liquid phase (soil water / soil solution)

- Occupies the **pore spaces** not filled with air; in an ideal soil about **25% by volume**.
- Contains dissolved salts, nutrients and gases - hence it is called the **soil solution**.
- Acts as a **carrier of nutrients** to plant roots and as the **medium for all soil chemical reactions**.
- Also called the **"master variable"** because it controls aeration, temperature and nutrient movement.

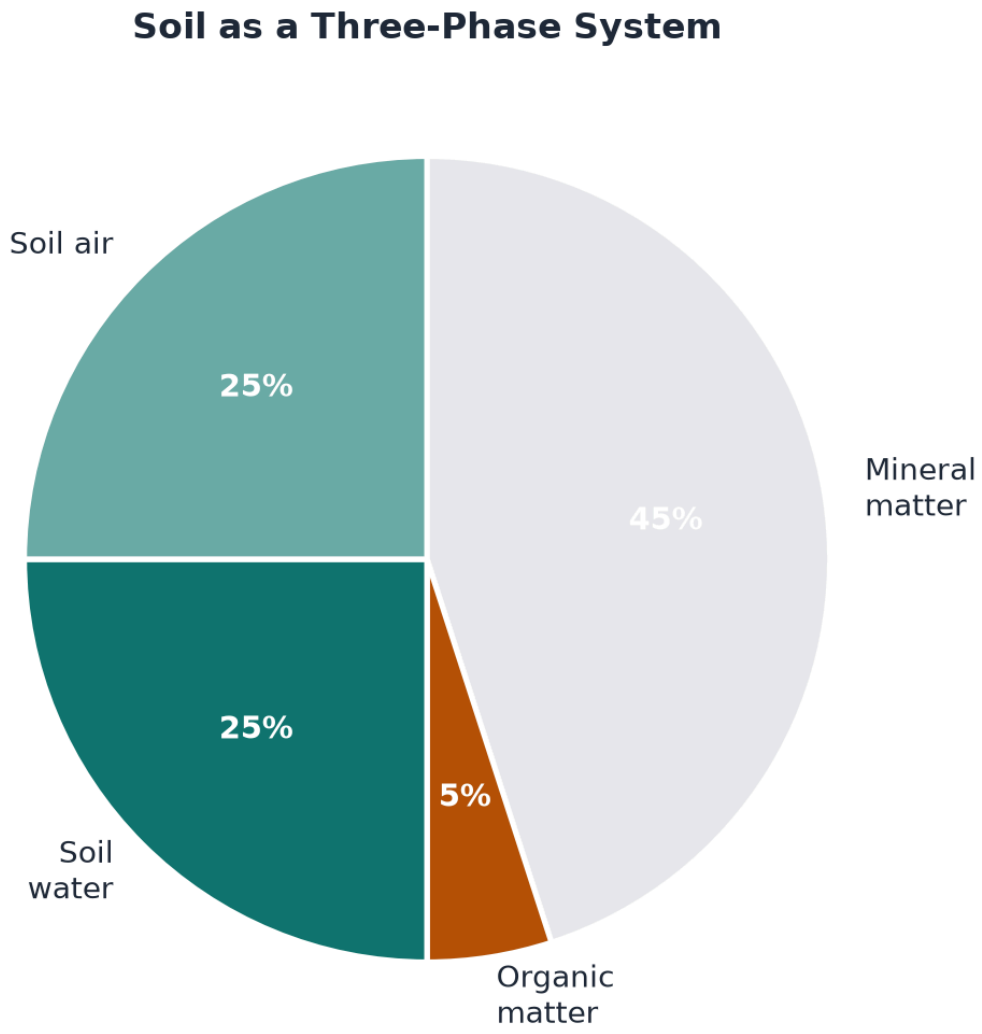


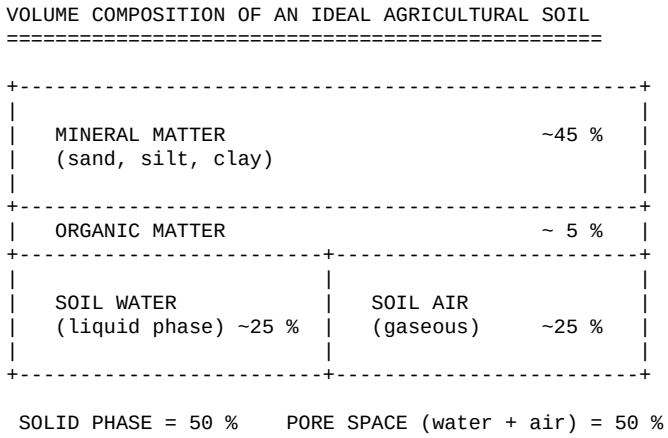
Figure 2.1 Soil as a three-phase system

Typical volumetric composition of a loam topsoil: mineral matter ~45%, organic matter ~5%, water ~25% and air ~25%.

3. Gaseous phase (soil air)

- Fills the remaining pore spaces; in an ideal soil about **25% by volume**.
- Similar to atmospheric air but with **more CO2 and less O2**, and nearly saturated with water vapour.
- Supplies **oxygen to roots and microbes** and removes CO2.

2.3 Diagram: Soil as a Three Phase System



If water content increases, air content falls correspondingly, and vice versa. The **sum of water and air always equals the pore space**, which is about 50% of the total volume.

2.4 Volume and Weight Composition (Approximate)

Component	Volume (%)	Weight (%)	Remarks
Mineral matter	~45	~ 95	From rocks and minerals
Organic matter	~ 5	~ 5	From plant and animal residues
Soil water	~25	varies	Carries dissolved nutrients
Soil air	~25	negligible	Supplies O2, removes CO2
Total	**100**	**100**	-

Note: Weight percentages vary widely. Organic soils (peat) may contain 60-90% organic matter by weight.

2.5 Functions of Each Phase

- **Solid phase:** gives structure, supports roots, holds colloids, stores nutrients, provides adsorption sites.
- **Liquid phase:** dissolves and transports nutrients, supplies water to plants, medium for microbial and chemical activity.
- **Gaseous phase:** supplies O2 for root respiration and microbial activity, carries away CO2.

2.6 Composition of the Earth's Crust

The Earth's crust is the outermost solid shell of the Earth. It is made mostly of rocks, which in turn are made of minerals. Only a small number of elements dominate the crust.

Average composition of the Earth's crust (by weight)

Element	Symbol	Approx. % by weight
Oxygen	O	46.6
Silicon	Si	27.7
Aluminium	Al	8.1
Iron	Fe	5.0
Calcium	Ca	3.6
Sodium	Na	2.8

Element	Symbol	Approx. % by weight
Potassium	K	2.6
Magnesium	Mg	2.1
Others (Ti, H, P, Mn, etc.)	-	~1.5

Key observations

- **Oxygen and silicon together make about 74%** of the crust - hence the crust is dominated by **silicate minerals**.
- The **eight elements** O, Si, Al, Fe, Ca, Na, K and Mg make up **more than 98%** of the crust.
- The most common minerals are **quartz (SiO2)**, **feldspars**, **micas**, **amphiboles** and **pyroxenes**.

2.7 Common Elements in Soil and Crust - Comparison

Element	In crust (%)	In soil (approx.)	Comment
Oxygen	46.6	~49	Most abundant in both
Silicon	27.7	~33	Rich in sandy soils
Aluminium	8.1	~7	Core of clay minerals
Iron	5.0	~4	Gives red/yellow colour
Calcium	3.6	~1.4	Leached in humid regions
Sodium	2.8	~0.6	Leached; high in sodic soils
Potassium	2.6	~1.4	Important plant nutrient
Magnesium	2.1	~0.6	Central atom of chlorophyll

Soils are generally richer in oxygen and silicon than the crust (due to quartz sand) and poorer in calcium, sodium and magnesium (because these are leached by water).

2.8 Rock - Mineral - Soil Relationship

- **Rocks** are aggregates of one or more minerals.
- **Minerals** are naturally occurring, inorganic substances with a definite chemical composition and crystal structure.
- **Soil** forms when rocks and minerals are broken down by weathering and mixed with organic matter.

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ROCK  --->  MINERALS  --->  WEATHERING  --->  SOIL
(aggregate)  (building units)   (breakdown)     (+ organic matter)
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2.9 Worked Example 1: Volume Percentages from a Given Soil Sample

Problem: A soil sample of 100 cm3 total volume contains 45 cm3 mineral matter, 5 cm3 organic matter, 25 cm3 water and the rest air. Calculate the percentage of each phase and the pore space.

Solution

- Total volume = 100 cm3
- Mineral matter = 45/100 x 100 = **45%**
- Organic matter = 5/100 x 100 = **5%**
- Water = 25/100 x 100 = **25%**
- Air = 100 - (45 + 5 + 25) = 100 - 75 = 25 cm3 = **25%**
- Solid phase = 45 + 5 = **50%**
- Pore space = water + air = 25 + 25 = **50%**

Answer: Solid 50% (mineral 45%, organic 5%), water 25%, air 25%, pore space 50%.

2.10 Worked Example 2: Crust Composition Check

Problem: Verify that the eight major elements make up over 98% of the crust.

Solution

- Sum = 46.6 + 27.7 + 8.1 + 5.0 + 3.6 + 2.8 + 2.6 + 2.1
- = 46.6 + 27.7 = 74.3
- 74.3 + 8.1 = 82.4
- 82.4 + 5.0 = 87.4
- 87.4 + 3.6 = 91.0
- 91.0 + 2.8 = 93.8
- 93.8 + 2.6 = 96.4
- 96.4 + 2.1 = **98.5%**

Answer: The eight elements make about 98.5% of the crust, confirming that they dominate its composition.

2.11 Exam Focus

Key points

- Soil is a three phase system - solid, liquid and gas.
- Ideal agricultural soil: 50% solid, 25% water, 25% air by volume.
- Solid phase = mineral matter (~45%) + organic matter (~5%).
- Pore space = water + air = about 50% of volume.
- The crust is dominated by oxygen (46.6%) and silicon (27.7%) - about 74% together.
- Eight elements make over 98% of the crust; hence silicates dominate.
- Soils are richer in Si and O but poorer in Ca, Na and Mg than the crust (leaching).
- Rock -> mineral -> weathering -> soil is the basic chain of soil formation.

Short-answer questions

1. *Explain soil as a three phase system.*

Model answer: Soil consists of a solid phase (mineral + organic matter, ~50% by volume), a liquid phase (soil water, ~25%) and a gaseous phase (soil air, ~25%). The solid phase forms the matrix, while water and air occupy the pore space and their proportions vary inversely with each other.

2. *Give the average composition of the Earth's crust.*

Model answer: By weight - oxygen 46.6%, silicon 27.7%, aluminium 8.1%, iron 5.0%, calcium 3.6%, sodium 2.8%, potassium 2.6%, magnesium 2.1%; remaining elements about 1.5%.

3. *Why is the Earth's crust dominated by silicate minerals?*

Model answer: Because oxygen (46.6%) and silicon (27.7%) together make about 74% of the crust by weight, and these two combine to form silicate minerals such as quartz, feldspars and micas.

Long-answer questions

1. *Describe the components of soil and explain the three phase system with a diagram.*

Outline: Introduction - solid phase (mineral and organic) - liquid phase - gaseous phase - volume composition diagram - ideal proportions - functions of each phase - effect of water content on air - conclusion.

2. *Explain the composition of the Earth's crust and its relation to soil formation.*

Outline: Definition of crust - major elements table - dominance of O and Si - silicate minerals - comparison of crust and soil composition - rock-mineral-soil chain - importance for agriculture - conclusion.

Lecture 3: Soil Genesis: Soil Forming Rocks - Definition, Formation, Classification of Rocks - Igneous, Sedimentary and Metamorphic Rocks

3.1 Introduction

Soil is formed from **rocks and minerals** through the process of weathering. To understand soil genesis (origin), we must first understand rocks - their definition, formation and classification. The rock from which a soil forms is called its **parent material**.

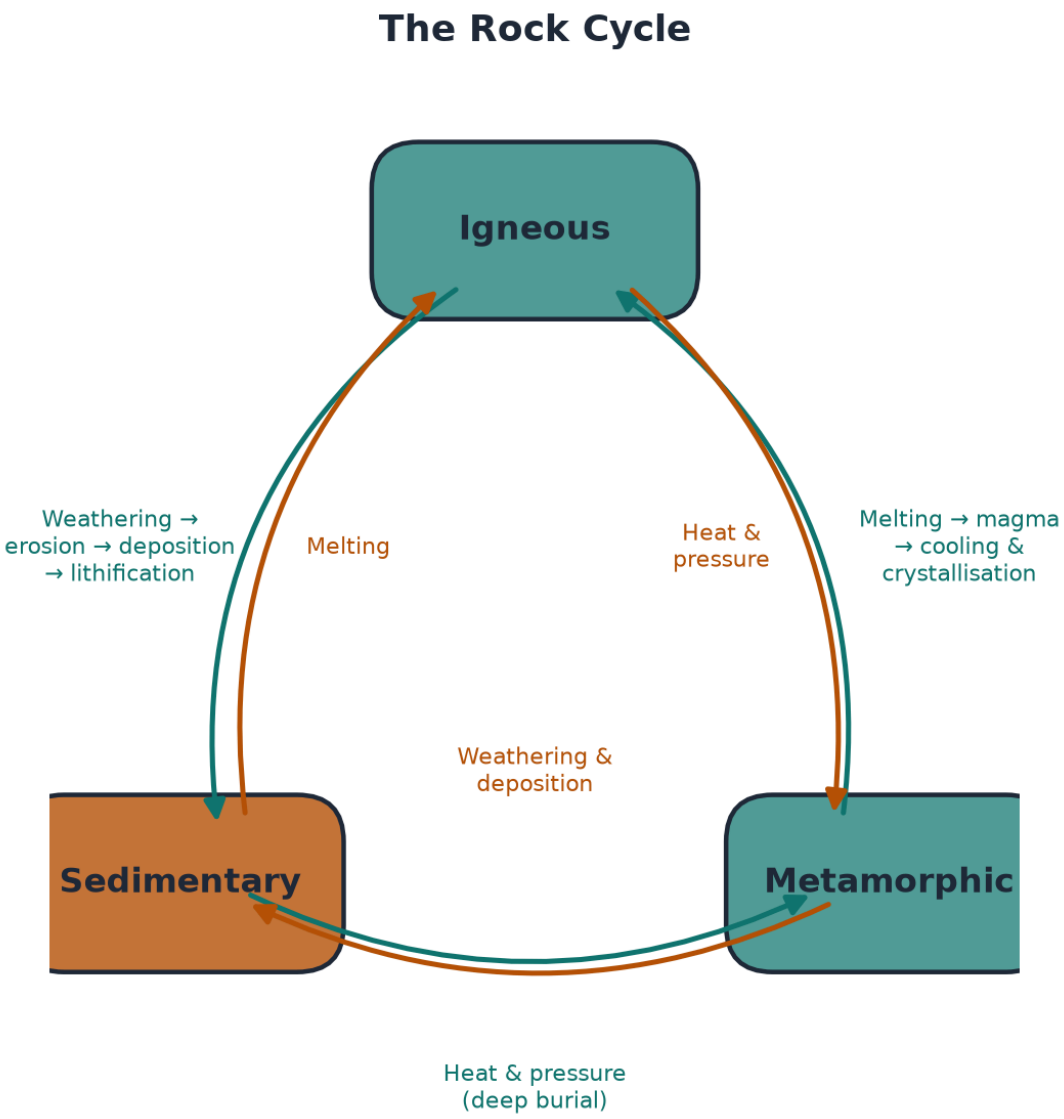


Figure 3.1 The rock cycle

Transformation pathways linking igneous, sedimentary and metamorphic rocks.

3.2 Definition of a Rock

A **rock** is a naturally occurring, hard, consolidated aggregate (mass) of one or more minerals.

- A rock made of a **single mineral** is called **monomineralic** (e.g., limestone - mostly calcite; rock salt - halite).
- A rock made of **several minerals** is called **polymineralic** (e.g., granite - quartz + feldspar + mica).

A **mineral** is a naturally occurring, inorganic, homogeneous solid with a definite chemical composition and a definite (usually crystalline) atomic structure.

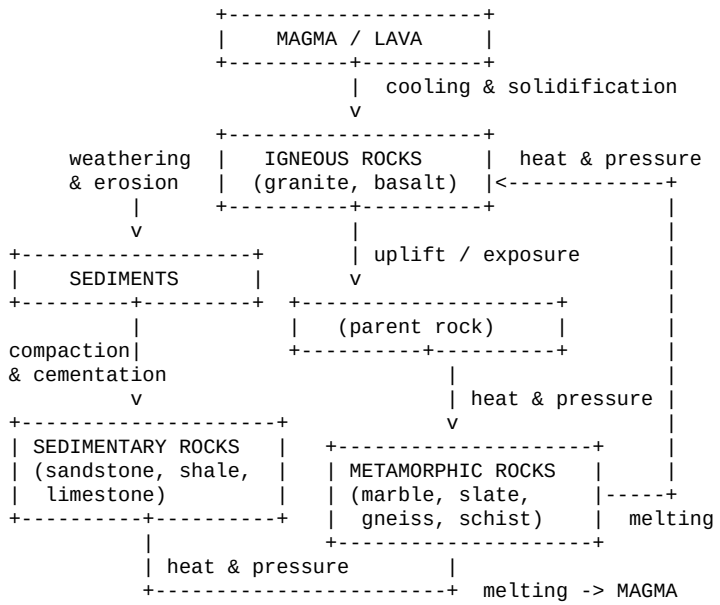
3.3 Formation of Rocks

Rocks are formed through three different processes, giving the three great classes:

1. **Igneous rocks** - formed by **cooling and solidification of magma or lava** (molten material). Also called **primary rocks**.
2. **Sedimentary rocks** - formed by **deposition, compaction and cementation of sediments** (fragments of older rocks and remains of organisms).
3. **Metamorphic rocks** - formed by the **alteration of pre-existing rocks** under heat, pressure and chemically active fluids.

3.4 The Rock Cycle

The rock cycle shows how the three rock types are interrelated and how one type can change into another over geological time.



3.5 Classification of Rocks

Rocks are classified mainly on the basis of their **mode of origin**:

Class	Mode of origin	Also called	Example
Igneous	Cooling and solidification of magma/lava	Primary rocks	Granite, basalt
Sedimentary	Deposition and cementation of sediments	Secondary/stratified rocks	Sandstone, limestone
Metamorphic	Alteration of existing rocks by heat/pressure	Altered rocks	Marble, slate, gneiss

3.6 Igneous Rocks

Definition: Rocks formed by the cooling and solidification of hot molten material (magma below the surface or lava on the surface).

Characteristics

- Called **primary rocks** - they were the first rocks formed on Earth.
- **No fossils** are found in them.
- Usually **crystalline** and massive; may be coarse or fine grained.
- Formed at **high temperature**; chemically they are rich in silicates.
- **Do not occur in layers (strata).**

Classification of igneous rocks

- On the basis of **silica content**:

- **Acidic** - >65% silica (e.g., granite).

- **Intermediate** - 55-65% silica (e.g., syenite, diorite).

- **Basic** - 45-55% silica (e.g., gabbro, basalt).

- **Ultrabasic** - <45% silica (e.g., peridotite).

- On the basis of **mode of occurrence**:

- **Plutonic (intrusive)** - solidified deep below the surface; **coarse grained** (e.g., granite, gabbro).

- **Volcanic (extrusive)** - solidified at the surface; **fine grained** (e.g., basalt, rhyolite).

Igneous rock	Type	Silica	Main minerals	Grain size
Granite	Acidic, intrusive	High	Quartz, feldspar, mica	Coarse
Rhyolite	Acidic, extrusive	High	Quartz, feldspar	Fine
Syenite	Intermediate, intrusive	Medium	Feldspar, little quartz	Coarse
Diorite	Intermediate, intrusive	Medium	Plagioclase, amphibole	Coarse
Gabbro	Basic, intrusive	Low	Plagioclase, pyroxene	Coarse
Basalt	Basic, extrusive	Low	Plagioclase, pyroxene	Fine

Soil significance: Granite weathers to **sandy, acidic soils** (rich in quartz). Basalt weathers to **dark, clayey, fertile soils** (black soils of Deccan). In Tamil Nadu, the **black cotton soils** over parts of the state are derived from basic igneous rocks.

3.7 Sedimentary Rocks

Definition: Rocks formed by the **deposition, compaction and cementation** of sediments (fragments of older rocks, or remains of plants and animals, or chemical precipitates).

Characteristics

- Called **secondary or stratified rocks** - they occur in **layers (strata)**.
- **Contain fossils** - evidence of past life.
- Usually **softer** than igneous rocks and often porous.
- Cover about **75% of the Earth's land surface** though only about 5% of the crust by volume.
- **Most soils form from sedimentary rocks** because these are exposed at the surface.

Classification of sedimentary rocks

- **Clastic (mechanical)** - from fragments of older rocks: sandstone (sand), shale (clay), conglomerate (gravel).
- **Organic** - from plant/animal remains: limestone (shells), coal (plants).
- **Chemical** - from chemical precipitation: rock salt, gypsum, some limestones.

Sedimentary rock	Type	Main composition	Soil formed
Sandstone	Clastic	Quartz grains	Sandy, well-drained soils
Shale	Clastic	Clay minerals	Clayey, poorly drained soils
Limestone	Organic/chemical	Calcium carbonate	Calcareous, alkaline soils
Conglomerate	Clastic	Pebbles + cement	Coarse, gravelly soils

Sedimentary rock	Type	Main composition	Soil formed
Rock salt	Chemical	Sodium chloride	Saline soils
Gypsum	Chemical	Calcium sulphate	Gypsiferous soils

Soil significance: Sandstone gives light sandy soils; shale gives heavy clay soils; limestone gives calcareous soils. In Tamil Nadu, **limestone occurs widely** (e.g., Ariyalur, Trichy) and gives calcareous black and red soils.

3.8 Metamorphic Rocks

Definition: Rocks formed by the **alteration (metamorphism)** of pre-existing igneous or sedimentary rocks under the action of **heat, pressure and chemically active fluids**.

Characteristics

- Called **altered rocks**.
- Formed in the **solid state** (without melting).
- Often show **foliation or banding** (parallel alignment of minerals).
- Usually **hard, compact and crystalline**.
- May contain fossils only if the original rock was sedimentary and metamorphism was mild.

Types of metamorphism

- **Thermal (contact) metamorphism** - due to heat from nearby magma.
- **Dynamic metamorphism** - due to pressure from earth movements.
- **Regional metamorphism** - due to both heat and pressure over large areas.

Original rock	Metamorphic rock	Change
Limestone	Marble	Recrystallised calcite
Sandstone	Quartzite	Fused quartz grains
Shale	Slate -> Schist	Clay -> mica
Granite	Gneiss	Banded minerals
Coal	Graphite / Diamond	Carbon recrystallised
Basalt	Schist / Amphibolite	Recrystallised

Soil significance: Metamorphic rocks are hard and resistant, weathering slowly. Quartzite gives sandy soils; marble and schist may give calcareous or micaceous soils.

3.9 Summary: The Three Rock Types at a Glance

Feature	Igneous	Sedimentary	Metamorphic
Origin	Cooling of magma	Deposition of sediments	Alteration of rocks
Other name	Primary	Secondary/stratified	Altered
Layering	Absent	Present (strata)	Foliation/banding
Fossils	Absent	Present	Usually absent
Hardness	Hard	Softer	Hard
Porosity	Low	High	Low-medium
Example	Granite, basalt	Sandstone, limestone	Marble, slate
Soil contribution	Parent of many soils	**Most soils**	Fewer soils

3.10 Exam Focus

Key points

- A rock is an aggregate of one or more minerals; the parent material of soil.
- Rocks are classified by origin into igneous, sedimentary and metamorphic.
- Igneous rocks form by cooling of magma/lava; they are primary, crystalline, fossil-free.

- Sedimentary rocks form by deposition and cementation; they are stratified, fossil-bearing, and give most soils.
- Metamorphic rocks form by alteration under heat and pressure; they show foliation/banding.
- The rock cycle links all three types through cooling, weathering, compaction and metamorphism.
- Granite -> sandy acidic soils; basalt -> dark fertile soils; limestone -> calcareous soils.
- Tamil Nadu: black soils from basic igneous rocks; calcareous soils from limestone belts.

Short-answer questions

1. *Define a rock and classify rocks on the basis of origin.*

Model answer: A rock is a naturally occurring aggregate of one or more minerals. On the basis of origin, rocks are classified as igneous (cooling of magma), sedimentary (deposition and cementation of sediments) and metamorphic (alteration of existing rocks by heat and pressure).

2. *Give two differences between igneous and sedimentary rocks.*

Model answer: (i) Igneous rocks form by cooling of magma while sedimentary rocks form by deposition and cementation of sediments. (ii) Igneous rocks are non-stratified and lack fossils, while sedimentary rocks occur in layers (strata) and contain fossils.

3. *What is the rock cycle?*

Model answer: The rock cycle is the continuous process by which the three rock types change into one another - magma cools to form igneous rocks; weathering and deposition form sedimentary rocks; heat and pressure form metamorphic rocks; and any rock may melt back into magma.

Long-answer questions

1. *Describe the classification of rocks with examples and their significance in soil formation.*

Outline: Definition of rock - basis of classification - igneous (formation, characteristics, types, examples) - sedimentary (formation, characteristics, types, examples) - metamorphic (formation, characteristics, examples) - soil significance of each - conclusion.

2. *Explain the formation of igneous, sedimentary and metamorphic rocks and draw the rock cycle.*

Outline: Introduction - igneous formation (magma, lava, intrusive/extrusive) - sedimentary formation (clastic, organic, chemical) - metamorphic formation (heat, pressure, types) - rock cycle diagram - conclusion.

Lecture 4: Brief Description of Important Rocks - Mineralogical Composition

4.1 Introduction

Each rock type has a characteristic **mineralogical composition** that decides the nature of the soil formed from it. This lecture describes the important rocks and the minerals they contain.

4.2 Minerals in Rocks - Basic Idea

- Rocks are made of **rock-forming minerals**.
- The most abundant rock-forming minerals are **silicates** - minerals built around the **silica tetrahedron (SiO_4)**.
- The **primary** rock-forming minerals are: **quartz, feldspars, micas, amphiboles, pyroxenes and olivine**.
- **Secondary** minerals (clay minerals, oxides, carbonates) form by the weathering of primary minerals.

4.3 Important Rocks and Their Mineralogical Composition

Rock	Class	Chief minerals	Colour	Soil formed
Granite	Igneous (acidic)	Quartz, orthoclase feldspar, mica	Light grey/pink	Sandy, acidic, well-drained
Basalt	Igneous (basic)	Plagioclase, pyroxene, olivine	Dark grey/black	Dark, clayey, fertile (black soils)
Gabbro	Igneous (basic)	Plagioclase, pyroxene	Dark	Clayey, iron-rich

Rock	Class	Chief minerals	Colour	Soil formed
Sandstone	Sedimentary	Quartz (cemented)	Variable	Sandy, porous
Shale	Sedimentary	Clay minerals, quartz	Grey	Clayey, impervious
Limestone	Sedimentary	Calcite (CaCO3)	White/grey	Calcareous, alkaline
Marble	Metamorphic	Recrystallised calcite	White	Calcareous
Slate	Metamorphic	Clay/mica (foliated)	Dark grey	Clayey
Gneiss	Metamorphic	Quartz, feldspar, mica (banded)	Banded	Sandy-clayey
Quartzite	Metamorphic	Quartz	White/pink	Sandy, poor in nutrients

4.4 Mineralogical Composition of Granite (Detail)

Granite is the most important **acidic intrusive** rock.

- **Quartz (SiO2)** - 20-40% - hard, resistant, gives sand.
- **Feldspars (orthoclase and plagioclase)** - 50-65% - weather to clay and release K, Na, Ca.
- **Mica (muscovite and biotite)** - 5-15% - source of K, Mg, Fe.
- **Accessory minerals** - hornblende, magnetite, apatite - small amounts.

Because granite is rich in quartz, it weathers to **light, sandy, often acidic soils** that are well drained but low in bases.

4.5 Mineralogical Composition of Basalt (Detail)

Basalt is the most important **basic extrusive** rock.

- **Plagioclase feldspar (calcium-rich)** - 40-60% - releases Ca, Na, Al.
- **Pyroxene (augite)** - 20-40% - source of Mg, Fe, Ca.
- **Olivine** - 0-20% in some basalts - rich in Mg and Fe.
- **Magnetite and ilmenite** - iron oxides giving dark colour.

Basalt is rich in **ferromagnesian minerals**, so it weathers rapidly to give **dark, clayey soils rich in bases** - the famous **black cotton soils (regur)**.

4.6 Minerals and the Elements They Supply to Soil

Mineral	Chemical formula (approx.)	Elements released	Weathers to
Quartz	SiO2	Si, O	Sand (resists weathering)
Orthoclase feldspar	KAlSi3O8	K, Al, Si	Clay + K
Plagioclase feldspar	(Na,Ca)AlSi3O8	Na, Ca, Al, Si	Clay + bases
Muscovite mica	KAl3Si3O10(OH)2	K, Al, Si	Clay + K
Biotite mica	K(Mg,Fe)3AlSi3O10(OH)2	K, Mg, Fe, Al	Clay + Fe oxides
Hornblende (amphibole)	Complex Ca-Mg-Fe silicate	Ca, Mg, Fe	Clay + Fe oxides
Augite (pyroxene)	Complex Ca-Mg-Fe silicate	Ca, Mg, Fe	Clay + Fe oxides
Olivine	(Mg,Fe)2SiO4	Mg, Fe	Serpentine, Fe oxides
Calcite	CaCO3	Ca	Ca bicarbonate (soluble)
Gypsum	CaSO4.2H2O	Ca, S	Soluble salts

4.7 Rock Colour and Soil Colour

- **Light-coloured (acidic) rocks** such as granite and quartzite give **light-coloured, sandy soils**.
- **Dark-coloured (basic) rocks** such as basalt and gabbro give **dark, iron-rich soils**.
- **Calcareous rocks** such as limestone give **light grey to white calcareous soils**.
- **Iron-rich rocks** give **red and yellow soils** after weathering (e.g., red soils of Tamil Nadu from gneiss and granite).

4.8 Tamil Nadu Examples

- **Red soils** - derived from **granite, gneiss and laterite**; occur in Coimbatore, Salem, Dharmapuri, Vellore.
- **Black soils** - derived from **basaltic/gneissic parent material**; occur in Coimbatore, Madurai, Tirunelveli, Ramanathapuram.
- **Laterite soils** - derived from **weathering of iron-rich rocks**; occur in Nilgiris, Kanyakumari, parts of Salem.
- **Alluvial soils** - derived from **transported sediments** of rivers; occur in the Cauvery delta (Thanjavur, Nagapattinam, Tiruvarur).
- **Coastal soils** - derived from **marine sediments and sand**; occur along the Coromandel and Gulf of Mannar coasts.

4.9 Exam Focus

Key points

- Each rock has a characteristic mineralogical composition that decides the soil formed.
- Granite: quartz + feldspar + mica -> sandy, acidic soils.
- Basalt: plagioclase + pyroxene + olivine -> dark, clayey, fertile soils.
- Sandstone -> sandy soils; shale -> clayey soils; limestone -> calcareous soils.
- Quartz resists weathering (gives sand); feldspars and micas weather to clay and release K, Na, Ca, Mg, Fe.
- Dark basic rocks give dark fertile soils; light acidic rocks give light poor soils.
- Tamil Nadu: red soils from granite/gneiss; black soils from basic rocks; laterite from iron-rich rocks.
- Ferromagnesian minerals (olivine, pyroxene, amphibole, biotite) supply Mg and Fe.

Short-answer questions

1. *Give the mineralogical composition of granite and the soil it forms.*

Model answer: Granite contains quartz (20-40%), feldspars (50-65%), mica (5-15%) and accessory minerals. Being rich in quartz, it weathers to light, sandy, well-drained and often acidic soils.

2. *Why do basaltic rocks give fertile black soils?*

Model answer: Basalt is rich in ferromagnesian minerals (plagioclase, pyroxene, olivine) that weather rapidly and release bases such as Ca, Mg, Fe and Na. This gives dark, clayey soils rich in bases and nutrients, i.e., the black cotton soils.

3. *Which mineral resists weathering and why is it important in soil?*

Model answer: Quartz (SiO_2) resists weathering and remains as sand, giving soils their skeleton, drainage and physical support to roots.

Long-answer questions

1. *Describe the important rocks and their mineralogical composition with examples from Tamil Nadu.*

Outline: Introduction - rock-forming minerals - granite, basalt, sandstone, shale, limestone, marble, slate, gneiss with compositions - soil formed from each - Tamil Nadu soil-rock relationships - conclusion.

2. *Explain how the mineralogical composition of a rock influences the nature and fertility of the soil formed from it.*

Outline: Introduction - acidic vs basic rocks - quartz-rich rocks give sandy poor soils - ferromagnesian-rich rocks give clayey fertile soils - calcareous rocks give alkaline soils - iron-rich rocks give red soils - examples - conclusion.

Lecture 5: Minerals - Definition, Occurrence, Classification of Important Soil Forming Primary Minerals - Silicate and Non-Silicate Minerals, Ferro and Non-Ferro Magnesium Minerals

5.1 Definition of a Mineral

A **mineral** is a naturally occurring, inorganic, homogeneous solid substance with a **definite chemical composition** and a **definite (usually crystalline) atomic structure**.

Key terms:

- **Naturally occurring** - not man-made.
- **Inorganic** - not formed by living organisms (though some, like calcite in shells, are biogenic, they are still mineral matter).
- **Definite chemical composition** - e.g., quartz is always SiO2.
- **Definite crystal structure** - atoms arranged in an orderly geometric pattern.

5.2 Occurrence of Minerals

- Minerals occur in **rocks** as their building units.
- They also occur as **soil particles** after weathering.
- They may occur as **primary minerals** (unchanged since the rock formed) or **secondary minerals** (formed by weathering or new formation in soil).
- In soil, minerals occur in three size fractions: **sand** (mostly primary), **silt** (mixed) and **clay** (mostly secondary).

5.3 Classification of Minerals

Minerals are classified in several ways:

A. On the basis of origin

- **Primary minerals** - formed at high temperature when the rock solidified (magma); e.g., quartz, feldspar, mica, olivine.
- **Secondary minerals** - formed by weathering or alteration of primary minerals; e.g., clay minerals, iron oxides, carbonates.

B. On the basis of chemical composition

- **Silicate minerals** - contain silicon and oxygen (SiO4 tetrahedron); e.g., quartz, feldspar, mica.
- **Non-silicate minerals** - do not contain the SiO4 tetrahedron; e.g., oxides, carbonates, sulphates, sulphides, halides.

C. On the basis of content of iron and magnesium

- **Ferro-magnesian (mafic) minerals** - rich in iron (Fe) and magnesium (Mg); e.g., olivine, pyroxene, amphibole, biotite.
- **Non-ferro-magnesian (felsic) minerals** - poor in Fe and Mg, rich in silica and aluminium; e.g., quartz, feldspar, muscovite.

5.4 Primary Minerals - Overview

Primary minerals are the original minerals of the rock. They are usually **coarse grained** and occur in the **sand and silt** fractions. They are the **storehouse of most plant nutrients** released slowly by weathering.

Important primary minerals:

- Quartz
- Feldspars (orthoclase, plagioclase)
- Micas (muscovite, biotite)
- Amphiboles (hornblende)
- Pyroxenes (augite)
- Olivine
- Accessory minerals (magnetite, apatite, zircon)

5.5 Silicate Minerals

Silicate minerals are built from the **silica tetrahedron** - one silicon atom surrounded by four oxygen atoms (SiO4)4-. Tetrahedra join in different ways to give different silicate structures.

Silicate group	Structure	Example	Weathering
Nesosilicates	Isolated tetrahedra	Olivine	Easily weathered
Sorosilicates	Double tetrahedra	Epidote	Moderate
Cyclosilicates	Rings	Beryl	Resistant
Inosilicates (single chain)	Single chains	Pyroxene (augite)	Moderate
Inosilicates (double chain)	Double chains	Amphibole (hornblende)	Moderate
Phyllosilicates	Sheets	Mica (biotite, muscovite)	Easily weathered
Tectosilicates	3-D framework	Quartz, feldspar	Quartz resistant; feldspar moderate

Key point: The more the tetrahedra are linked, the more resistant the mineral. **Quartz (framework, fully linked)** is the most resistant; **olivine (isolated tetrahedra)** is the least resistant.

5.6 Non-Silicate Minerals

Non-silicate minerals do not contain the SiO4 tetrahedron. They are grouped by their anion.

Group	Anion	Examples	Occurrence in soil
Oxides	O2-	Haematite (Fe2O3), magnetite (Fe3O4), gibbsite (Al(OH)3)	Red/yellow soil colour
Carbonates	CO3^2-	Calcite (CaCO3), dolomite	Calcareous soils
Sulphates	SO4^2-	Gypsum (CaSO4.2H2O)	Saline/gypsiferous soils
Sulphides	S2-	Pyrite (FeS2)	Acid sulphate soils
Halides	Cl-, F-	Halite (NaCl)	Saline soils
Phosphates	PO4^3-	Apatite	Source of P

Soil significance: Non-silicate minerals are mostly **secondary** and give soil its colour (iron oxides), its carbonate content (calcite) and its salt content (halite, gypsum).

5.7 Comparison: Silicate vs Non-Silicate Minerals

Feature	Silicate minerals	Non-silicate minerals
Basic unit	SiO4 tetrahedron	No SiO4 tetrahedron
Abundance	About 90% of crust	About 10% of crust
Origin	Mostly primary	Mostly secondary
Examples	Quartz, feldspar, mica	Haematite, calcite, gypsum
Colour contribution	Little	Strong (red, yellow, black)
Nutrient role	K, Na, Ca, Mg, Fe	Fe, Ca, S, P
Weathering	Variable	Variable; carbonates soluble

5.8 Ferro-Magnesian (Mafic) vs Non-Ferro-Magnesian (Felsic) Minerals

Feature	Ferro-magnesian (mafic)	Non-ferro-magnesian (felsic)
Rich in	Fe and Mg	Si and Al
Colour	Dark (green, black, brown)	Light (white, pink, grey)
Specific gravity	High	Lower
Examples	Olivine, pyroxene, amphibole, biotite	Quartz, feldspar, muscovite
Weathering	Weather rapidly	More resistant
Soil produced	Dark, clayey, fertile	Light, sandy, less fertile
Nutrients released	Fe, Mg, Ca	K, Na, Al

5.9 Weathering Sequence of Primary Minerals (Goldich Stability Series)

Minerals weather in a definite order - the **least stable weather first**:

```
LEAST STABLE (weathers first)
     |
     v
Olivine
Pyroxene (augite)
Amphibole (hornblende)
Biotite mica
Plagioclase feldspar (Ca-rich)
Plagioclase feldspar (Na-rich)
Orthoclase feldspar
Muscovite mica
Quartz
     ^
     |
MOST STABLE (resists weathering, remains as sand)
```

This is why **quartz dominates the sand fraction** of mature soils and **olivine and pyroxene disappear quickly**.

5.10 Primary vs Secondary Minerals

Feature	Primary minerals	Secondary minerals
Formed	When magma/rock solidified	By weathering or new formation
Temperature	High	Low (surface conditions)
Size fraction	Sand and silt	Clay and fine silt
Examples	Quartz, feldspar, mica, olivine	Clay minerals, iron oxides, calcite
Nutrient supply	Slow (weathering)	Faster (large surface area)
Solubility	Generally low	Variable

5.11 Exam Focus

Key points

- A mineral is a naturally occurring, inorganic solid with definite composition and crystal structure.
- Primary minerals form at high temperature in rocks; secondary minerals form by weathering.
- Silicates contain the SiO4 tetrahedron and make about 90% of the crust.
- Non-silicates (oxides, carbonates, sulphates, halides) make about 10% and control colour and salts.
- Ferro-magnesian minerals are dark, dense and weather rapidly to give fertile soils.
- Non-ferro-magnesian minerals are light and more resistant.
- Goldich's stability series: olivine least stable, quartz most stable.
- Quartz dominates sand; clay minerals dominate the clay fraction.

Short-answer questions

1. *Define a mineral and give two examples of primary and secondary minerals.*

Model answer: A mineral is a naturally occurring, inorganic, homogeneous solid with a definite chemical composition and crystal structure. Primary: quartz, feldspar. Secondary: kaolinite clay, haematite.

2. *Distinguish ferro-magnesian and non-ferro-magnesian minerals.*

Model answer: Ferro-magnesian minerals are rich in iron and magnesium, are dark coloured and weather rapidly (olivine, pyroxene, biotite). Non-ferro-magnesian minerals are rich in silica and aluminium, are light coloured and more resistant (quartz, feldspar, muscovite).

3. *Why is quartz abundant in the sand fraction of soils?*

Model answer: Quartz is a framework silicate with fully linked tetrahedra, making it the most stable and resistant mineral (Goldich series). It therefore survives weathering and accumulates as sand.

Long-answer questions

1. *Classify the important soil-forming primary minerals and describe them.*

Outline: Definition of mineral - primary vs secondary - silicate minerals (groups with examples) - non-silicate minerals (groups with examples) - ferro-magnesian vs non-ferro-magnesian - Goldich stability series - soil significance - conclusion.

2. *Explain the classification of minerals with special reference to silicate and non-silicate minerals.*

Outline: Introduction - basis of classification - silicates (SiO_4 tetrahedron, structural groups, examples) - non-silicates (oxide, carbonate, sulphate, halide groups) - comparison table - importance in soil - conclusion.

Lecture 6: Formation of Secondary Minerals - Clay Minerals and Amorphous Minerals

6.1 Introduction

Secondary minerals are those formed by the **weathering and alteration of primary minerals**, or by **new formation (neoformation)** from the products of weathering in the soil. They are mainly **clay-sized** (<2 micrometres) and are the most chemically active part of the soil. Two broad groups exist: **crystalline clay minerals** and **amorphous minerals**.

6.2 How Secondary Minerals Form

Secondary minerals form by two main pathways:

1. **Alteration (transformation)** - the primary mineral changes structure without complete dissolution; e.g., mica -> illite.
2. **Neoformation (precipitation)** - primary minerals dissolve completely, and the dissolved silica and alumina recombine to precipitate new minerals; e.g., kaolinite forming from solution.

The general weathering reaction is **hydrolysis**:

- Feldspar + water + CO_2 -> clay mineral + soluble bases + silica

Example:

- $2\ KAlSi_3O_8 + 2\ H^+ + 9\ H_2O \rightarrow Al_2Si_2O_5(OH)_4$ (kaolinite) $+ 4\ H_4SiO_4 + 2\ K^+$

6.3 Factors Controlling the Type of Secondary Mineral Formed

Factor	Effect
Climate (rainfall, temperature)	High rainfall + high temperature -> kaolinite, oxides (laterite)
Drainage / leaching	Poor drainage + high bases -> smectite (montmorillonite)
Parent material	Mica-rich rocks -> illite; basic rocks -> smectite
Concentration of Si and bases in solution	High Si + bases -> 2:1 clays; low Si -> 1:1 clays and oxides
pH	Alkaline -> 2:1 clays; acidic -> kaolinite/oxides
Time	Long time -> more stable minerals (kaolinite, oxides)

6.4 Clay Minerals - Definition

Clay minerals are **crystalline secondary hydrous aluminium (or magnesium) silicates** of very small particle size (usually <2 micrometres) with a **layered (sheet) structure**. They are the products of weathering of primary silicates.

They are called **layer silicates** or **phyllosilicates** because their structure is made of sheets.

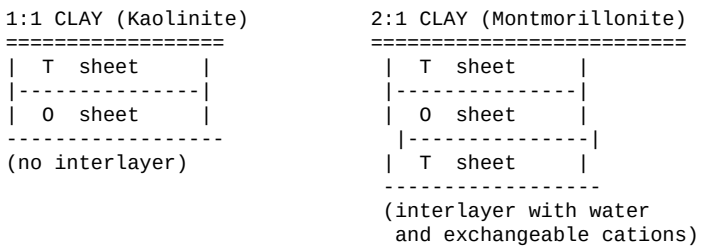
6.5 Basic Building Blocks of Clay Minerals

Clay minerals are built from two types of sheets:

- **Silica tetrahedral sheet** - made of SiO_4 tetrahedra sharing three oxygens, forming a hexagonal sheet. (T sheet)
- **Alumina octahedral sheet** - made of Al (or Mg) octahedra, with Al in the centre and OH/O at the corners. (O sheet)

A **1:1 clay** has one tetrahedral sheet + one octahedral sheet (e.g., kaolinite).

A **2:1 clay** has one octahedral sheet sandwiched between two tetrahedral sheets (e.g., montmorillonite, illite).



6.6 Types of Clay Minerals (Brief)

- **1:1 type** - kaolinite, halloysite. Non-expanding, low CEC (3-15 cmol/kg), low surface area.
- **2:1 expanding type** - montmorillonite (smectite), vermiculite. High CEC (80-150 cmol/kg), high surface area, shrink-swell.
- **2:1 non-expanding type** - illite (mica-like), with K in the interlayer. Moderate CEC (20-40 cmol/kg).
- **2:2 (or 2:1:1) type** - chlorite, with a brucite (Mg(OH)2) interlayer. Low-moderate CEC.
- **Amorphous** - allophane, imogolite, and oxides/hydroxides of Fe and Al.

(Structure and classification are treated in detail in Lecture 25.)

6.7 Amorphous Minerals

Amorphous minerals are secondary minerals that **lack a definite crystal structure** - their atoms are arranged randomly, not in an orderly lattice.

Examples

- **Allophane** and **imogolite** - formed from volcanic ash; hydrous aluminium silicates.
- **Iron oxides/hydroxides** - ferrihydrite (poorly crystalline), goethite, haematite.
- **Aluminium hydroxides** - gibbsite, boehmite.
- **Silica** - opaline silica (diatoms, phytoliths).
- **Manganese oxides** - birnessite.

Properties of amorphous minerals

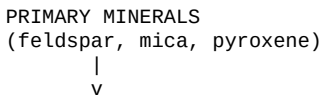
- Very high **specific surface area**.
- High **pH-dependent charge** (variable charge).
- High **water-holding capacity** and high **phosphate fixation**.
- Form weak, non-sticky **crumb** aggregates.

6.8 Crystalline vs Amorphous Secondary Minerals

Feature	Crystalline clay minerals	Amorphous minerals
Structure	Ordered lattice	Random, no lattice
Examples	Kaolinite, montmorillonite	Allophane, ferrihydrite
Surface area	High	Very high
Charge	Permanent + variable	Mainly variable (pH-dependent)
CEC	Varies (3-150)	High but pH-dependent
Occurrence	Most soils	Volcanic soils (Andisols), young soils
X-ray diffraction	Sharp peaks	Broad/absent peaks

6.9 Formation Sequence (Weathering Pathway)

As weathering proceeds, minerals change in a definite order:



```
INTERMEDIATE / ILLITE, SMECTITE
(2:1 clays - moderate weathering)
       |
       v
KAOLINITE (1:1 clay - strong weathering)
       |
       v
OXIDES & HYDROXIDES (Fe2O3, Al2O3)
(extreme weathering / laterite)
```

This sequence is controlled mainly by the **intensity of leaching** (rainfall) and **temperature**. In the humid tropics (parts of Tamil Nadu, Western Ghats), weathering proceeds to kaolinite and iron/aluminium oxides, giving **red and laterite soils**.

6.10 Worked Example: Percent Base Loss during Weathering

Problem: A feldspar contains 12% K2O. After weathering, the resulting clay contains only 1.5% K2O. Calculate the percentage loss of K2O during weathering.

Solution

- Initial K2O = 12%
- Final K2O in clay = 1.5%
- Loss = 12 - 1.5 = **10.5%**
- Percentage loss = (10.5 / 12) x 100 = **87.5%**

Answer: About 87.5% of the K2O is lost (leached) during weathering, and the rest is retained in the clay.

6.11 Tamil Nadu Context

- **Red soils** of Coimbatore, Salem and Vellore contain mainly **kaolinite** (1:1 clay) - formed by strong leaching in the sub-humid climate.
- **Black soils** of Madurai and Coimbatore contain **montmorillonite (smectite)** - formed under poor drainage with high bases.
- **Laterite soils** of the Nilgiris and Kanyakumari contain **kaolinite + iron and aluminium oxides** - extreme weathering.
- **Alluvial soils** of the Cauvery delta contain a **mixture of illite, kaolinite and smectite** carried from upstream.

6.12 Exam Focus

Key points

- Secondary minerals form by weathering (transformation or neoformation) of primary minerals.
- Clay minerals are crystalline hydrous aluminium/magnesium silicates with a layered structure.
- Building blocks: tetrahedral (T) sheet and octahedral (O) sheet.
- 1:1 clays (kaolinite) - non-expanding, low CEC; 2:1 clays (montmorillonite, illite) - higher CEC.
- Amorphous minerals (allophane, ferrihydrite) lack crystal structure; have high surface area and pH-dependent charge.
- Weathering sequence: primary -> 2:1 clays -> kaolinite -> oxides (laterite).
- High rainfall + high temperature -> kaolinite and oxides; poor drainage + bases -> smectite.
- Tamil Nadu: red soils = kaolinite; black soils = smectite; laterite = kaolinite + oxides.

Short-answer questions

1. *What are secondary minerals? How do they form?*

Model answer: Secondary minerals are those formed by the weathering and alteration of primary minerals or by new formation in soil. They form by transformation (e.g., mica to illite) or neoformation (dissolved silica and alumina recombine, e.g., kaolinite), mainly through hydrolysis.

2. *Define clay minerals and name the two building sheets.*

Model answer: Clay minerals are crystalline hydrous aluminium (or magnesium) silicates of very small size with a layered structure. They are built from the silica tetrahedral sheet and the alumina octahedral sheet.

3. *What are amorphous minerals? Give two examples.*

Model answer: Amorphous minerals are secondary minerals without a definite crystal structure, having randomly arranged atoms. Examples: allophane and ferrihydrite.

Long-answer questions

1. *Explain the formation of secondary minerals with special reference to clay minerals and amorphous minerals.*

Outline: Introduction - weathering pathways (transformation, neoformation) - hydrolysis reaction - factors controlling type of clay - clay minerals definition and building blocks - types of clays - amorphous minerals and properties - weathering sequence - conclusion.

2. *Describe clay minerals and amorphous minerals and compare their properties.*

Outline: Introduction - clay minerals (definition, structure, types) - amorphous minerals (definition, examples) - comparison table (structure, surface area, charge, CEC) - soil significance - Tamil Nadu examples - conclusion.

Lecture 7: Weathering - Rocks and Minerals - Physical, Chemical and Biological Weathering

7.1 Definition of Weathering

Weathering is the **disintegration (physical breakdown) and decomposition (chemical breakdown) of rocks and minerals in situ** (in place) by the action of physical, chemical and biological agents.

Key points:

- Weathering occurs **in place** - the products are not transported (that is erosion).
- It is the **first step** in soil formation.
- It converts **rock** into **regolith** (the loose, weathered mantle) and finally into soil.

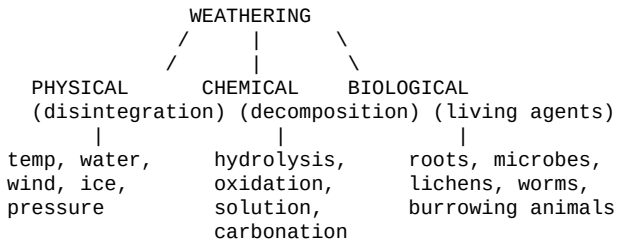
7.2 Difference between Weathering and Erosion

Feature	Weathering	Erosion
Process	Breakdown in place	Removal and transport
Movement	No transport	Material is transported
Agents	Temperature, water, organisms	Water, wind, ice, gravity
Result	Regolith / soil	Sediment elsewhere

7.3 Types of Weathering

Weathering is of three types:

1. **Physical (mechanical) weathering** - breakdown into smaller pieces without change in chemical composition.
2. **Chemical weathering** - decomposition with change in chemical composition; new minerals form.
3. **Biological weathering** - breakdown by the action of living organisms (plants, animals, microbes).



7.4 Physical Weathering

Physical weathering breaks rock into smaller fragments **without any chemical change**. The main agents are:

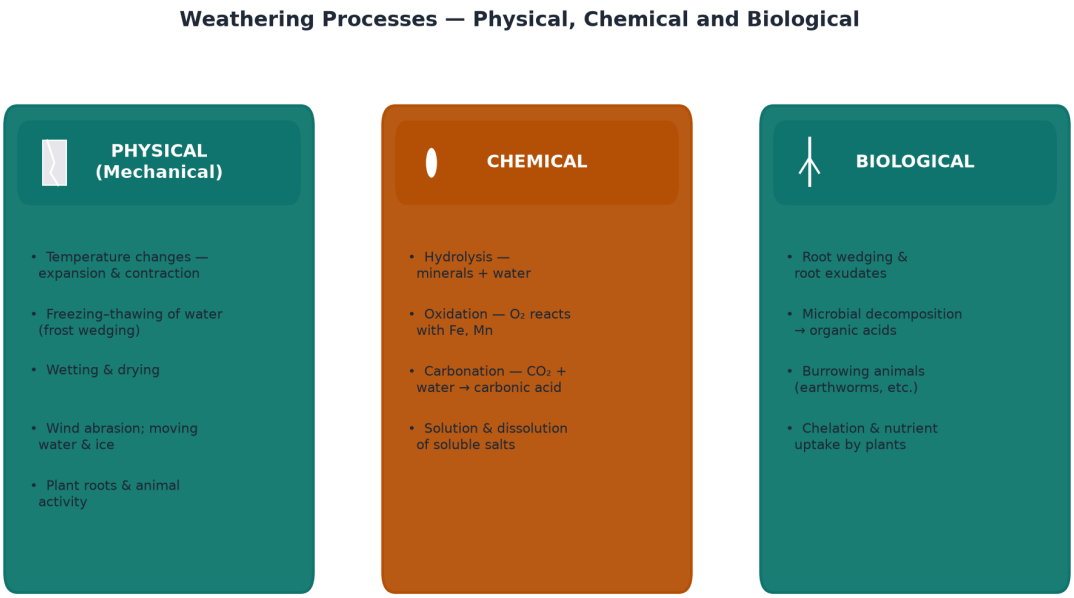


Figure 7.1 Weathering processes

Physical (mechanical), chemical and biological weathering of rocks and minerals.

A. Temperature changes

- **Exfoliation / thermal expansion** - rocks expand when heated and contract when cooled; repeated cycles cause the outer layers to peel off (exfoliation).
- **Block disintegration** - in deserts, large day-night temperature differences crack rocks into blocks.

B. Frost action (freeze-thaw)

- Water enters cracks, freezes, expands by about 9% and forces the rock apart. Repeated freeze-thaw cycles shatter rocks (frost wedging).

C. Crystal growth (salt weathering)

- Salt crystals growing in pores exert pressure and break the rock (salt crystallisation).

D. Water and ice

- **Raindrop impact**, **running water** and **glacial abrasion** wear away rock surfaces.

E. Wind

- Wind-blown sand abrades rock surfaces (abrasion / deflation).

F. Organic / biotic (also listed under biological)

- Growing roots and burrowing animals widen cracks.

Factors promoting physical weathering: large temperature range, water availability (freezing), low vegetation, exposed rock surface.

7.5 Chemical Weathering

Chemical weathering **changes the chemical composition** of minerals and forms new (secondary) minerals. The main processes are:

A. Hydrolysis (most important)

- Reaction of minerals with water; H+ replaces cations in the mineral.
- Example: Orthoclase feldspar -> kaolinite + silica + K+
- 2 KAlSi3O8 + 2 H+ + 9 H2O -> Al2Si2O5(OH)4 + 4 H4SiO4 + 2 K+

B. Oxidation

- Reaction with oxygen; iron (Fe2+) is oxidised to Fe3+.
- Example: 4 FeO + O2 -> 2 Fe2O3 (gives red colour)
- Example: 2 FeS2 + 7 O2 + 2 H2O -> 2 FeSO4 + 2 H2SO4 (pyrite oxidation - acid sulphate soils)

C. Carbonation

- Reaction with carbonic acid (H2CO3) formed from CO2 and water.
- Example: CaCO3 + H2CO3 -> Ca(HCO3)2 (soluble; limestone dissolves)

D. Solution / Dissolution

- Simple dissolving of soluble minerals.
- Example: NaCl (halite) dissolves in water; gypsum dissolves slowly.

E. Hydration

- Water combines with a mineral to form a new hydrate.
- Example: 2 Fe2O3 + 3 H2O -> 2 Fe2O3.3H2O (limonite); anhydrite -> gypsum

F. Acid attack (H+ from organic acids, CO2, acid rain)

- Accelerates hydrolysis and solution.

Factors promoting chemical weathering: high temperature, high rainfall, high CO2, high organic acids, fine particles, long time, good drainage.

7.6 Biological Weathering

Biological weathering is the breakdown of rocks by **living organisms** - both physical and chemical effects.

A. Higher plants

- **Roots** grow into cracks and split rocks (physical).

- Roots release **organic acids and CO2** that dissolve minerals (chemical).
- **Root respiration** produces CO2 and carbonic acid.

B. Microorganisms

- **Lichens and mosses** growing on rock surfaces release acids and chelating compounds that dissolve minerals.
- **Bacteria** (e.g., nitrifying, sulphur-oxidising) produce strong acids (HNO3, H2SO4).
- **Fungi** produce organic acids and chelate Fe and Al.

C. Animals

- **Earthworms**, ants, termites and burrowing animals mix and break soil material.
- **Cattle and other animals** trample and loosen material.

D. Human activity

- Ploughing, mining, quarrying, construction break rock and soil.

7.7 Comparison of the Three Types of Weathering

Feature	Physical weathering	Chemical weathering	Biological weathering
Nature of change	Size only	Chemical composition	Both (physical + chemical)
Agents	Temp, frost, wind, salt	Water, O2, CO2, acids	Plants, microbes, animals
Products	Fragments of same mineral	New (secondary) minerals	Both fragments and new minerals
Main regions	Cold, dry (deserts)	Hot, humid (tropics)	Where life is active
Speed	Rapid for weak rocks	Slow but continuous	Variable

7.8 Relative Resistance of Minerals to Weathering

Using **Goldich's stability series** (Lecture 5):

- **Least resistant** (weather first): olivine, pyroxene, amphibole, biotite.
- **Intermediate**: plagioclase, orthoclase, muscovite.
- **Most resistant** (weather last): quartz.

Hence sandy soils (rich in quartz) are common in highly weathered regions.

7.9 Weathering and Soil Formation

- Physical weathering **increases surface area**, speeding up chemical weathering.
- Chemical weathering **produces clay minerals**, which give soil its fertility and water-holding capacity.
- Biological weathering **adds organic matter** and nutrients.
- Together, weathering converts **rock -> regolith -> soil**.

7.10 Worked Example: Volume Increase by Frost Action

Problem: Water freezes and expands by 9% in volume. If 10 cm3 of water fills a crack, what is the volume of ice formed, and by how much does the crack widen?

Solution

- Volume of water = 10 cm3
- Expansion = 9% of 10 = 0.9 cm3
- Volume of ice = 10 + 0.9 = **10.9 cm3**
- Increase in volume = **0.9 cm3**

Answer: The ice occupies 10.9 cm3, an increase of 0.9 cm3, which exerts pressure and widens the crack.

7.11 Exam Focus

Key points

- Weathering = disintegration and decomposition of rocks and minerals in place.
- Weathering occurs in situ; erosion involves transport.
- Physical weathering: temperature, frost, salt, wind, water - size change only.
- Chemical weathering: hydrolysis, oxidation, carbonation, solution, hydration - composition change.
- Hydrolysis is the most important chemical weathering process and produces clay.
- Biological weathering: roots, microbes, lichens, animals, humans - both physical and chemical.
- Physical weathering increases surface area; chemical weathering produces clay.
- Goldich series: olivine least resistant, quartz most resistant.

Short-answer questions

1. *Define weathering and distinguish it from erosion.*

Model answer: Weathering is the in-place disintegration and decomposition of rocks and minerals by physical, chemical and biological agents. Erosion involves the removal and transport of the weathered material. Weathering does not move material; erosion does.

2. *Name four processes of chemical weathering with one example each.*

Model answer: Hydrolysis (feldspar to kaolinite), oxidation (FeO to Fe2O3), carbonation (CaCO3 to soluble Ca(HCO3)2) and solution (halite dissolving in water).

3. *How does frost action weather rocks?*

Model answer: Water enters cracks in rock, freezes, expands by about 9% in volume and exerts pressure that widens and splits the rock. Repeated freeze-thaw cycles cause frost wedging and shatter the rock.

Long-answer questions

1. *Describe physical, chemical and biological weathering with examples.*

Outline: Definition - physical weathering (agents and processes) - chemical weathering (hydrolysis, oxidation, carbonation, solution, hydration with equations) - biological weathering (plants, microbes, animals) - comparison table - significance in soil formation - conclusion.

2. *Explain the role of weathering in soil formation.*

Outline: Introduction - weathering vs erosion - physical weathering increases surface area - chemical weathering produces clay and releases nutrients - biological weathering adds organic matter - rock-regolith-soil sequence - factors - Goldich series - conclusion.

Lecture 8: Factors of Soil Formation - Passive and Active Soil Forming Factors

8.1 Introduction

Soil is formed by the combined action of several factors acting on parent material over time. The study of these factors is central to pedology. The classical framework was given by **V.V. Dokuchaev** and later expressed in the **Jenny equation** by **Hans Jenny (1941)**.

8.2 The Jenny Equation (Fundamental Equation of Soil Formation)

s = f (cl, o, r, p, t, ...)

Where:

- s = the soil (or any soil property)
- cl = **climate**
- o = **organisms** (biota)
- r = **relief** (topography)
- p = **parent material**
- t = **time**
- ... = additional factors (e.g., groundwater, human activity)

This means soil is a **function** of these factors. Change any one factor and the soil changes.

8.3 Classification of Soil Forming Factors: Passive vs Active

Soil forming factors are grouped into **passive** and **active** factors.

- **Passive factors** - they supply the **material** or the **setting** for soil formation but do not themselves provide the energy to drive the process. They are: **parent material** and **relief (topography)**.
- **Active factors** - they provide the **energy and the driving force** for soil formation. They are: **climate**, **organisms** and **time**.

Type	Factors	Role
Passive	Parent material, Relief/topography	Provide material and setting
Active	Climate, Organisms, Time	Provide energy and drive the process

Note: Time is sometimes treated separately as it is not an "agent" but a dimension; many texts list climate and organisms as the truly active agents.

8.4 Parent Material (Passive Factor)

Definition: The parent material is the **unconsolidated, weathered mineral or organic material** from which the soil develops.

Types

- **Residual (in situ)** - formed from the underlying rock; soil lies over its own parent rock.
- **Transported** - carried and deposited elsewhere:

- **Colluvial** - by gravity (hill slopes).

- **Alluvial** - by rivers (Cauvery delta).

- **Glacial** - by ice.

- **Aeolian** - by wind (loess).

- **Marine** - by sea (coastal soils).

Influence

- Determines the **initial mineral composition** and nutrient reserve.
- Influences **texture** (sandstone -> sandy; shale -> clayey) and **colour**.
- Controls **base status** (limestone -> calcareous; granite -> acidic).
- Its influence **decreases** as soil matures (climate becomes dominant).

8.5 Relief / Topography (Passive Factor)

Definition: Relief refers to the **lay of the land** - its slope, elevation, aspect and position in the landscape.

Influence

- **Slope steepness:** steep slopes -> rapid runoff, thin soils, erosion; gentle slopes -> deep soils.
- **Aspect:** slope facing the sun is warmer and drier; the shaded slope is cooler and wetter.
- **Elevation:** temperature falls with altitude, changing vegetation and soil (e.g., Nilgiris).
- **Drainage position:** uplands are well drained (leached); lowlands are poorly drained (accumulation of salts and clay).
- **Groundwater:** a shallow water table gives waterlogged, grey (hydric) soils.

Relief causes **local differences** in soil even within the same climate and parent material - this is called a **catena** (a sequence of soils down a slope).

8.6 Climate (Active Factor)

Definition: Climate is the **long-term average weather** - mainly **temperature and rainfall** - of a region.

Why it is the most important active factor

- Climate **supplies the energy** (heat) and the **water** that drive weathering and leaching.
- It controls the **type and rate** of weathering.

- It controls **vegetation** (which adds organic matter).
- It determines whether soil is leached (humid) or accumulates salts (arid).

Effects of rainfall

- High rainfall -> strong leaching -> acidic, kaolinite-rich, base-poor soils.
- Low rainfall -> little leaching -> base-rich, calcareous, saline soils.

Effects of temperature

- High temperature -> faster chemical reactions and microbial activity.
- Low temperature -> slow weathering, accumulation of organic matter (peat).

Climate and soil zones

Climate	Dominant process	Soil formed
Hot & humid	Strong leaching, laterization	Laterite, red soils
Sub-humid	Moderate leaching	Red, brown soils
Semi-arid	Moderate accumulation	Black soils, calcification
Arid	Evaporation, salinization	Saline, desert soils

8.7 Organisms / Biota (Active Factor)

Definition: Organisms include **vegetation, microbes, animals and humans** that live in or on the soil.

Influence

- **Vegetation:** roots weather minerals, add organic matter, recycle nutrients. Grassland -> dark, humus-rich soils; forest -> more acidic, podzolic soils.
- **Microorganisms:** decompose organic matter, fix nitrogen, solubilise minerals.
- **Earthworms and burrowing animals:** mix soil, improve structure and aeration.
- **Humans:** ploughing, irrigation, fertilization, liming, drainage - strongly modify soils (a "human" factor).

The **O, A and B horizons** owe much of their character to organisms. Organic matter is the main product of the biotic factor.

8.8 Time (Active Factor)

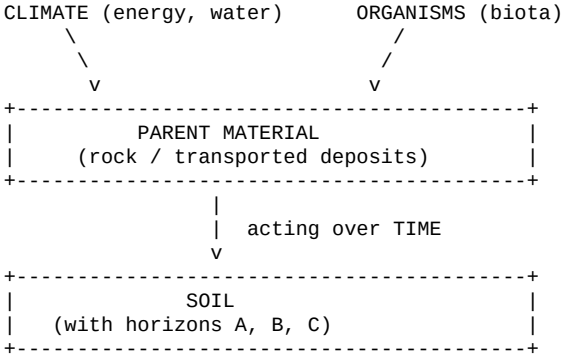
Definition: Time is the **duration** over which the other factors have acted.

Influence

- Young soils (e.g., alluvial soils of the Cauvery delta) are **shallow, weakly developed**, retain parent material features.
- Old soils (e.g., laterite of the Nilgiris) are **deeply weathered**, leached and rich in oxides.
- Soil development passes through stages: **young -> mature -> old (senile)**.
- Given enough time, soils reach a **steady state** with the climate.

Rate: Soil formation is **very slow** - roughly **1 cm of soil may take 100 to 1000 years** to form (a widely quoted range; actual rates vary greatly).

8.9 Diagram: The Five Factors Acting on Parent Material



Mounish Raj · MouRa · Agri AI

```
              ^
              |
      RELIEF / TOPOGRAPHY
(slope, drainage, aspect, elevation)
```

8.10 Soil Forming Factors Summary Table

Factor	Type	Main role	Example effect
Parent material	Passive	Supplies material	Sandstone -> sandy soil
Relief	Passive	Setting, drainage	Steep slope -> thin soil
Climate	Active	Energy and water	Humid -> leached acidic soil
Organisms	Active	Organic matter, mixing	Grassland -> dark soil
Time	Active	Duration of action	Old -> laterite; young -> alluvium

8.11 Tamil Nadu Context

- **Cauvery delta alluvial soils** are **young** (time) and derived from **transported parent material** (alluvium).
- **Laterite soils of the Nilgiris** are **old** and formed under a **humid, cool climate** (climate + time).
- **Black soils** form where **relief** gives poor drainage and high bases (relief + parent material).
- **Coastal sandy soils** are **young marine deposits** (parent material + time).

8.12 Exam Focus

Key points

- Jenny equation: s = f (cl, o, r, p, t, ...).
- Passive factors: parent material and relief; active factors: climate, organisms, time.
- Parent material decides initial minerals, texture and base status.
- Relief decides slope, drainage and depth; causes catena.
- Climate is the most important active factor - controls weathering and leaching.
- Organisms add organic matter and mix soil; humans modify soil strongly.
- Time decides maturity - young soils are shallow, old soils are deeply weathered.
- Soil formation is slow: roughly 1 cm in 100-1000 years (standard range).

Short-answer questions

1. *List the five factors of soil formation and give the Jenny equation.*

Model answer: The factors are climate, organisms, relief, parent material and time. Jenny's equation is s = f (cl, o, r, p, t), meaning soil is a function of these factors.

2. *Distinguish passive and active soil forming factors.*

Model answer: Passive factors (parent material and relief) supply the material and setting for soil formation but do not provide energy. Active factors (climate, organisms and time) provide the energy and drive the process of soil formation.

3. *How does relief affect soil formation?*

Model answer: Relief controls slope, drainage, aspect and elevation. Steep slopes give thin, eroded soils; gentle slopes give deep soils. Uplands are leached and acidic; lowlands are poorly drained and accumulate salts and clay, producing a catena down the slope.

Long-answer questions

1. *Describe the factors of soil formation, distinguishing passive and active factors.*

Outline: Introduction - Jenny equation - passive factors (parent material, relief) with types and effects - active factors (climate, organisms, time) with effects - diagram - summary table - Tamil Nadu examples - conclusion.

2. *"Climate is the most important soil forming factor." Discuss.*

Outline: Introduction - list factors - why climate is active - effect of rainfall (leaching) - effect of temperature - climate-soil zones table - comparison with other factors - Tamil Nadu examples - conclusion.

Lecture 9: Soil Forming Process - Fundamental - Simenson's Four Fold Soil Forming Process: Eluviation, Illuviation, Translocation and Humification

9.1 Introduction

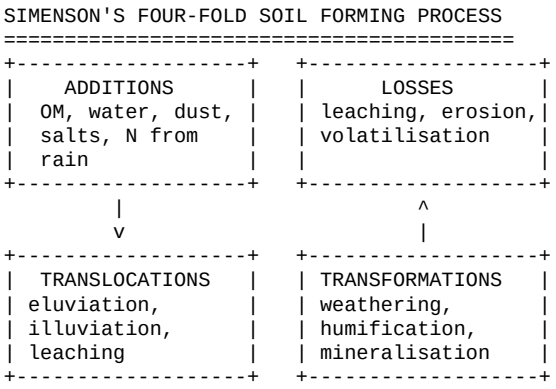
Soil forming **processes** are the actual operations by which the soil body is built from the parent material. The great Russian soil scientist **Simenson** (also written Simonson) described soil formation as a combination of **fundamental (general) processes** and **specific (special) processes**. The fundamental processes operate in almost all soils; the specific processes give each soil its distinctive character.

9.2 Simenson's Four-Fold Fundamental Soil Forming Process

Simenson grouped the fundamental processes into **four groups**:

1. **Additions** - material added to the soil (organic matter, water, dust, salts).
2. **Losses** - material removed from the soil (leaching, erosion, volatilisation).
3. **Translocations** - material moved within the soil (eluviation, illuviation).
4. **Transformations** - material changed in form within the soil (weathering, humification, mineralisation).

Note: Eluviation, illuviation and humification are the widely examined terms. Eluviation and illuviation are the two parts of **translocation**.



9.3 Additions

Material **added** to the soil from outside:

- **Organic matter** from plant litter, roots and animals.
- **Water** from rainfall and irrigation.
- **Dissolved salts and nutrients** from rain, dust and groundwater.
- **Nitrogen** fixed from the atmosphere by microbes.
- **Oxygen and CO2** exchanged with the atmosphere.

9.4 Losses

Material **removed** from the soil:

- **Leaching** - soluble salts and colloids washed downward out of the soil.
- **Erosion** - removal of surface soil by water or wind.
- **Volatilisation** - loss of N as ammonia or nitrogen gas.
- **Crop removal** - nutrients taken up by harvested crops.
- **Decomposition losses** - CO2 released to the atmosphere.

9.5 Transformations

Material **changed in form** within the soil:

- **Weathering** - primary minerals change to secondary minerals (clay).

- **Humification** - fresh organic matter changes to humus.
- **Mineralisation** - organic matter decomposes to release inorganic nutrients.
- **Synthesis** - new minerals (clays, oxides) form.

9.6 Translocation

Translocation is the **movement of material within the soil** from one horizon to another, either downward (eluviation) or upward (capillary rise), or in solution or suspension.

Two important parts of translocation:

- **Eluviation** - movement of material **out of** a horizon.
- **Illuviation** - movement of material **into** a horizon.

9.7 Eluviation

Definition: Eluviation is the **movement (washing out) of soil material from an upper horizon downward** by water, in solution or in suspension. The horizon that loses material is called the **eluviated horizon (E horizon or A2 horizon)**.

What is eluviated

- **Clay** (in suspension).
- **Iron and aluminium oxides** (in suspension/solution).
- **Organic matter** (in solution).
- **Soluble salts and bases** (Ca, Mg, Na, K in solution).

Result: The eluviated horizon is **light coloured, sandy, low in clay and low in bases** - often pale grey or whitish.

Example: In a podzol, the **ash-grey E horizon** is a strongly eluviated horizon.

9.8 Illuviation

Definition: Illuviation is the **deposition and accumulation of material that has been eluviated from an upper horizon** into a lower horizon. The horizon that gains material is called the **illuvial horizon (B horizon)**.

What is illuviated

- **Clay** -> **argillic (Bt) horizon**.
- **Iron and aluminium oxides** -> **spodic (Bs) horizon** (in podzols).
- **Organic matter** -> **humic horizon**.
- **Calcium carbonate** -> **calcic (Bk) horizon**.
- **Salts** -> **salic horizon**.

Result: The illuvial horizon is **darker, richer in clay, iron or carbonates** than the horizon above.

9.9 Eluviation vs Illuviation

Feature	Eluviation	Illuviation
Meaning	Washing out of material	Washing in of material
Direction	Downward removal from a horizon	Downward deposition into a horizon
Horizon	E (eluviated) horizon	B (illuvial) horizon
Result	Light, sandy, base-poor	Dark, clayey, base/oxide-rich
Material	Clay, Fe/Al oxides, OM, salts	Clay, Fe/Al oxides, OM, CaCO3
Colour	Pale grey/white	Red, brown, dark

9.10 Humification

Definition: Humification is the **process by which fresh organic matter (plant and animal residues) is decomposed and converted into humus** - a dark, amorphous, complex, resistant organic substance.

Steps

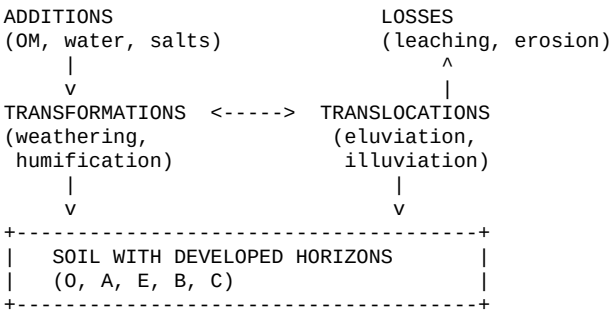
1. **Decomposition** - microbes break down complex organic compounds.
2. **Synthesis** - simple products recombine to form new complex compounds (humic and fulvic acids, humin).
3. **Accumulation** - humus builds up in the A horizon.

Factors affecting humification

- **Temperature** - optimum 25-35 degrees C favours rapid humification.
- **Moisture** - adequate moisture is needed.
- **Aeration** - aerobic conditions favour humification; anaerobic conditions give peat.
- **C:N ratio** - narrow C:N (legumes) humifies fast; wide C:N (straw) humifies slowly.
- **Microbes** - bacteria and fungi carry out the process.

Product: Humus - dark brown/black, colloidal, high CEC, high water-holding capacity, improves structure. (Detailed in Lecture 28.)

9.11 Relationship between the Fundamental Processes



9.12 Worked Example: Clay Movement by Eluviation

Problem: An A horizon originally contained 30% clay. After eluviation, it now contains 18% clay, and the B horizon below now contains 42% clay (originally 30%). Calculate the clay lost from A and gained by B (assuming equal soil masses).

Solution

- Clay lost from A = 30 - 18 = **12%**
- Clay gained by B = 42 - 30 = **12%**

Answer: 12% of clay has been eluviated from the A horizon and illuviated into the B horizon, showing translocation of clay.

9.13 Exam Focus

Key points

- Simenson described four fundamental processes: additions, losses, translocations, transformations.
- Additions add OM, water, salts, N; losses remove material by leaching and erosion.
- Transformations change material form (weathering, humification).
- Translocation moves material within the soil: eluviation (out) and illuviation (in).
- Eluviation creates the pale E horizon; illuviation creates the B horizon.
- Humification converts fresh organic matter to humus.
- Humification is favoured by warm, moist, aerated conditions and narrow C:N ratio.
- Eluviation + illuviation together build the textural and chemical contrast of horizons.

Short-answer questions

1. *Define eluviation and illuviation.*

Model answer: Eluviation is the movement (washing out) of soil material from an upper horizon downward, creating the eluviated (E) horizon. Illuviation is the deposition of that material into a lower horizon, creating the illuvial (B) horizon.

2. *What is humification?*

Model answer: Humification is the process by which fresh plant and animal residues are decomposed and converted into humus, a dark, complex, resistant colloidal organic substance, by soil microorganisms.

3. *List Simenson's four fundamental soil forming processes.*

Model answer: Additions, losses, translocations (eluviation and illuviation) and transformations (weathering and humification).

Long-answer questions

1. *Explain Simenson's four-fold soil forming process with a diagram.*

Outline: Introduction - additions - losses - transformations - translocations (eluviation, illuviation) - diagram - humification - how horizons form - conclusion.

2. *Describe eluviation and illuviation and their role in soil profile development.*

Outline: Definition - material moved - eluviated (E) horizon - illuvial (B) horizon - argillic, spodic, calcic horizons - comparison table - worked example - significance - conclusion.

Lecture 10: Specific Soil Forming Processes - Podzolization, Laterization, Salinization, Alkalization, Calcification, Decalcification and Pedoturbation

10.1 Introduction

While the fundamental processes operate in all soils, **specific (special) soil forming processes** operate under particular conditions and give each soil its distinctive character. These processes produce the great soil groups such as podzols, laterites, saline soils, sodic soils and calcareous soils.

10.2 Podzolization

Definition: Podzolization is the process of **intense eluviation of iron, aluminium and organic matter** from the surface horizon under cool, humid, coniferous forest conditions, leaving a pale ash-grey horizon and depositing Fe-Al-humus in the B horizon below.

Conditions required

- **Cool and humid climate** (boreal/temperate).
- **Coniferous (pine/spruce) vegetation** - litter rich in acidic compounds.
- **Acidic organic acids** (fulvic acid) that chelate and mobilise Fe and Al.
- **Acidic, well-drained, sandy parent material** poor in bases.

Process

1. Conifer litter decomposes to produce acidic organic compounds (fulvic acid).
2. These acids **chelate and dissolve iron and aluminium oxides** and organic matter.
3. The dissolved material is **eluviated** downward, leaving a bleached **E (A2) horizon**.
4. It is **illuviated** in the B horizon as a **dark or reddish-brown spodic (Bs) horizon**.

Resulting soil: Podzol (Spodosol) - profile O-A-E-Bs-C; highly acidic, low in bases, low in fertility.

Note: Podzolization is rare in Tamil Nadu (climate is too warm and dry); it occurs in cool, humid regions.

10.3 Laterization (Laterisation)

Definition: Laterization is the process of **intense weathering and leaching** under hot, humid tropical conditions, in which **silica is removed** and **iron and aluminium oxides accumulate**, giving a red, iron-rich, hard material called **laterite**.

Conditions required

- **Hot and humid tropical climate** with alternating wet and dry seasons.
- **Good drainage** (removal of silica and bases).
- **Long time** (old land surfaces).

- **Basic/iron-rich parent rocks** help but not essential.

Process

1. Heavy rainfall and high temperature cause **rapid hydrolysis** of silicates.
2. **Silica and bases (Ca, Mg, Na, K) are leached** away in solution.
3. **Iron and aluminium oxides (Fe2O3, Al2O3) remain and accumulate** in the soil.
4. On exposure and drying, the iron-rich material **hardens** into laterite (a brick-like crust).

Resulting soil: Laterite soil - red/brown, rich in Fe and Al oxides, low in silica, low in bases, poor in fertility, often gravelly. Used as building brick.

Tamil Nadu: Laterite soils occur in the **Nilgiris, Kanyakumari, parts of Salem and Coimbatore**.

10.4 Salinization (Salinisation)

Definition: Salinization is the **accumulation of soluble salts** (mainly chlorides and sulphates of Na, Ca and Mg) in the soil, usually in the root zone, making the soil **saline**.

Conditions required

- **Arid or semi-arid climate** where evaporation exceeds rainfall.
- **High water table** with salty groundwater, or **irrigation with saline water**.
- **Poor drainage** so salts are not leached out.
- **Capillary rise** bringing salts to the surface and their accumulation as evaporation dries the soil.

Process

1. Groundwater or irrigation water contains dissolved salts.
2. **Evaporation** at the surface draws water upward by capillarity.
3. Water evaporates, **leaving salts behind** at the surface.
4. Salts accumulate as a **white crust**.

Resulting soil: Saline soil (Solonchak / Salorthid) - ECe more than 4 dS/m, pH usually below 8.5, white salt crust, poor crop growth.

Reclamation: Leaching with good-quality water + drainage + growing salt-tolerant crops.

10.5 Alkalization (Alkalinisation / Sodication)

Definition: Alkalization is the process by which **sodium ions (Na+) accumulate on the exchange complex** of the clay, replacing Ca2+ and Mg2+, making the soil **sodic (alkaline)**.

Conditions required

- Presence of **sodium salts** (Na2CO3, NaHCO3) or sodium-rich water.
- **Poor drainage**.
- High **exchangeable sodium percentage (ESP)** - more than 15%.

Process

1. Sodium salts dissolve and supply Na+.
2. **Na+ replaces Ca2+ and Mg2+** on the clay exchange complex.
3. The clay becomes **sodium-saturated**, swells, disperses and seals the pores.
4. The soil becomes **impervious, black (due to dispersed organic matter), and highly alkaline** (pH more than 8.5).

Resulting soil: Sodic (alkali) soil (Solonetz / Natrargid) - ESP more than 15%, pH 8.5-10.5, poor structure, poor drainage.

Reclamation: Application of **gypsum (CaSO4.2H2O)** to replace Na+ with Ca2+, followed by leaching and drainage.

10.6 Saline vs Sodic (Alkali) Soils

Feature	Saline soil	Sodic (alkali) soil
Main problem	Excess soluble salts	Excess exchangeable sodium
ECe	> 4 dS/m	< 4 dS/m (often)
ESP	< 15%	> 15%
pH	Usually < 8.5	Usually 8.5-10.5
Appearance	White salt crust	Black, dispersed surface
Structure	Usually good	Poor, impervious
Reclamation	Leaching + drainage	Gypsum + leaching + drainage
Crops	Salt-tolerant crops	Gypsum then tolerant crops

(Soils with both high salts and high ESP are **saline-sodic soils**.)

10.7 Calcification

Definition: Calcification is the **accumulation of calcium carbonate (CaCO3)** in the soil profile, usually in the B or C horizon, under semi-arid conditions where leaching is insufficient to remove the carbonate.

Conditions required

- **Semi-arid climate** with moderate rainfall.
- **Calcareous parent material** (limestone) or Ca-rich dust/water.
- **Insufficient leaching** so CaCO3 is not removed.

Process

1. Ca2+ and bicarbonate are released by weathering.
2. Water moves down, carrying Ca(HCO3)2 in solution.
3. As water is used up or evaporates, **CaCO3 precipitates** as **lime nodules (kankar)**.
4. The horizon of accumulation is the **calcic (Bk) horizon**.

Resulting soil: Calcareous soil - light coloured, alkaline (pH 7.5-8.5), with visible white nodules of lime. Common in the black soils and semi-arid red soils.

10.8 Decalcification

Definition: Decalcification is the **removal (leaching) of calcium carbonate** from the soil profile by percolating water, usually under humid conditions.

Conditions required

- **Humid climate** with enough rainfall to leach.
- **Good drainage**.
- **Acidic conditions** (carbonic acid helps dissolve CaCO3).

Process

1. Carbonic acid (H2CO3) dissolves CaCO3: CaCO3 + H2CO3 -> Ca(HCO3)2 (soluble).
2. The soluble bicarbonate is **leached downward** and out of the profile.
3. The soil becomes **free of carbonates**, and if leaching continues, **acidic**.

Resulting soil: Non-calcareous, sometimes acidic soil. Decalcification is essentially the opposite of calcification and is common in humid regions.

10.9 Calcification vs Decalcification

Feature	Calcification	Decalcification
Climate	Semi-arid	Humid
Water movement	Insufficient leaching	Strong leaching
CaCO3	Accumulates (lime nodules)	Removed
Soil reaction	Alkaline	Neutral to acidic

Feature	Calcification	Decalcification
Horizon	Calcic (Bk) horizon	No carbonate horizon
Result	Calcareous soil	Non-calcareous/acidic soil

10.10 Pedoturbation

Definition: Pedoturbation is the **mixing of soil material** within the profile by physical, biological or other agencies, disturbing the normal horizon sequence.

Types

- **Faunal pedoturbation** - mixing by animals: earthworms, ants, termites, rodents (crayfish, moles). Earthworm mixing is called **vermimixis**; termite/ant mixing is called **bioturbation**.
- **Floral pedoturbation** - mixing by plant roots and uprooted trees (tree-throw).
- **Cryoturbation** - mixing by freeze-thaw action in cold regions (frost heaving).
- **Argilliturbation (argillipedoturbation)** - mixing by shrinking and swelling of expanding clays (vertisols); forms **slickensides** and **gilgai** micro-relief.
- **Aeroturbation** - mixing by trapped air or gas.
- **Crystal pedoturbation** - by growing salt crystals.
- **Human pedoturbation** - ploughing, digging.

Result: Pedoturbation produces **mixed horizons (A/B mixtures)** and can make soils appear uniform (e.g., vertisols). It is common in **black soils (vertisols)** of Tamil Nadu.

10.11 Summary of Specific Soil Forming Processes

Process	Key condition	What accumulates/removed	Resulting soil
Podzolization	Cool humid, conifer	Fe-Al-OM moved to B	Podzol (Spodosol)
Laterization	Hot humid tropical	Fe-Al oxides accumulate; Si removed	Laterite soil
Salinization	Arid, poor drainage	Soluble salts accumulate	Saline soil
Alkalization	Na salts, poor drainage	Na+ on exchange complex	Sodic soil
Calcification	Semi-arid	CaCO3 accumulates	Calcareous soil
Decalcification	Humid	CaCO3 removed	Acidic soil
Pedoturbation	Mixing agencies	Horizon mixing	Mixed/uniform profile

10.12 Worked Example: ESP and Gypsum Requirement (Basics)

Problem: A sodic soil has a CEC of 25 cmol/kg and an exchangeable sodium of 5 cmol/kg. Calculate the exchangeable sodium percentage (ESP).

Solution

- ESP = (exchangeable Na / CEC) x 100
- ESP = (5 / 25) x 100 = **20%**

Answer: ESP is 20%, which is above 15%, so the soil is **sodic** and needs gypsum for reclamation. (Gypsum requirement is computed from the amount of Na to be replaced - see Lecture 27.)

10.13 Exam Focus

Key points

- Specific processes give each great soil group its character.
- Podzolization: cool humid conifer forest; Fe-Al-OM eluviated to B; pale E horizon; podzol.
- Laterization: hot humid tropics; silica leached, Fe-Al oxides accumulate; laterite.
- Salinization: arid, poor drainage; soluble salts accumulate; saline soil (ECe > 4 dS/m).
- Alkalization: Na+ replaces Ca/Mg on the complex; sodic soil (ESP > 15%, pH > 8.5).

- Calcification: semi-arid; CaCO3 accumulates as lime nodules (kankar).
- Decalcification: humid; CaCO3 leached out; soil becomes acidic.
- Pedoturbation: mixing by animals, roots, shrink-swell clays, frost; common in vertisols.
- Reclamation: saline - leaching + drainage; sodic - gypsum + leaching + drainage.

Short-answer questions

1. *Define laterization and state the soil formed.*

Model answer: Laterization is the intense weathering and leaching of soil under hot, humid tropical conditions, in which silica and bases are removed and iron and aluminium oxides accumulate. It forms laterite soil, which is red, iron-rich and low in fertility.

2. *Distinguish saline and sodic soils.*

Model answer: Saline soils have excess soluble salts (ECe > 4 dS/m), pH usually below 8.5 and good structure; sodic soils have excess exchangeable sodium (ESP > 15%), pH 8.5-10.5 and poor, impervious structure. Saline soils are reclaimed by leaching; sodic soils need gypsum.

3. *What is pedoturbation? Give two examples.*

Model answer: Pedoturbation is the mixing of soil material within the profile by physical or biological agencies. Examples: earthworm mixing (vermimixis) and shrink-swell mixing in vertisols (argilliturbation).

Long-answer questions

1. *Describe the important specific soil forming processes with their conditions and resulting soils.*

Outline: Introduction - podzolization - laterization - salinization - alkalization - calcification - decalcification - pedoturbation - summary table - Tamil Nadu examples - conclusion.

2. *Explain salinization and alkalization and how saline and sodic soils are reclaimed.*

Outline: Introduction - salinization (conditions, process, result) - alkalization (conditions, process, result) - comparison table - reclamation (leaching, drainage, gypsum) - worked ESP example - conclusion.

Lecture 11: Soil Profile - Horizons, Master Horizons and Subordinate Horizons, Subdivisions, Lithological Discontinuity

11.1 Definition of Soil Profile

The **soil profile** is a **vertical section of the soil** from the surface down to the parent material, showing the different **horizons** arranged one above the other.

A soil profile is studied by digging a **soil pit** (usually 1.5-2 m deep) and examining the exposed face. The profile is the basic unit for soil description and classification.

11.2 Definition of a Soil Horizon

A **soil horizon** is a **layer within the soil profile that is roughly parallel to the surface and has distinct characteristics** (colour, texture, structure, organic matter, chemistry) different from the layers above and below.

Horizons are the result of the soil forming processes (additions, losses, translocations, transformations).

11.3 Master Horizons

Master horizons are the main, large layers of the profile, designated by **capital letters**: **O, A, E, B, C and R** (and L for a limnic layer).

Horizon	Name	Description
O	Organic horizon	Dominated by organic matter; topmost (forest litter). Oi (fibric), Oe (hemic), Oa (sapric).
A	Topsoil / mineral surface	Mineral horizon at the surface, dark, mixed with humus, zone of root activity and biological activity.

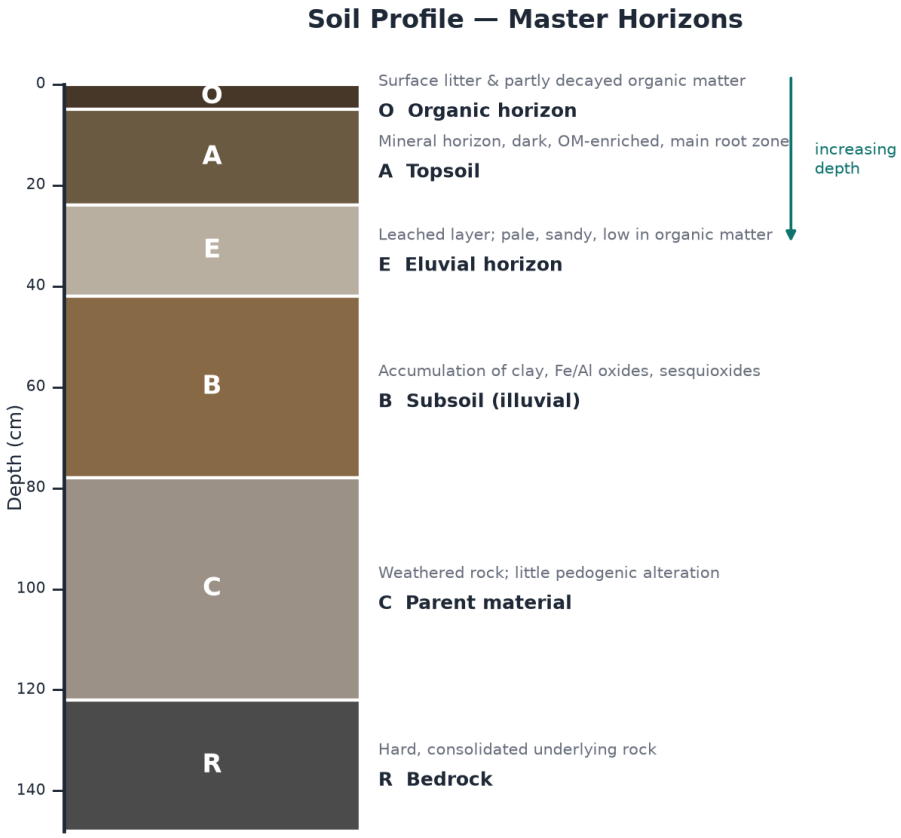


Figure 11.1 Soil profile — master horizons

Master horizons O, A, E, B, C and R shown to an approximate depth scale.

Horizon	Name	Description
E	Eluvial horizon	Light-coloured, leached of clay, Fe, Al and OM; sandy and pale.
B	Subsoil	Illuvial horizon of accumulation of clay, Fe/Al oxides, carbonate or humus; blocky structure.
C	Parent material	Weathered parent material, little affected by soil forming processes; no pedogenic structure.
R	Bedrock	Hard, consolidated rock; cannot be dug by hand.
L	Limnic layer	Organic material deposited in water.

The standard horizon sequence in a well-developed soil is **O - A - E - B - C - R**.

11.4 Diagram: Ideal Soil Profile

```
SOIL PROFILE (ideal sequence)
=============================================
 O   Organic layer (litter, humus)
---------------------------------------------
 A   Topsoil - dark, humus-rich, roots
---------------------------------------------
 E   Eluvial - pale, leached of clay/Fe
---------------------------------------------
 B   Subsoil - illuvial; clay/Fe/CaCO3
---------------------------------------------
 C   Parent material - weathered rock
---------------------------------------------
 R   Bedrock - hard rock
=============================================
 (Depth increases downward)
```

11.5 Subordinate (Sub) Horizons

Subordinate horizons are subdivisions of a master horizon, indicated by **lower-case letters** added as suffixes. They describe specific features.

Common suffix symbols

Suffix	Meaning	Example
a	Highly decomposed organic matter (sapric)	Oa
b	Buried horizon	Ab
c	Concretions (nodules)	Bc
d	Dense, compact (physical root restriction)	Cd
e	Moderately decomposed organic matter (hemic)	Oe
f	Frozen soil (permafrost)	Cf
g	Strong gleying (mottles/grey, waterlogged)	Bg
h	Illuvial humus accumulation	Bh
i	Slightly decomposed organic matter (fibric)	Oi
j	Jarosite (yellow mottles, acid sulphate)	Bj
k	Carbonate (CaCO3) accumulation	Bk
m	Strong cementation (petroferric)	Bkm
n	Exchangeable sodium accumulation	Bn
o	Residual sesquioxides (Fe, Al oxides)	Bo

Suffix	Meaning	Example
p	Plough layer (tillage disturbance)	Ap
q	Silica accumulation	Bq
r	Weathered bedrock	Cr
s	Illuvial sesquioxides (Fe, Al oxides)	Bs
t	Illuvial clay accumulation	Bt
v	Plinthite (iron-rich, hardens on exposure)	Bv
w	Colour/structure development (weathered)	Bw
y	Gypsum accumulation	By
z	Soluble salts (salic) accumulation	Az

11.6 Combining Symbols

A horizon may carry more than one suffix to show several features:

- **Ap** - plough layer.
- **Bt** - clay-enriched subsoil (argillic).
- **Btk** - clay AND carbonate accumulation.
- **Bs** - sesquioxide (Fe-Al) accumulation (spodic).
- **Bg** - gleyed subsoil.
- **Ck** - carbonate-rich parent material.
- **2Bt** - an argillic horizon that belongs to a **second (different) parent material** (see lithological discontinuity).

11.7 Transitional and Combined Horizons

- **Transitional horizons** have properties of two master horizons, shown by two letters: **AB** (more like A), **BA** (more like B), **BC** (more like B).
- **Combined horizons** show two distinct features in the same layer, shown by a slash: **E/B**, **B/C** - meaning distinct parts of E and B within the same layer.

11.8 Lithological Discontinuity

Definition: A **lithological discontinuity** is a **change in the parent material** within the soil profile, i.e., a boundary where the material above differs from the material below in origin, not just in pedogenic development.

How it is recognised

- A **sharp, abrupt change** in texture, mineralogy, or coarse fragments that cannot be explained by soil forming processes.
- The presence of **stone lines** or gravel layers.
- A marked change in the content of resistant minerals (e.g., zircon, tourmaline).

Notation: Horizons below a discontinuity are given the prefix **"2"** (then "3", "4", etc.). For example, an **A-Bt-2Bt-2C** profile has the lower Bt and C formed from a **different** parent material than the upper part.

Causes

- Deposition of a **new layer** of material over an older soil (e.g., alluvium over an old soil, volcanic ash over a soil, colluvium over a buried soil).
- **Buried soils** (palaeosols) - old soil surfaces covered by new deposits.

Importance: It affects water movement, root growth, nutrient distribution and the interpretation of the soil profile for classification and management.

11.9 Horizon Boundary Types

Horizons may have different boundary **distinctness** and **form**:

- Distinctness: **abrupt** (<2 cm), **clear** (2-5 cm), **gradual** (5-15 cm), **diffuse** (>15 cm).

- Form: **smooth**, **wavy**, **irregular**, **broken**.

11.10 Tamil Nadu Examples

- **Red soils** typically show A-Bt-C with kaolinite in the Bt and iron oxides giving red colour.
- **Black soils (vertisols)** show A-Bss-C with **slickensides** and **pedoturbation**; often deep and uniform.
- **Laterite soils** show A-Bo(v)-C with a hard iron-rich plinthite/laterite layer.
- **Alluvial soils of the Cauvery delta** show A-C profiles (weakly developed, young) often with **lithological discontinuities** from successive flood deposits.
- **Coastal sandy soils** show A-C with little horizon development.

11.11 Worked Example: Identifying Horizon Symbols

Problem: A profile is described as Oi - A - E - Bt - Btk - 2C. Explain each horizon.

Solution

- **Oi** - organic horizon, slightly decomposed (fibric) litter.
- **A** - mineral topsoil, humus-rich.
- **E** - eluvial horizon, leached pale layer.
- **Bt** - illuvial horizon with accumulated clay (argillic).
- **Btk** - illuvial horizon with clay AND carbonate accumulation.
- **2C** - parent material that is a **different (second) material** (lithological discontinuity) - weathered rock.

11.12 Exam Focus

Key points

- A soil profile is the vertical section from surface to parent material, showing horizons.
- Master horizons: O, A, E, B, C, R (and L).
- A = topsoil; E = eluvial (leached); B = illuvial (accumulation); C = parent material; R = bedrock.
- Subordinate horizons are shown by lower-case suffixes (t = clay, k = carbonate, g = gley, s = sesquioxides, p = plough).
- Transitional horizons: AB, BA, BC; combined horizons: E/B.
- Lithological discontinuity is a change in parent material; horizons below get the prefix "2".
- Horizon boundaries differ in distinctness (abrupt to diffuse) and form.
- Tamil Nadu: red soils A-Bt-C; black soils A-Bss-C; alluvium A-C with discontinuities.

Short-answer questions

1. *Define soil profile and soil horizon.*

Model answer: A soil profile is the vertical section of soil from the surface down to the parent material, showing horizons. A soil horizon is a layer roughly parallel to the surface with characteristics (colour, texture, structure) different from the layers above and below.

2. *Name the master horizons and give one feature of each.*

Model answer: O - organic; A - mineral topsoil rich in humus; E - eluvial, pale and leached; B - illuvial accumulation; C - weathered parent material; R - hard bedrock.

3. *What is a lithological discontinuity?*

Model answer: A lithological discontinuity is a change in the parent material within the soil profile. It is marked by an abrupt change in texture or mineralogy, and horizons below it are given the prefix "2" (e.g., 2Bt, 2C).

Long-answer questions

1. *Describe the soil profile with its master and subordinate horizons.*

Outline: Definition of profile and horizon - master horizons O, A, E, B, C, R with description - diagram - subordinate suffix symbols with meanings - transitional and combined horizons - horizon boundaries - conclusion.

2. *Explain lithological discontinuity and its importance in soil study.*

Outline: Definition - how recognised (texture, stone lines, resistant minerals) - notation (prefix 2, 3) - causes (new deposits, buried soils) - effect on water, roots, nutrients - Tamil Nadu example (Cauvery alluvium) - conclusion.

Lecture 12: Soil Physical Properties: Soil Texture - Particle Size Distribution - Textural Classes - Textural Triangular Diagram - Significance of Soil Texture

12.1 Introduction

Soil texture refers to the **relative proportions of the different sized mineral particles** - sand, silt and clay - in a soil. It is a basic physical property that cannot be easily changed and strongly influences water, air, nutrient and root behaviour.

Soil separates are the groups of particles of a given size range (sand, silt, clay).

12.2 Particle Size Distribution (Mechanical Analysis)

Particle size distribution (also called **mechanical composition**) is the determination of the proportion of each size fraction in a soil.

Methods of determination

- **Sieve method** - for sand and coarser fractions (dry or wet sieving).
- **Sedimentation method (Stokes' law)** - for silt and clay; particles settle according to size.

- **Hydrometer method** (Bouyoucos) - measures suspension density.

- **Pipette method** - samples suspension at a fixed depth and time.

Stokes' law: The settling velocity (v) of a spherical particle is:

- v = (2 r^2 (d1 - d2) g) / (9 n)

Where r = particle radius, d1 = particle density, d2 = density of liquid, g = acceleration due to gravity, n = viscosity of liquid.

12.3 Particle Size Classes (ISSS and USDA Systems)

Different systems use different size limits. The two common systems are the **International Society of Soil Science (ISSS)** and the **United States Department of Agriculture (USDA)**.

Fraction	ISSS (mm)	USDA (mm)
Coarse sand	2.0 - 0.2	2.0 - 0.5 (very coarse + coarse)
Fine sand	0.2 - 0.02	0.5 - 0.05 (medium + fine)
Silt	0.02 - 0.002	0.05 - 0.002
Clay	< 0.002	< 0.002

Key differences:

- The **ISSS** divides sand at 0.2 mm and 0.02 mm; **silt** is 0.02-0.002 mm.
- The **USDA** divides sand at 0.05 mm; **silt** is 0.05-0.002 mm.
- Both agree that **clay is < 0.002 mm**.
- The USDA further subdivides sand into very coarse, coarse, medium, fine and very fine sand.

12.4 Comparison: ISSS vs USDA Particle Size Classes

Class	ISSS	USDA
Sand	2.0 - 0.02 mm	2.0 - 0.05 mm
Silt	0.02 - 0.002 mm	0.05 - 0.002 mm
Clay	< 0.002 mm	< 0.002 mm
Basis	International	United States

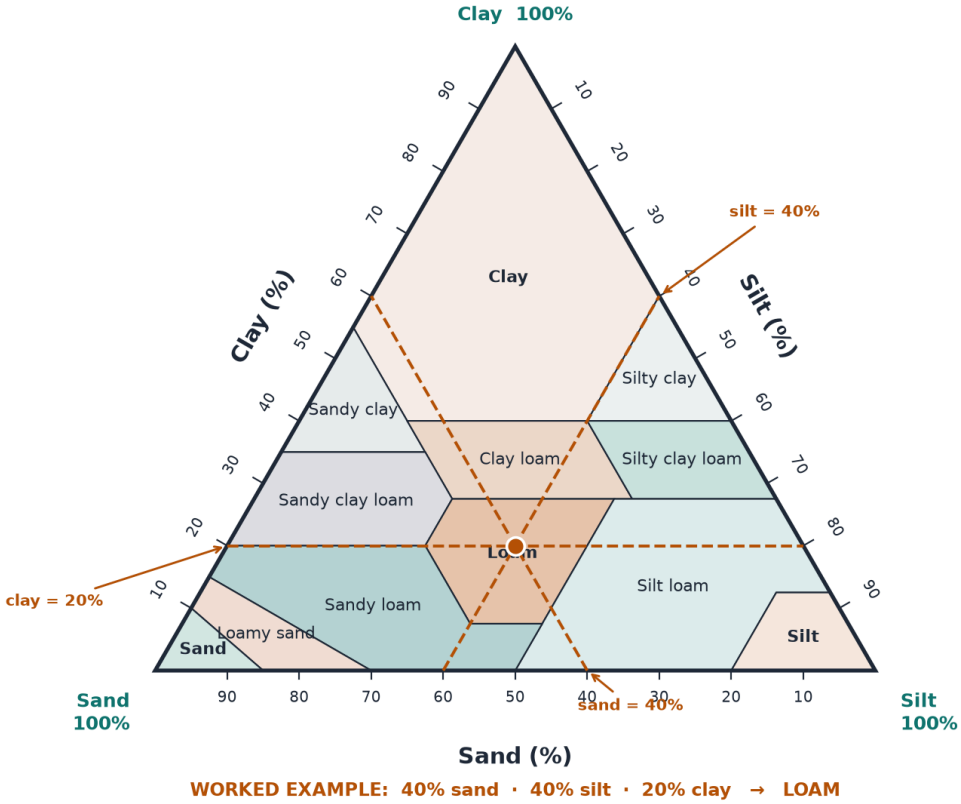


Figure 12.1 Soil textural triangle

The USDA textural triangle with the twelve textural classes; a worked reading example is marked.

Class	ISSS	USDA
Used in	Many countries incl. India (historically)	USA and widely now

12.5 Soil Textural Classes

A **textural class** is a name given to a soil based on its proportions of sand, silt and clay. The USDA recognises **12 textural classes**:

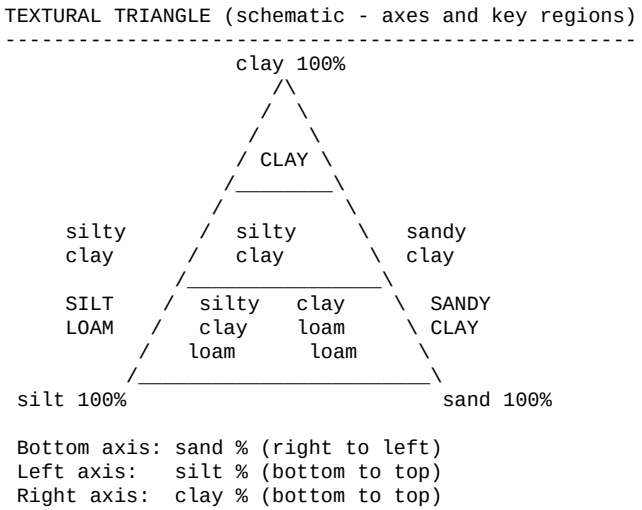
1. Sand
2. Loamy sand
3. Sandy loam
4. Loam
5. Silt loam
6. Silt
7. Sandy clay loam
8. Clay loam
9. Silty clay loam
10. Sandy clay
11. Silty clay
12. Clay

12.6 The Textural Triangle

The **textural triangle** (or **textural triangular diagram**) is a triangular graph used to find the **textural class** of a soil from its percentages of sand, silt and clay.

How to read it

- Each side of the triangle represents 0-100% of one fraction: **sand on the bottom-right side** (increasing to the left), **silt on the left side**, **clay on the right side** (increasing upward).
- Lines parallel to each side are drawn for the given percentage.
- The **point where the three lines intersect** lies in a region labelled with a textural class.
- The three percentages must sum to **100%**.



(For exact boundary lines use the USDA textural triangle chart; the schematic above shows the relative positions of the main classes.)

12.7 Textural Class Boundaries (USDA - approximate key limits)

Class	Clay (%)	Silt (%)	Sand (%)
Sand	0-10	0-15	85-100

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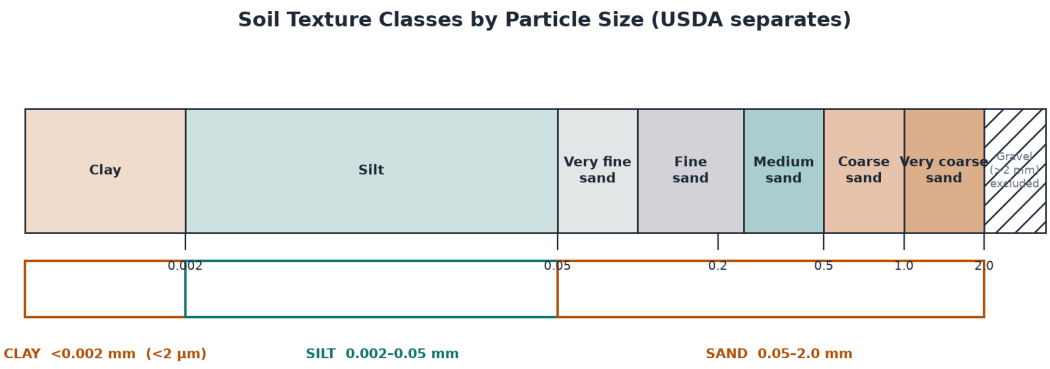


Figure 12.2 Soil texture classes by particle size

USDA soil separates: clay (< 0.002 mm), silt (0.002–0.05 mm) and sand (0.05–2.0 mm).

Class	Clay (%)	Silt (%)	Sand (%)
Loamy sand	0-15	0-30	70-90
Sandy loam	0-20	0-50	43-85
Loam	7-27	28-50	23-52
Silt loam	0-27	50-88	0-50
Silt	0-12	80-100	0-20
Sandy clay loam	20-35	0-28	45-80
Clay loam	27-40	15-53	20-45
Silty clay loam	27-40	40-73	0-20
Sandy clay	35-55	0-20	45-65
Silty clay	40-60	40-60	0-20
Clay	40-100	0-40	0-45

(These are approximate limits from the USDA triangle; use the triangle for exact placement.)

12.8 Diagram: Textural Triangle by Silt/Clay/Sand Percentage (Tabulated Guide)

If clay is	and silt is	the class is likely
<10	<15	Sand
10-15	low	Loamy sand
<20	<50	Sandy loam
7-27	28-50	Loam
<27	>50	Silt loam
27-40	15-53	Clay loam
35-55	<20	Sandy clay
>40	40-60	Silty clay
>40	<40	Clay

12.9 Worked Example 1: Determining Textural Class from the Triangle

Problem: A soil has 60% sand, 20% silt and 20% clay. Determine its textural class.

Solution

- Sand = 60%, Silt = 20%, Clay = 20%. Sum = 100%. (Check: 60+20+20 = 100 - OK.)
- Using the triangle: with clay 20% and sand 60%, the point falls in the **sandy clay loam** region (sandy clay loam has clay 20-35%, sand 45-80%).
- Verify against the table: clay 20% is at the lower limit of sandy clay loam; sand 60% is within 45-80%.

Answer: Textural class = **Sandy clay loam**.

12.10 Worked Example 2: Determining Textural Class

Problem: A soil has 30% sand, 40% silt and 30% clay. Determine its textural class.

Solution

- Sand = 30%, Silt = 40%, Clay = 30%. Sum = 100%.
- Clay 30% and silt 40% place the point in the **clay loam** region (clay loam: clay 27-40%, silt 15-53%, sand 20-45%).
- Sand 30% is within 20-45%, confirming clay loam.

Answer: Textural class = **Clay loam**.

12.11 Worked Example 3: Percentages from Weights

Problem: A 50 g soil sample contains 30 g sand, 12 g silt and 8 g clay. Find the percentage of each and the textural class.

Solution

- Sand % = (30/50) x 100 = **60%**
- Silt % = (12/50) x 100 = **24%**
- Clay % = (8/50) x 100 = **16%**
- Sum = 100%.
- Clay 16%, sand 60% -> **sandy loam** (sandy loam: clay 0-20%, sand 43-85%).

Answer: Sand 60%, silt 24%, clay 16% -> **Sandy loam**.

12.12 Worked Example 4: Stokes' Law Settling (Concept)

Problem: Using Stokes' law, compare the settling velocity of a 0.02 mm particle with that of a 0.002 mm particle (same density and liquid).

Solution

- v is proportional to r^2.
- Ratio of radii = 0.02 / 0.002 = 10.
- Ratio of velocities = 10^2 = **100**.

Answer: The 0.02 mm particle settles **100 times faster** than the 0.002 mm particle. This is the basis of separating silt from clay by sedimentation.

12.13 Significance of Soil Texture

- **Water holding capacity:** clay soils hold more water; sandy soils hold less.
- **Drainage and aeration:** sandy soils drain fast and are well aerated; clay soils drain slowly and may waterlog.
- **Nutrient holding:** clay and humus hold nutrients (high CEC); sandy soils are poor.
- **Workability (tillage):** loams are easiest to plough; clays are sticky when wet and hard when dry; sands are easy but droughty.
- **Root growth:** loams and sandy loams are best; heavy clays restrict roots.
- **Erosion:** fine sands and silts are easily eroded; clays resist but runoff is high.
- **Crop suitability:** groundnut and tuber crops prefer sandy loams; paddy prefers clayey soils; most crops do best on loams.
- **Fertility management:** sandy soils need frequent, light fertilizer doses; clay soils hold nutrients better.

12.14 Texture and Soil Behaviour Table

Property	Sand	Silt	Clay
Particle size	Largest	Intermediate	Smallest
Surface area	Low	Medium	Very high
Water holding	Low	Medium	High
Drainage	Fast	Moderate	Slow
Aeration	Good	Moderate	Poor when wet
CEC	Very low	Low	High
Nutrient supply	Poor	Moderate	High
Workability	Easy	Silty, crusts	Sticky/hard
Erodibility	Low (coarse)	High	Low (but runoff)

12.15 Tamil Nadu Context

- **Red sandy loams** of Coimbatore, Salem - good for groundnut, millets.

- **Black clay soils (vertisols)** of Madurai, Coimbatore - heavy, self-mulching, good for cotton.
- **Alluvial loams of the Cauvery delta** - ideal for paddy.
- **Coastal sands** of Ramanathapuram - droughty, need organic matter and irrigation.

12.16 Exam Focus

Key points

- Soil texture = relative proportions of sand, silt and clay.
- ISSS: sand 2.0-0.02 mm, silt 0.02-0.002 mm, clay <0.002 mm.
- USDA: sand 2.0-0.05 mm, silt 0.05-0.002 mm, clay <0.002 mm.
- Twelve USDA textural classes; read them from the textural triangle.
- Stokes' law governs settling; velocity proportional to radius squared.
- Clay holds most water and nutrients; sand drains fast and holds least.
- Loam is the best all-round texture for most crops.
- Sandy soils need frequent light fertilization; clay soils hold nutrients well.

Short-answer questions

1. *Define soil texture and name the three soil separates.*

Model answer: Soil texture is the relative proportion of the different sized mineral particles in a soil. The three separates are sand (2.0-0.05 mm), silt (0.05-0.002 mm) and clay (<0.002 mm) in the USDA system.

2. *How is the textural triangle used?*

Model answer: The textural triangle is used to find the textural class of a soil from its sand, silt and clay percentages. Lines are drawn parallel to each side for the given percentages; the point where the three lines intersect lies in a region labelled with a textural class.

3. *Why do clay soils hold more water and nutrients than sandy soils?*

Model answer: Clay particles are very small with a very large surface area and carry negative charges, giving high water-holding capacity and high cation exchange capacity. Sand particles are large with little surface area and negligible charge.

Long-answer questions

1. *Explain soil texture, particle size classes and the textural triangle with examples.*

Outline: Definition - soil separates - ISSS vs USDA classes table - textural classes - textural triangle (how to read, schematic) - worked examples - conclusion.

2. *Discuss the significance of soil texture in crop production.*

Outline: Introduction - water holding, drainage, aeration - nutrient holding - workability - root growth - erosion - crop suitability - management of sandy and clayey soils - Tamil Nadu examples - conclusion.

Lecture 13: Soil Structure - Classification - Genesis - Factors Influencing Structural Stability - Significance of Soil Structure

13.1 Definition of Soil Structure

Soil structure is the **arrangement of soil particles (sand, silt and clay) into aggregates (peds)**, together with the size, shape and stability of these aggregates.

- **Texture** = the size of individual particles.
- **Structure** = the way those particles are grouped together.
- Structure can be **changed by management** (unlike texture).

13.2 Terms Used in Describing Structure

- **Ped** - a natural soil aggregate (unit of structure).
- **Aggregate** - a group of particles held together.
- **Clod** - an artificially formed lump (by ploughing).

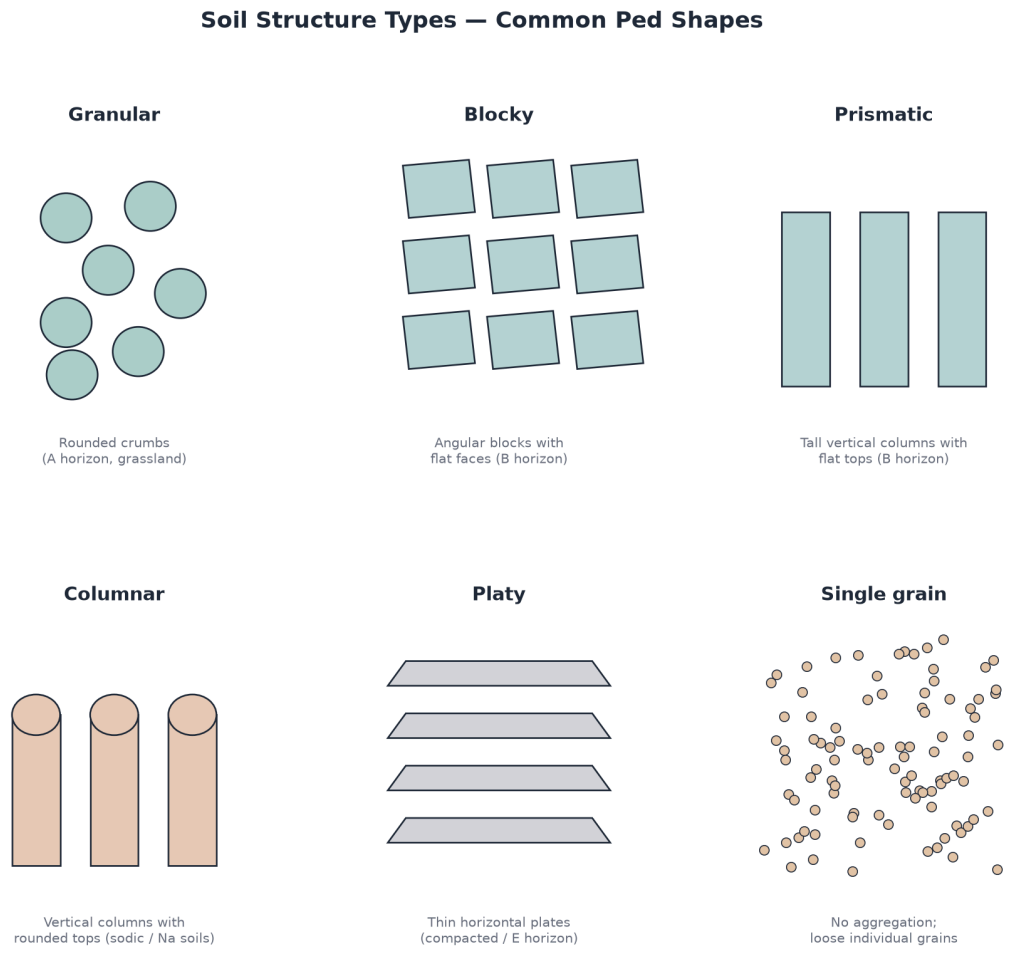


Figure 13.1 Soil structure types

Common ped shapes: granular, blocky, prismatic, columnar, platy and single grain.

- **Concretion** - a hard, cemented body (e.g., lime or iron).
- **Grade** - the strength of aggregation: **structureless, weak, moderate, strong**.
- **Size** - fine, medium, coarse (varies with the type).
- **Shape** - the type of structure (see classification).

13.3 Classification of Soil Structure

Soil structure is classified by **shape** into four main types:

A. Spheroidal (granular and crumb)

- Rounded aggregates.
- **Granular** - porous, rounded; typical of A horizons; ideal for plant growth.
- **Crumb** - very porous, fine, unstable; common in grassland surface soils.

B. Plate-like (platy)

- Thin, flat, horizontal plates.
- Common in compacted horizons (plough pan) and in some B horizons.
- Poor for water and root movement.

C. Block-like (blocky and subangular blocky)

- Blocky - angular faces; subangular blocky - rounded faces.
- Common in **B horizons**; medium to fine; good structure.

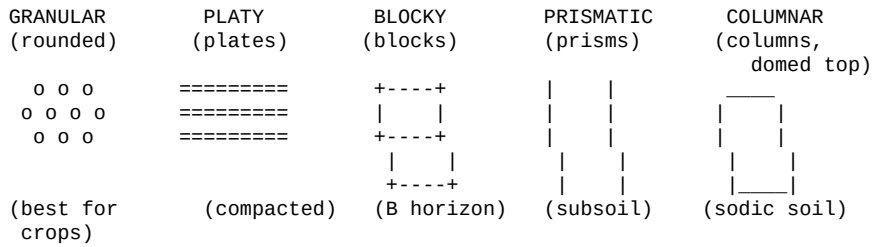
D. Prism-like (prismatic and columnar)

- Vertical columns.
- **Prismatic** - flat tops; common in subsoils.
- **Columnar** - rounded/domed tops; typical of **sodic (alkali) soils**.
- Poor structure; restricts drainage.

E. Structureless

- **Single grain** - all particles separate (sands).
- **Massive** - no aggregation, a large coherent mass (heavy clay when wet).

13.4 Diagram: Types of Soil Structure



13.5 Classification of Structure - Summary Table

Type	Shape	Size classes	Where found	Quality
Granular	Rounded, porous	Very fine to coarse	A horizon	Excellent
Crumb	Very porous, fine	Very fine	Grassland topsoil	Excellent
Platy	Thin horizontal plates	Very fine to coarse	Compacted layers	Poor
Blocky	Angular blocks	Fine to coarse	B horizon	Good
Subangular blocky	Rounded blocks	Fine to coarse	B horizon	Good
Prismatic	Vertical prisms, flat top	Fine to coarse	Subsoil	Fair
Columnar	Vertical columns, domed top	Fine to coarse	Sodic soils	Poor
Single grain	Loose, separate	-	Sands	Poor (no structure)

Type	Shape	Size classes	Where found	Quality
Massive	Coherent mass	-	Heavy clay wet	Poor

13.6 Genesis (Formation) of Soil Structure

Aggregates form in two main ways:

A. Physico-chemical (binding) processes

- **Flocculation** - clay particles clump together when cations (especially Ca2+) neutralise their negative charge; Na+ causes dispersion, Ca2+ causes flocculation.
- **Cementation** - by organic matter, iron oxides, carbonates and silica acting as cement between particles.

B. Physical (mechanical) processes

- **Wetting and drying** - shrink-swell creates cracks and peds.
- **Freezing and thawing** - ice forms aggregates.
- **Tillage** - breaks soil into clods and peds.
- **Root and faunal action** - roots and earthworms create and stabilise aggregates.

Order of aggregate formation (hierarchical)

- Primary particles (sand, silt, clay) -> microaggregates (<250 micrometres) -> macroaggregates (>250 micrometres).

13.7 Agents that Bind (Aggregating Agents)

- **Organic matter / humus** - most important; binds particles and is called the "glue" of soil.
- **Clay minerals** - act as cementing agents.
- **Polyvalent cations (Ca2+, Mg2+)** - flocculate clay; **Na+** disperses it.
- **Iron and aluminium oxides** - cement in red soils.
- **Calcium carbonate** - cements in calcareous soils.
- **Microbial polysaccharides** - sticky products of microbes and roots.

13.8 Factors Influencing Structural Stability

A. Soil management

- **Tillage** - excessive tillage destroys aggregates; appropriate tillage improves them.
- **Organic matter additions** - manure, compost, green manure, crop residues improve structure.
- **Fertilizers and amendments** - gypsum improves sodic soils; lime improves acidic soils.
- **Cropping systems** - grass/legume leys improve structure; continuous clean cultivation degrades it.
- **Irrigation** - waterlogging and sodium in water degrade structure.

B. Soil properties

- **Clay type** - smectite swells and shrinks (unstable); kaolinite is more stable.
- **Exchangeable cations** - Ca2+ and Mg2+ stabilise; Na+ disperses.
- **Organic matter content** - high OM stabilises.
- **Iron and aluminium oxides** - stabilise aggregates.

C. Environmental factors

- **Wetting and drying cycles** - can either build or break structure.
- **Freezing and thawing** - improves structure in temperate regions.
- **Raindrop impact** - breaks surface aggregates, causing crusting.
- **Biological activity** - earthworms, roots and microbes stabilise structure.

13.9 Significance of Soil Structure

- **Water movement:** good structure gives rapid infiltration and drainage; poor structure causes runoff and waterlogging.

- **Aeration:** well-aggregated soils have good pore space for air.
- **Root growth:** granular and blocky structures allow roots to penetrate easily.
- **Nutrient availability:** good structure supports microbial activity and nutrient cycling.
- **Erosion control:** stable aggregates resist water and wind erosion.
- **Tillage:** good structure reduces draft and improves seedbed.
- **Seed germination:** a fine, crumb surface (tilth) gives good germination.
- **Crop yield:** good structure gives higher, more stable yields.

13.10 Management of Soil Structure

- Add **organic matter** (FYM, compost, green manure, residues).
- Practise **minimum tillage** and avoid working wet soil.
- Use **crop rotations** with grasses and legumes.
- Apply **gypsum** to sodic soils and **lime** to acidic soils.
- Ensure **proper drainage** and avoid using high-sodium irrigation water.
- Use **mulching** and **cover crops** to protect surface aggregates.
- **Contour ploughing** and **conservation practices** to reduce erosion.

13.11 Worked Example: Aggregate Stability Index

Problem: A soil sample contains 60% of aggregates larger than 0.25 mm after wet sieving, out of a total of 80% water-stable aggregates before sieving. Calculate the aggregate stability.

Solution

- Water-stable aggregates (>0.25 mm) = 60%
- This is expressed as a percentage of the whole soil: **60%**.
- If the total possible (from the same soil, dry) is 80%, then stability index = (60/80) x 100 = **75%**.

Answer: The aggregate stability index is 75%, indicating moderately stable structure.

13.12 Exam Focus

Key points

- Soil structure = arrangement of particles into aggregates (peds).
- Texture is particle size; structure is particle grouping; structure can be changed.
- Types by shape: spheroidal (granular, crumb), plate-like (platy), block-like (blocky), prism-like (prismatic, columnar), and structureless (single grain, massive).
- Granular/crumb structure is best for crops; platy and massive are poor.
- Columnar structure is typical of sodic soils.
- Structure forms by flocculation, cementation, wetting-drying, freezing-thawing and biological activity.
- Organic matter and Ca2+ stabilise structure; Na+ disperses it.
- Good structure improves water, air, roots, nutrients, erosion control and yield.

Short-answer questions

1. *Define soil structure and name the main structural types.*

Model answer: Soil structure is the arrangement of soil particles into aggregates (peds). The main types by shape are spheroidal (granular, crumb), plate-like (platy), block-like (blocky, subangular blocky) and prism-like (prismatic, columnar), plus structureless (single grain, massive).

2. *How does organic matter improve soil structure?*

Model answer: Organic matter and humus act as a binding (cementing) agent, gluing particles into stable aggregates, improving porosity, water infiltration, aeration and root growth, and reducing erosion.

3. *Why is columnar structure a problem?*

Model answer: Columnar structure occurs in sodic soils; the dispersed sodium-clay swells and seals pores, so water and roots cannot move freely, drainage is poor and the soil is difficult to manage.

Long-answer questions

1. *Classify soil structure and describe the factors influencing structural stability.*

Outline: Definition - terms (ped, aggregate) - classification by shape with diagram and table - genesis of structure - aggregating agents - factors influencing stability (management, soil, environmental) - management measures - conclusion.

2. *Discuss the significance of soil structure in crop production.*

Outline: Introduction - water movement and drainage - aeration - root growth - nutrient availability - erosion - tillage and tilth - crop yield - management practices - Tamil Nadu examples - conclusion.

Lecture 14: Soil Bulk Density, Particle Density and Porosity - Factors Influencing - Significance

14.1 Introduction

The physical condition of a soil is often expressed through its **density** and **porosity**. These properties describe how tightly the solid particles are packed and how much pore space exists for water and air.

14.2 Particle Density (Dp)

Definition: Particle density is the **mass per unit volume of the soil solids only** (excluding pore space).

- Also called **true density** or **specific gravity** of soil.
- Formula: **Dp = Mass of oven-dry soil / Volume of soil solids**
- Units: g/cm3 (or Mg/m3).
- Typical value: **2.6 - 2.7 g/cm3** (average about 2.65 g/cm3).
- For most mineral soils it is taken as **2.65 g/cm3**.
- Organic soils have lower particle density (about 1.4-1.8 g/cm3) because organic matter has low density.

14.3 Bulk Density (Db)

Definition: Bulk density is the **mass per unit volume of the whole soil**, including the pore space.

- Formula: **Db = Mass of oven-dry soil / Total volume of soil (solids + pores)**
- Units: g/cm3 (or Mg/m3).
- Typical value: **1.0 - 1.8 g/cm3** for mineral soils.
- Bulk density is **always less than particle density** because it includes pore space.

14.4 Relationship between Bulk Density, Particle Density and Porosity

- **Porosity (%) = (1 - Db/Dp) x 100**
- Also: **Db = Dp x (1 - porosity fraction)**

Where Db = bulk density, Dp = particle density, porosity fraction = pore volume / total volume.

14.5 Bulk Density Ranges by Soil Type

Soil type / condition	Typical bulk density (g/cm3)
Sandy soil	1.5 - 1.8
Sandy loam	1.4 - 1.6
Loam	1.3 - 1.5
Silt loam	1.2 - 1.4
Clay loam	1.1 - 1.4
Clay soil	1.0 - 1.3
Organic / peat soil	0.2 - 0.8
Compacted / plough pan	1.6 - 2.0
Ideal for root growth	1.1 - 1.5

General rule: Bulk density **increases** with compaction and coarse texture, and **decreases** with organic matter and good aggregation. Root growth is restricted above about **1.6-1.7 g/cm3** for most crops (varies with texture).

14.6 Particle Density of Common Materials

Material	Particle density (g/cm3)
Quartz	2.65
Feldspar	2.55 - 2.76
Mica	2.8 - 3.1
Clay minerals	2.6 - 2.8
Iron oxides	3.0 - 5.2
Organic matter	1.3 - 1.5
Average mineral soil	2.65

14.7 Factors Influencing Bulk Density and Porosity

A. Texture

- Sandy soils have **higher bulk density** (fewer small pores, tight packing).
- Clayey soils have **lower bulk density** (more pore space, but also swelling).

B. Organic matter

- Increases pore space, lowers bulk density.
- Manured soils have lower bulk density than unmanured soils.

C. Soil structure and aggregation

- Well-aggregated soils have more pore space and lower bulk density.
- Poorly structured, massive soils have higher bulk density.

D. Compaction and tillage

- Traffic, ploughing at the same depth and animal trampling increase bulk density (form plough pan).
- Tillage loosens the soil temporarily and lowers bulk density.

E. Depth in the profile

- Bulk density generally **increases with depth** (less organic matter, more compaction).

F. Soil moisture

- Swelling clays have lower bulk density when wet; but measurement is always on an oven-dry basis.

G. Cropping and management

- Continuous paddy (wetland) cultivation can increase bulk density in some layers; grasses and legumes reduce it.

14.8 Porosity

Definition: Porosity (pore space) is the **percentage of the total soil volume occupied by pores** (the space not filled by solids).

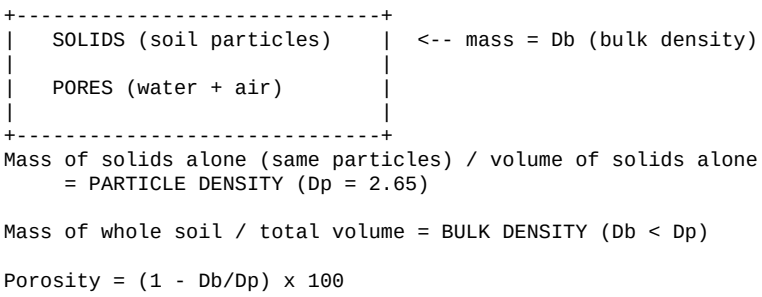
- Formula: **Porosity (%) = (pore volume / total volume) x 100 = (1 - Db/Dp) x 100**
- Typical values: **sandy soils 35-45%**, **loams 45-55%**, **clay soils 50-60%**, **organic soils up to 80-90%**.

Types of pores

- **Macropores** (>0.08 mm) - drain freely, carry air (aeration).
- **Mesopores** (0.03-0.08 mm) - hold plant-available water.
- **Micropores** (<0.03 mm) - hold water tightly (unavailable).

14.9 Diagram: Solids, Pores and Densities

```
TOTAL SOIL VOLUME (say 1 cm3)
```



Mass of solids alone (same particles) / volume of solids alone
= PARTICLE DENSITY (Dp = 2.65)

Mass of whole soil / total volume = BULK DENSITY (Db < Dp)

Porosity = (1 - Db/Dp) x 100

14.10 Worked Example 1: Bulk Density and Porosity

Problem: A soil core of volume 100 cm3 has an oven-dry mass of 140 g. Particle density is 2.65 g/cm3. Calculate the bulk density and porosity.

Solution

- Bulk density Db = mass / total volume = 140 / 100 = **1.40 g/cm3**
- Porosity = (1 - Db/Dp) x 100 = (1 - 1.40/2.65) x 100
- 1.40/2.65 = 0.528
- 1 - 0.528 = 0.472
- Porosity = 0.472 x 100 = **47.2%**

Answer: Bulk density = 1.40 g/cm3; porosity = 47.2%.

14.11 Worked Example 2: Particle Density

Problem: The volume of solids in a soil sample is 40 cm3 and the mass of the solids is 106 g. Calculate the particle density.

Solution

- Dp = mass of solids / volume of solids = 106 / 40 = **2.65 g/cm3**

Answer: Particle density = 2.65 g/cm3 (a typical mineral soil value).

14.12 Worked Example 3: Porosity from Bulk and Particle Density

Problem: A compacted soil has a bulk density of 1.65 g/cm3 and particle density of 2.65 g/cm3. Find the porosity and comment.

Solution

- Porosity = (1 - 1.65/2.65) x 100
- 1.65/2.65 = 0.623
- 1 - 0.623 = 0.377
- Porosity = **37.7%**

Answer: Porosity is only 37.7%, which is low (compaction). Such a soil will have poor aeration and restricted root growth.

14.13 Worked Example 4: Mass of Soil in a Layer

Problem: Calculate the mass of soil (in tonnes) in the top 15 cm layer of one hectare if the bulk density is 1.4 g/cm3.

Solution

- Volume = 1 ha x 0.15 m = 10,000 m2 x 0.15 m = 1,500 m3
- Bulk density = 1.4 g/cm3 = 1.4 t/m3 (since 1 g/cm3 = 1 t/m3)
- Mass = volume x bulk density = 1,500 x 1.4 = **2,100 tonnes**

Answer: The top 15 cm of one hectare weighs about 2,100 tonnes. (A useful figure: an acre-furrow-slice, 1 acre x 15 cm, weighs about 2 million pounds; a hectare-furrow-slice is about 2.24 million kg.)

14.14 Significance of Bulk Density, Particle Density and Porosity

- **Bulk density** indicates compaction; it is used to convert weight to volume for fertilizer and lime recommendations.
- **High bulk density** restricts root growth, reduces infiltration and aeration.
- **Porosity** controls water and air movement; it determines available water capacity.
- **Particle density** is used with bulk density to compute porosity.
- These properties are used in **irrigation scheduling, soil erosion studies and soil survey**.
- They help in judging the **suitability of a soil for crops and for tillage**.

14.15 Exam Focus

Key points

- Particle density = mass of solids / volume of solids; average 2.65 g/cm3.
- Bulk density = mass of oven-dry soil / total volume (solids + pores).
- Bulk density is always less than particle density.
- Porosity (%) = (1 - Db/Dp) x 100.
- Sandy soils have higher bulk density; clay and organic soils lower.
- Bulk density increases with compaction and depth; decreases with organic matter.
- Root growth is restricted above about 1.6-1.7 g/cm3.
- Typical porosity: sand 35-45%, loam 45-55%, clay 50-60%, peat up to 80-90%.

Short-answer questions

1. *Define bulk density and particle density.*

Model answer: Bulk density is the mass of oven-dry soil per unit total volume (solids plus pores), typically 1.0-1.8 g/cm3. Particle density is the mass of soil solids per unit volume of solids alone, typically 2.65 g/cm3 for mineral soils.

2. *Give the relationship between bulk density, particle density and porosity.*

Model answer: Porosity (%) = (1 - bulk density / particle density) x 100. Porosity is the fraction of the total volume occupied by pores.

3. *Why is bulk density higher in sandy soils?*

Model answer: Sand particles are large and pack closely with fewer, smaller pore spaces, so the same volume contains more solid mass, giving a higher bulk density than clay or organic soils.

Long-answer questions

1. *Explain bulk density, particle density and porosity, and the factors influencing them.*

Outline: Definitions and formulae - units and typical values - relationship - factors (texture, organic matter, structure, compaction, depth, management) - worked examples - significance - conclusion.

2. *Discuss the significance of bulk density and porosity in agriculture.*

Outline: Introduction - definitions - compaction indicator - conversion of weight to volume for fertilizer/lime - effect on root growth, water and air - irrigation and erosion studies - management to maintain optimum - conclusion.

Lecture 15: Soil Colour - Significance - Measurement (Munsell Colour Chart) - Factors Influencing Soil Colour

15.1 Introduction

Soil colour is the most obvious visual property of soil. It is a useful indicator of several soil conditions such as organic matter content, drainage, iron content and parent material. Though it does not directly affect plant growth, it is a key field characteristic used in soil description and classification.

15.2 Significance of Soil Colour

- **Indicator of organic matter:** dark colours usually mean high organic matter.

- **Indicator of drainage:** grey and mottled (bluish-grey with rust spots) colours indicate poor drainage (gleying).
- **Indicator of iron oxides:** red and yellow colours indicate oxidised iron (well-drained); bluish-grey indicates reduced iron (waterlogged).
- **Indicator of parent material:** red soils from iron-rich rocks; black from basaltic rocks.
- **Indicator of climate and weathering:** red and lateritic soils indicate humid tropical weathering.
- **Use in soil classification:** colour is part of horizon description and soil naming.
- **Use in land evaluation:** helps judge suitability and management needs.
- **Aesthetic and practical:** dark soils warm up faster; colour affects heat absorption.

15.3 Causes of Soil Colour

Colour	Chief cause	Example soil
Black / dark brown	Organic matter; manganese oxides	Black soils, peat
Red	Haematite (Fe2O3), anhydrous iron oxide	Red soils
Yellow / brown	Goethite (FeOOH), hydrated iron oxide	Yellow soils
Grey / bluish-grey	Reduced iron (Fe2+), gleying	Waterlogged soils
White / pale	Silica, calcium carbonate, gypsum, salts	Calcareous, saline soils
Mottled (grey + red)	Alternate oxidation and reduction	Imperfectly drained soils

15.4 The Munsell Colour System

Soil colour is measured objectively using the **Munsell Soil Colour Chart**. Each colour is described by **three components**:

1. **Hue** - the dominant spectral colour (the "colour name"), such as red, yellow, or red-yellow. Written as 2.5YR, 5YR, 10YR, etc.
2. **Value** - the lightness or darkness of the colour, from 0 (black) to 10 (white).
3. **Chroma** - the strength (saturation or purity) of the colour, from 0 (grey) upward (e.g., 1, 2, 3, 4...).

A colour is written as **Hue Value/Chroma**, for example **10YR 4/3** (a dark yellowish-brown).

Common hues: 2.5YR, 5YR (red), 7.5YR, 10YR (yellowish), 2.5Y (yellow), 5Y (grey), and neutral (N).

15.5 How to Read the Munsell Chart

- Hold the soil sample (moist and dry) next to the colour chips.
- Match the chip that best fits under natural daylight.
- Record the notation: **Hue Value/Chroma**.
- Record both **moist** and **dry** colours (moist soil appears darker).

Example notations

Notation	Meaning
5YR 4/6	Reddish-brown (red hue, medium value, strong chroma)
10YR 4/3	Dark yellowish-brown
10YR 7/2	Pale (light) greyish-brown
2.5Y 8/2	Very pale (light) grey (calcareous)
5Y 6/1	Grey (gleyed)
N 3/0	Very dark grey (neutral, no hue)

15.6 Factors Influencing Soil Colour

A. Organic matter

- High organic matter -> **dark brown to black**.

- The dark colour is due to humus and organic coatings on particles.

B. Iron compounds (state of oxidation)

- **Oxidised (Fe3+)**, well-drained -> **red (haematite) and yellow/brown (goethite)**.
- **Reduced (Fe2+)**, waterlogged -> **grey, bluish-grey or greenish**.
- **Mottling** - alternate wet and dry -> rust spots on a grey background.

C. Mineral composition of parent material

- Iron-rich rocks -> red soils; basaltic rocks -> black soils; limestone -> pale/white.

D. Calcium carbonate, gypsum and salts

- Give **white or light grey** colours (calcareous, saline soils).

E. Drainage and aeration

- Well-drained -> red/yellow; poorly drained -> grey (gley).

F. Moisture content

- Moist soil looks darker than dry soil; value decreases when wet.

G. Climate and weathering

- Warm, humid climates -> red and lateritic colours; cool, humid -> podzolic greyish colours.

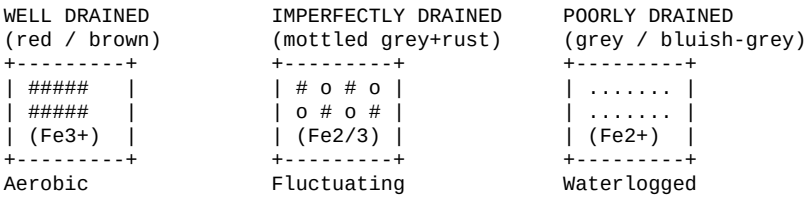
H. Manganese oxides

- Give **dark black** coatings (along with organic matter).

15.7 Relationship between Colour and Drainage

Colour pattern	Drainage condition	Iron state
Uniform red / yellow-brown	Well drained	Fe3+ (oxidised)
Grey with red/yellow mottles	Imperfectly drained	Mixed Fe2+/Fe3+
Uniform grey / bluish-grey	Poorly drained (gleyed)	Fe2+ (reduced)
Pale white	Calcareous / saline	Carbonate / salts

15.8 Diagram: Colour vs Drainage



15.9 Worked Example: Interpreting Munsell Notations

Problem: Three horizons have colours 5YR 4/6, 10YR 7/2 and 5Y 6/1. Interpret each.

Solution

- **5YR 4/6** - red hue, medium value, strong chroma -> **reddish-brown**, indicating well-drained, iron-oxide-rich (oxidised) soil.
- **10YR 7/2** - yellow hue, high value, low chroma -> **pale greyish-brown**, indicating a leached or calcareous horizon.
- **5Y 6/1** - yellow/grey hue, medium-high value, very low chroma -> **grey**, indicating a poorly drained (gleyed) horizon with reduced iron.

Answer: The first is well-drained oxidised soil; the second a pale leached/calcareous horizon; the third a gleyed, poorly drained horizon.

15.10 Tamil Nadu Context

- **Red soils** - 2.5YR to 5YR hues, reddish-brown (well-drained, iron oxides).
- **Black soils (vertisols)** - 10YR 2/1 to 10YR 3/1, very dark grey to black (organic matter + smectite).

- **Laterite soils** - 2.5YR to 5YR, red to dark red, often with a hard crust.
- **Alluvial soils of the Cauvery delta** - 10YR brown to dark brown, sometimes grey when waterlogged.
- **Coastal sandy soils** - 10YR pale yellowish-brown to white.

15.11 Exam Focus

Key points

- Soil colour indicates organic matter, drainage, iron state and parent material.
- Dark = organic matter; red/yellow = oxidised iron (well drained); grey/bluish = reduced iron (waterlogged).
- Munsell system uses three components: hue, value, chroma.
- Notation is written as Hue Value/Chroma (e.g., 10YR 4/3).
- Mottling indicates fluctuating (imperfect) drainage.
- White/pale colours indicate carbonate, gypsum or salts.
- Moist soil is darker than dry soil.
- Colour is a standard field characteristic for soil description and classification.

Short-answer questions

1. *What is the significance of soil colour?*

Model answer: Soil colour indicates organic matter content (dark), drainage (grey = poor, red = good), iron oxidation state, parent material and climate. It is used in soil description and classification.

2. *Explain the three components of the Munsell colour system.*

Model answer: Hue is the dominant colour (e.g., 10YR), value is the lightness (0-10), and chroma is the colour strength (0 upward). Soil colour is written as Hue Value/Chroma, e.g., 10YR 4/3.

3. *What causes red and grey colours in soil?*

Model answer: Red colour is caused by oxidised iron (haematite, Fe2O3) in well-drained soils; grey or bluish-grey colour is caused by reduced iron (Fe2+) in poorly drained, waterlogged (gleyed) soils.

Long-answer questions

1. *Describe soil colour, its measurement and the factors influencing it.*

Outline: Introduction - significance - causes of different colours (table) - Munsell system (hue, value, chroma) - how to read the chart - factors influencing colour (OM, iron, minerals, drainage, moisture, climate) - colour and drainage table - Tamil Nadu examples - conclusion.

2. *Explain how soil colour indicates drainage and soil conditions.*

Outline: Introduction - red/yellow = well drained - grey = poorly drained - mottling = imperfect drainage - pale = calcareous/saline - dark = organic matter - diagram - practical use in field description - conclusion.

Lecture 16: Soil Consistency - Cohesion, Adhesion, Plasticity, Atterberg's Constants - Upper and Lower Plastic Limits, Plasticity Number - Significance

16.1 Definition of Soil Consistency

Soil consistency is the **resistance of soil to deformation or rupture** under applied forces (pressure, tension, shear). It describes how the soil behaves when forces are applied, especially at different moisture contents.

Consistency is expressed by terms such as **loose, soft, firm, hard, friable, sticky, plastic** - depending on the moisture content.

16.2 The Forces Involved: Cohesion and Adhesion

A. Cohesion

- **Cohesion** is the **attraction between soil particles themselves** (particle-to-particle).
- It holds soil particles together.

- Cohesion is due to:

- **Van der Waals forces** between particles.

- **Surface tension of water films** pulling particles together.

- **Electrostatic attraction** of clay and cations.

- Cohesion increases as water films thin (in moist soil) but decreases when the soil is very wet or very dry.

B. Adhesion

- **Adhesion** is the **attraction between soil particles and other surfaces** (e.g., water, implements, plant roots).
- It causes soil to stick to ploughs, tyres and tools.
- Adhesion depends on moisture content - maximum at an intermediate water content.

16.3 Soil Consistency and Moisture States

Consistency changes with moisture content. Four states are recognised:

1. **Dry state** - soil is hard or soft; described as **hard, very hard, soft, loose**.
2. **Moist state** - soil is described as **loose, friable, firm** (friable is ideal for tillage).
3. **Wet state** - soil is described as **sticky** (adhesion) and **plastic** (moulding).
4. **Cementation** - when cemented by lime, iron or silica, described as **weakly, strongly or indurated**.

16.4 Plasticity

Definition: Plasticity is the **ability of a soil to be moulded or deformed (into any shape) and to retain that shape** after the moulding force is removed.

- Plasticity requires **clay and water**.
- Sandy soils are **non-plastic**; clayey soils are **highly plastic**.
- Plasticity depends on:

- **Clay content** - more clay, more plasticity.

- **Type of clay** - montmorillonite is more plastic than kaolinite.

- **Type of cations** - Na-clay is more plastic than Ca-clay.

- **Organic matter** - reduces plasticity.

16.5 Atterberg's Constants (Limits)

The Swedish scientist **Atterberg** defined the moisture contents at which a soil changes from one consistency state to another. These are called **Atterberg's limits (constants)**.

The main limits are:

1. **Liquid limit (upper plastic limit, LL)** - the **minimum moisture content at which the soil will flow like a liquid** (or: the moisture content at which the soil passes from the plastic to the liquid state).
2. **Plastic limit (lower plastic limit, PL)** - the **minimum moisture content at which the soil can be rolled into a thread of about 3 mm diameter without breaking** (or: the moisture content at which the soil changes from a semi-solid to a plastic state).
3. **Shrinkage limit (SL)** - the **maximum moisture content at which further drying does not cause further decrease in volume** (moisture content below which the soil is completely dry and does not shrink further).

16.6 Plasticity Number (Plasticity Index)

Plasticity number (PI) is the **difference between the liquid limit and the plastic limit**:

- **PI = LL - PL**

It represents the **range of moisture content over which the soil remains plastic**. A high PI means the soil is highly plastic (stays workable over a wide moisture range); a low PI means it is less plastic.

Related indices

- **Flow index** - slope of the flow curve.

- **Toughness index** - ratio of plasticity index to flow index.
- **Liquidity index** - (natural water content - PL) / PI.
- **Shrinkage index** - LL - SL.

16.7 Atterberg Limits - Summary Table

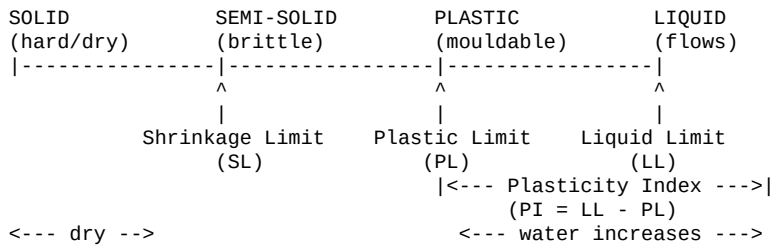
Limit	Symbol	Definition	Significance
Liquid limit	LL	Moisture content at which soil flows like a liquid	Upper limit of plasticity
Plastic limit	PL	Moisture content at which soil can be rolled to a 3 mm thread	Lower limit of plasticity
Shrinkage limit	SL	Moisture content below which no further shrinkage	Volume stability
Plasticity index	PI = LL - PL	Range of plastic moisture content	Measures plasticity
Shrinkage index	SI = LL - SL	Range of shrinkage	Measures shrink-swell

16.8 Typical Values of Atterberg Limits

Soil type	LL (%)	PL (%)	PI (%)
Sand	Non-plastic	-	0
Sandy loam	15-25	12-20	3-8
Loam	25-35	18-25	7-12
Silt loam	30-40	20-28	10-15
Clay loam	40-50	20-28	15-25
Kaolinite clay	40-60	25-35	15-25
Montmorillonite clay	100-500	40-60	60-400

(Values are standard ranges; montmorillonite has extremely high LL and PI due to its shrink-swell nature.)

16.9 Diagram: Consistency States and Atterberg Limits



16.10 Worked Example 1: Plasticity Index

Problem: A soil has a liquid limit of 45% and a plastic limit of 22%. Calculate the plasticity index and state its meaning.

Solution

- PI = LL - PL
- PI = 45 - 22 = **23%**

Answer: The plasticity index is 23%. The soil remains plastic over a moisture range of 23%, indicating a moderately to highly plastic (clayey) soil.

16.11 Worked Example 2: Shrinkage Index

Problem: A soil has a liquid limit of 60% and a shrinkage limit of 15%. Calculate the shrinkage index.

Solution

- Shrinkage index (SI) = LL - SL = 60 - 15 = **45%**

Answer: The shrinkage index is 45%, indicating a wide range of shrinkage - a soil that shrinks and swells considerably (e.g., montmorillonitic black soil).

16.12 Worked Example 3: Liquidity Index

Problem: A soil has a natural water content of 30%, a plastic limit of 20% and a plasticity index of 25%. Calculate the liquidity index and interpret.

Solution

- Liquidity index (LI) = (natural water content - PL) / PI
- LI = (30 - 20) / 25 = 10 / 25 = **0.40**

Answer: LI = 0.40 (between 0 and 1), so the soil is in a plastic, workable state; a value near 1 would mean the soil is close to its liquid limit (too wet).

16.13 Significance of Soil Consistency and Atterberg's Limits

- **Tillage and workability:** soil should be ploughed at the right (friable) moisture - between PL and LL it is plastic and sticky, so ploughing then is harmful.
- **Engineering and construction:** Atterberg limits are used to classify soils for roads, foundations and embankments.
- **Traffic and compaction:** working wet plastic soil causes compaction and structure damage.
- **Shrink-swell behaviour:** high PI and SI indicate vertisols that crack when dry and swell when wet.
- **Erosion:** plastic soils resist water erosion but are easily damaged by tillage.
- **Crop growth:** very plastic soils are poorly aerated and hard to manage; well-aggregated loams are friable and ideal.
- **Soil survey and classification:** consistency is a standard field property for describing horizons.

16.14 Management Implications

- **Do not plough when the soil is too wet** (plastic/sticky) or too dry (hard).
- **Add organic matter** to reduce plasticity and improve friability.
- **Apply gypsum** to sodic soils to reduce dispersion and improve structure.
- **Avoid heavy traffic** on wet soil to prevent compaction.
- **Manage vertisols** with timely tillage and moisture control.

16.15 Exam Focus

Key points

- Soil consistency is the resistance of soil to deformation under force.
- Cohesion = particle-to-particle attraction; adhesion = particle-to-other-surface attraction.
- Plasticity = ability to be moulded and retain shape; requires clay and water.
- Atterberg's limits: liquid limit, plastic limit, shrinkage limit.
- Plasticity index = LL - PL; shrinkage index = LL - SL.
- High LL and PI indicate montmorillonitic (black) soils; sands are non-plastic.
- Plough at friable moisture (between PL and LL is bad).
- Atterberg limits are used in agriculture and civil engineering.

Short-answer questions

1. *Define soil consistency and name the Atterberg limits.*

Model answer: Soil consistency is the resistance of soil to deformation under applied force. The Atterberg limits are the liquid limit (LL), the plastic limit (PL) and the shrinkage limit (SL).

2. *What is the plasticity index?*

Model answer: The plasticity index is the difference between the liquid limit and the plastic limit (PI = LL - PL). It is the range of moisture content over which the soil remains plastic.

3. *Distinguish cohesion and adhesion.*

Model answer: Cohesion is the attraction between soil particles themselves (particle-to-particle), holding the soil together. Adhesion is the attraction between soil particles and other surfaces (such as water films, implements or roots), causing soil to stick to tools.

Long-answer questions

1. *Explain soil consistency, cohesion, adhesion and the Atterberg constants.*

Outline: Definition of consistency - cohesion and adhesion - consistency states - plasticity - Atterberg limits (LL, PL, SL) with definitions - plasticity index - diagram - typical values - worked examples - significance - conclusion.

2. *Discuss the significance of Atterberg's limits in agriculture.*

Outline: Introduction - definition of limits - tillage and workability - shrink-swell behaviour - traffic and compaction - soil survey - crop growth - management implications - conclusion.

Lecture 17: Soil Water - Classification, Units of Expression and pF Scale

17.1 Introduction

Soil water is the liquid phase of the soil and is essential for plant growth, microbial activity and nutrient transport. It occupies the pore spaces not filled with air. The study of soil water includes its forms, the energy with which it is held, and its movement.

17.2 Importance of Soil Water

- **Essential for plant growth** - water is the medium for nutrient uptake and a raw material for photosynthesis.
- **Solvent and carrier** - dissolves and transports nutrients (the soil solution).
- **Temperature regulation** - buffers soil temperature.
- **Microbial activity** - needed for decomposition and nutrient cycling.
- **Weathering and soil formation** - drives hydrolysis and leaching.

17.3 Classification of Soil Water

Soil water is classified in several ways.

A. On the basis of retention/availability to plants (most common)

1. **Gravitational water** - water that drains freely through macropores under gravity. It is **not available** to plants (it drains away quickly) and may cause waterlogging.
2. **Capillary water** - water held in the micropores and capillary spaces by surface tension; it is the **main form available** to plants.
3. **Hygroscopic water** - water held very tightly as thin films around particles by adhesion (hygroscopic coefficient); it is **not available** to plants.
4. (Sometimes) **Water vapour** - water present as vapour in soil air.

B. On the basis of movement

- **Percolating (gravitational) water** - moves downward.
- **Capillary water** - moves in any direction by capillarity.
- **Hygroscopic water** - immobile films.

C. On the basis of origin/position

- **Pellicular water** - films held around particles.
- **Funicular water** - water in continuous films in pores.
- **Solvate water** - water bound to ions.

17.4 Diagram: Forms of Soil Water and Availability

```
SOIL WATER FORMS
================================================
```

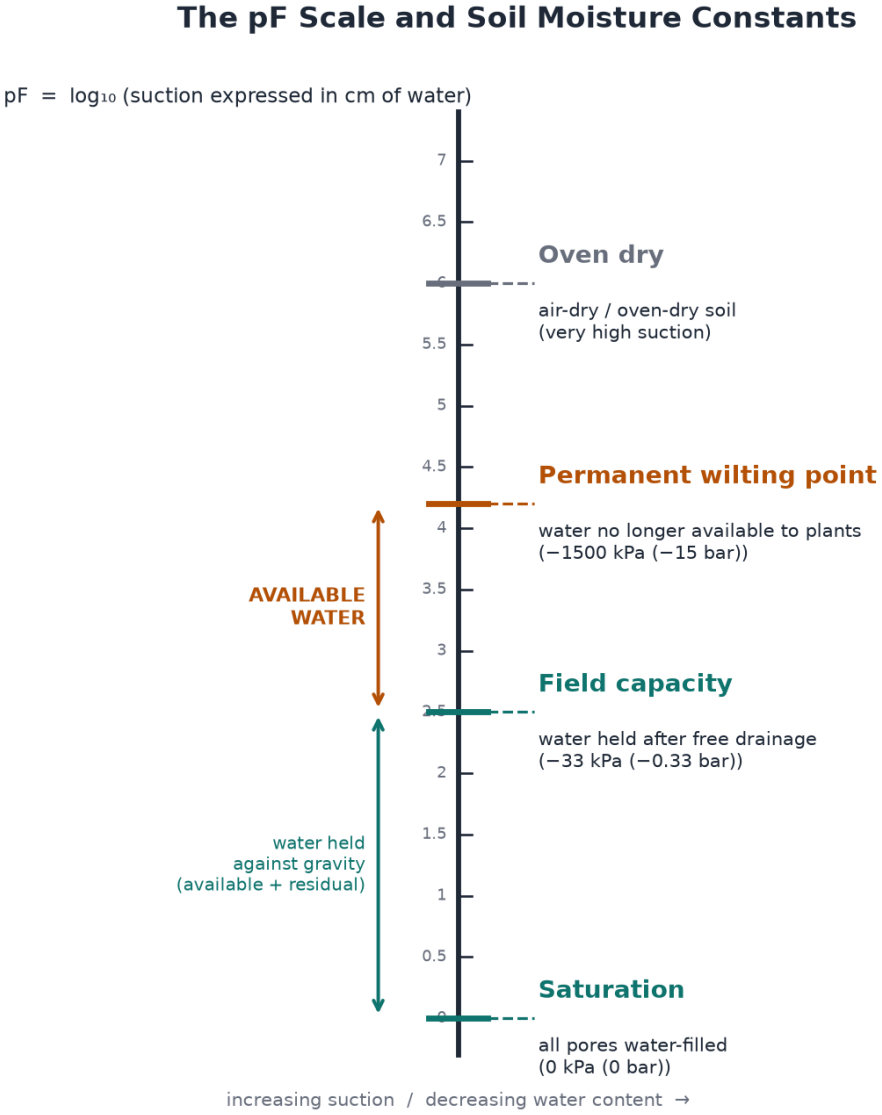


Figure 17.1 The pF scale and soil moisture constants

Saturation, field capacity, permanent wilting point and oven dry placed at their approximate pF values.

```
GRAVITATIONAL WATER   ---> drains away (not available)
(large pores, free)
-------------------------------------------------
CAPILLARY WATER       ---> AVAILABLE to plants
(small pores, held by
 surface tension)
-------------------------------------------------
HYGROSCOPIC WATER     ---> tightly held (not available)
(thin films, adhesion)
=================================================
Available water = capillary water (between FC and PWP)
```

17.5 Soil Moisture Constants

Soil moisture constants are the characteristic water contents at particular energy states:

- **Saturation** - all pores filled with water (water content = porosity).
- **Field capacity (FC)** - the water content after excess gravitational water has drained (usually 2-3 days after rain/irrigation). It is held at about **-33 kPa (1/3 bar)**. Also called field moisture capacity, normal moisture capacity.
- **Permanent wilting point (PWP)** - the water content at which plants can no longer extract water and wilt permanently. Held at about **-1500 kPa (15 bar)**.
- **Hygroscopic coefficient** - water content of air-dry soil, held at about **-3100 kPa (31 bar)**.
- **Oven-dry** - water content at **105 degrees C**, about -1000 MPa (essentially zero water).

Available water capacity (AWC) = Field capacity - Permanent wilting point.

17.6 Units of Expression of Soil Water

Soil water is expressed in several ways:

A. On a mass (weight) basis

- **Gravimetric water content (W)** = (mass of water / mass of oven-dry soil) x 100 (%)
- Also called percentage moisture on weight basis.

B. On a volume basis

- **Volumetric water content (theta)** = (volume of water / total volume of soil) x 100 (%)
- theta = W x Db (where W is the gravimetric fraction and Db is bulk density).

C. As depth of water

- **Depth of water (cm)** = volumetric water content x depth of soil (cm).
- Used in irrigation scheduling.

D. As energy (potential)

- Expressed in **bars, atmospheres, kPa or pF** (see below).

17.7 Soil Water Potential - Units

Water moves from a region of **higher potential (less negative)** to a region of **lower potential (more negative)**. The potential of free pure water is zero.

Common units:

- **Atmosphere (atm)** and **bar** (1 bar = 0.987 atm, taken as equal).
- **Pascal (Pa)** and its multiples: **kPa** (kilopascal) and **MPa** (megapascal).
- **1 bar = 100 kPa = 0.1 MPa = 10^5 Pa.**
- **pF** - the logarithm (base 10) of the negative pressure head expressed in cm of water.

17.8 The pF Scale

The **pF** value is defined as:

- **pF = log10 (height of water column in cm)** that corresponds to the suction (tension) holding the water.

It is a convenient logarithmic scale because the tensions range from 0 to about 1,000,000 cm of water.

Key conversions

- 1 bar = 1020 cm of water (approximately 1000 cm).
- pF = log10 (cm of water).

17.9 pF Scale and Moisture Constants Table

Condition	Tension (bar)	Tension (cm water)	pF	Availability
Saturation	0	0	0	-
Field capacity	0.33 (1/3)	330	2.54	Available
Field capacity (approx.)	0.1-0.3	100-300	2.0-2.5	Available
Capillary / available	0.33-15	330-15,000	2.5-4.2	Available
Permanent wilting point	15	15,000	4.2	Limit of available
Hygroscopic coefficient	31	31,000	4.5	Not available
Air-dry soil	~1000	~1,000,000	~6.0	Not available
Oven-dry (105 C)	very high	-	~7.0	Not available

(Values are standard published ranges; the exact pF for field capacity is often quoted as 2.5 and for PWP as 4.2.)

17.10 Diagram: pF Scale

```
pF SCALE
================================================
pF 7.0 | Oven-dry (105 C) - essentially no water
pF 6.0 | Air-dry soil (~1000 bar) - hygroscopic
pF 4.5 | Hygroscopic coefficient (31 bar)
pF 4.2 | PERMANENT WILTING POINT (15 bar) <-- plants wilt
       |     ... available water zone ...
pF 2.5 | FIELD CAPACITY (1/3 bar) <-- after drainage
pF 0.0 | SATURATION (all pores filled)
================================================
Available water = between pF 2.5 and pF 4.2
```

17.11 Worked Example 1: Gravimetric Water Content

Problem: A soil sample weighs 120 g when moist and 100 g when oven-dry. Calculate the gravimetric water content.

Solution

- Mass of water = 120 - 100 = 20 g
- Mass of oven-dry soil = 100 g
- Gravimetric water content W = (20 / 100) x 100 = **20%**

Answer: The gravimetric water content is 20%.

17.12 Worked Example 2: Volumetric Water Content

Problem: Using the soil above, if the bulk density is 1.4 g/cm3, calculate the volumetric water content.

Solution

- Gravimetric water content W = 0.20 (as a fraction)
- Volumetric water content theta = W x Db = 0.20 x 1.4 = 0.28 = **28%**

Answer: The volumetric water content is 28%.

17.13 Worked Example 3: Depth of Water

Problem: If a soil has a volumetric water content of 28% in the top 30 cm, how much water (in cm depth) is stored?

Solution

- Depth of water = volumetric water content x soil depth
- = 0.28 x 30 cm = **8.4 cm**

Answer: About 8.4 cm of water is stored in the top 30 cm. (1 cm of water over 1 hectare = 100,000 litres.)

17.14 Worked Example 4: pF from Tension

Problem: Calculate the pF value for a suction of 330 cm of water.

Solution

- pF = log10 (cm of water) = log10 (330)
- log10 (330) = log10 (3.3) + log10 (100) = 0.519 + 2 = **2.52**

Answer: pF = 2.52, which corresponds to field capacity (1/3 bar).

17.15 Worked Example 5: Available Water Capacity

Problem: A soil has a field capacity of 28% (volumetric) and a permanent wilting point of 12% (volumetric) in the top 30 cm. Calculate the available water capacity in cm.

Solution

- AWC (as fraction) = FC - PWP = 0.28 - 0.12 = 0.16
- AWC in depth = 0.16 x 30 cm = **4.8 cm**

Answer: The available water capacity is 4.8 cm of water in the top 30 cm.

17.16 Tamil Nadu Context

- **Sandy coastal soils** have low FC and low AWC (need frequent irrigation).
- **Black clay soils (vertisols)** have high FC and high AWC (hold much water but drain slowly).
- **Cauvery delta alluvial soils** have good AWC and are suited to paddy.
- TNAU recommends **irrigation scheduling based on soil moisture** (e.g., IW/CPE ratio) to avoid stress.

17.17 Exam Focus

Key points

- Soil water is the liquid phase, occupying pores not filled with air.
- Forms: gravitational (not available), capillary (available), hygroscopic (not available).
- Field capacity = -33 kPa (1/3 bar); permanent wilting point = -1500 kPa (15 bar).
- Available water = field capacity - permanent wilting point.
- Units: gravimetric (%), volumetric (%), depth (cm), potential (bar, kPa, pF).
- pF = log10 of the cm of water suction; FC = pF 2.5, PWP = pF 4.2.
- Volumetric water content = gravimetric x bulk density.
- Depth of water = volumetric water content x soil depth.

Short-answer questions

1. *Classify soil water on the basis of availability to plants.*

Model answer: Gravitational water (drains freely, not available), capillary water (held in small pores, available to plants) and hygroscopic water (held as thin films, not available).

2. *Define field capacity and permanent wilting point.*

Model answer: Field capacity is the water content after gravitational water has drained (about -33 kPa or 1/3 bar). Permanent wilting point is the water content at which plants wilt and cannot recover (about -1500 kPa or 15 bar).

3. *What is pF? Give the pF of field capacity and PWP.*

Model answer: pF is the logarithm (base 10) of the suction expressed in cm of water. Field capacity is about pF 2.5 and permanent wilting point about pF 4.2.

Long-answer questions

1. *Describe the classification of soil water and the units of its expression with the pF scale.*

Outline: Introduction - forms of soil water (gravitational, capillary, hygroscopic) - soil moisture constants - units of expression (gravimetric, volumetric, depth, potential) - pF scale and table - diagram - worked examples -

conclusion.

2. *Explain soil moisture constants and their importance in irrigation.*

Outline: Introduction - saturation, field capacity, PWP, hygroscopic coefficient - available water capacity - pF values - use in irrigation scheduling - Tamil Nadu examples - conclusion.

Lecture 18: Soil Water Potentials - Gravitational, Matric, Osmotic; Soil Moisture Constants

18.1 Introduction

Water in soil is held and moved by **energy**, not simply by "amount". The concept of **soil water potential** describes the energy state of soil water and predicts the direction and rate of its movement. Water always moves from a **higher (less negative) potential to a lower (more negative) potential**.

18.2 Definition of Soil Water Potential

Soil water potential is the **amount of work (energy) required to move a unit quantity of water from the soil to a reference state of free, pure water at the same temperature and pressure**. It is expressed as energy per unit mass (J/kg), per unit volume (pressure, kPa/bar), or per unit weight (head, cm).

- The reference state (free pure water) has **zero potential**.
- Soil water potential is usually **negative** (water in soil is held, so work must be done to remove it).
- Total water potential is the **sum** of several components.

18.3 Components of Total Soil Water Potential

psi_t = psi_g + psi_m + psi_o + psi_p + psi_a

Where:

- **psi_g = gravitational potential**
- **psi_m = matric potential**
- **psi_o = osmotic potential**
- **psi_p = pressure (hydrostatic) potential**
- **psi_a = pneumatic (air pressure) potential**

For most agricultural soils, the dominant components are **matric** and **osmotic** potentials; gravitational matters mainly for drainage.

18.4 Gravitational Potential (psi_g)

- Due to the **position of water in the gravitational field** (its height relative to a reference).
- Depends on **elevation**; water at a higher elevation has higher gravitational potential and tends to move downward.
- **Sign:** positive above the reference, negative below.
- **Effect:** causes **downward drainage** of free (gravitational) water.
- **Magnitude:** psi_g = z (height in cm of water).

18.5 Matric Potential (psi_m) - Also Called Capillary Potential

- Due to the **attraction of water by the soil matrix** (solid particles) through **adhesion and capillarity (surface tension)**.
- It is the main potential holding water against gravity in unsaturated soil.
- **Sign:** negative (water is held, so its potential is reduced).
- **Magnitude:** becomes more negative as the soil dries; near zero at saturation.
- **Controls:** water retention, available water, and water movement in unsaturated soil.
- **Examples of matric tension:** field capacity at -0.33 bar; PWP at -15 bar.

18.6 Osmotic Potential (psi_o) - Also Called Solute Potential

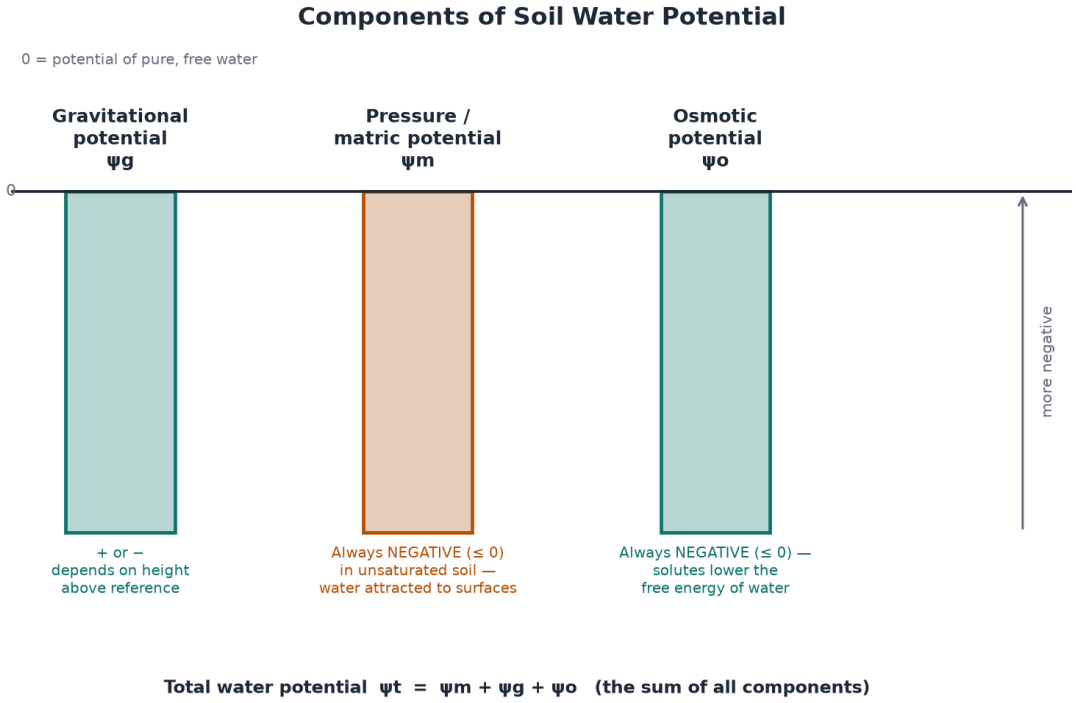


Figure 18.1 Components of soil water potential

Gravitational, matric (pressure) and osmotic components of the total soil water potential.

- Due to **dissolved salts (solutes)** in the soil solution.
- **Sign:** negative (solutes lower the potential of water).
- **Effect:** salts reduce the availability of water to plants (osmotic effect) - the "physiological drought".
- **Important in:** saline soils, where high salt content lowers osmotic potential and makes water harder for plants to extract.
- **Note:** the osmotic component affects **plant uptake** but does not itself move water through the soil matrix.

18.7 Comparison of the Potential Components

Component	Cause	Sign	Main effect
Gravitational	Height/position	+/-	Downward drainage
Matric	Soil matrix (adhesion, capillarity)	Negative	Holds water; unsaturated flow
Osmotic	Dissolved salts	Negative	Reduces water uptake by roots
Pressure (hydrostatic)	External pressure (water table)	Positive	Drives saturated flow
Pneumatic	Trapped air pressure	+/-	Air/gas movement

18.8 Diagram: Components of Soil Water Potential

```
TOTAL SOIL WATER POTENTIAL (psi_t)
==========================================
psi_t = psi_g + psi_m + psi_o + psi_p + psi_a
------------------------------------------
| psi_g  GRAVITATIONAL   (elevation, drainage)  |
| psi_m  MATRIC          (matrix, holds water)  |
| psi_o  OSMOTIC         (salts, -ve)           |
| psi_p  PRESSURE        (water table)          |
| psi_a  PNEUMATIC       (trapped air)          |
==========================================
Main players in agriculture: psi_m and psi_o
Water moves: high potential -> low potential
```

18.9 Soil Moisture Constants (Revisited)

Soil moisture constants are the water contents at defined potentials:

- **Saturation:** all pores full; potential = 0; water content = porosity.
- **Field capacity (FC):** after free drainage; potential about -0.33 bar (-33 kPa); pF 2.5.
- **Permanent wilting point (PWP):** potential about -15 bar (-1500 kPa); pF 4.2.
- **Hygroscopic coefficient:** potential about -31 bar; pF 4.5.
- **Air-dry soil:** about -1000 bar; pF 6.
- **Oven-dry:** 105 degrees C; essentially no water; pF about 7.

Available water capacity (AWC) = FC - PWP (the water held between -0.33 and -15 bar).

18.10 Water Movement and Potential Gradient

- Water moves **from higher potential to lower potential**.
- The driving force is the **potential gradient** (difference in potential per unit distance).
- In **saturated soil**, flow is driven mainly by **pressure (and gravitational) potential** - Darcy's law applies.
- In **unsaturated soil**, flow is driven mainly by **matric potential** gradients.
- In **saline soil**, the **osmotic potential** reduces plant water uptake.

18.11 Worked Example 1: Total Potential

Problem: A soil has a matric potential of -0.5 bar, an osmotic potential of -0.2 bar, and a gravitational potential of +0.1 bar. Calculate the total water potential (neglecting minor components).

Solution

- psi_t = psi_g + psi_m + psi_o

- psi_t = (+0.1) + (-0.5) + (-0.2) = **-0.6 bar**

Answer: The total water potential is -0.6 bar.

18.12 Worked Example 2: Comparing Two Points

Problem: Point A has a total potential of -0.5 bar and Point B has -0.8 bar. In which direction will water move?

Solution

- Water moves from higher (less negative) to lower (more negative) potential.
- -0.5 bar > -0.8 bar, so Point A has the higher potential.
- Water moves from **A to B**.

Answer: Water moves from A (higher potential) to B (lower potential).

18.13 Worked Example 3: Osmotic Effect on Available Water

Problem: A saline soil has an osmotic potential of -2 bar. If the plant can extract water only down to a total potential of -15 bar, what effective matric tension can the plant work against before the osmotic component is added?

Solution

- Total potential at wilting = psi_m + psi_o = -15 bar
- Given psi_o = -2 bar
- So psi_m = -15 - (-2) = -13 bar

Answer: The plant can only extract water down to a matric tension of about -13 bar (instead of -15 bar), so the available water is reduced by the salt effect - a "physiological drought".

18.14 Soil Moisture Constants Summary Table

Constant	Tension (bar)	pF	Availability
Saturation	0	0	-
Field capacity	0.33	2.5	Upper limit of available
Available range	0.33-15	2.5-4.2	Available
Permanent wilting point	15	4.2	Lower limit of available
Hygroscopic coefficient	31	4.5	Not available
Air-dry	~1000	~6.0	Not available

18.15 Exam Focus

Key points

- Soil water potential is the energy state of soil water; free water = 0.
- Water moves from higher (less negative) to lower (more negative) potential.
- Total potential = gravitational + matric + osmotic + pressure + pneumatic.
- Matric potential (negative) holds water in the matrix; main component in unsaturated soil.
- Osmotic potential (negative) is due to salts; causes physiological drought in saline soils.
- Gravitational potential causes downward drainage.
- Field capacity at -0.33 bar; PWP at -15 bar; AWC = FC - PWP.
- In saline soils, osmotic potential reduces available water.

Short-answer questions

1. *Define soil water potential and name its main components.*

Model answer: Soil water potential is the work required to move a unit quantity of water from the soil to free pure water at the same temperature and pressure. Its main components are gravitational, matric, osmotic (and pressure and pneumatic) potentials.

2. *What is matric potential?*

Model answer: Matric potential is the potential due to the attraction of water by the soil matrix through adhesion and capillarity. It is negative and is the main potential holding water in unsaturated soil.

3. *How does osmotic potential affect plant water uptake?*

Model answer: Dissolved salts lower the osmotic potential (make it more negative), so plants must work against a greater total tension to extract water. This reduces the available water and causes "physiological drought" in saline soils.

Long-answer questions

1. *Explain the components of soil water potential and their significance.*

Outline: Definition - total potential equation - gravitational, matric, osmotic, pressure, pneumatic potentials with cause, sign and effect - comparison table - diagram - significance in irrigation and salinity - worked examples - conclusion.

2. *Describe soil moisture constants and their relation to water potential.*

Outline: Introduction - saturation, field capacity, PWP, hygroscopic coefficient with tensions and pF - available water capacity - significance in irrigation scheduling - saline soil effect - conclusion.

Lecture 19: Movement of Soil Water - Saturated and Unsaturated Flow - Infiltration, Hydraulic Conductivity, Percolation, Permeability and Drainage

19.1 Introduction

Water moves through soil in two broad conditions: when all pores are filled with water (**saturated flow**) and when pores are partly filled with air (**unsaturated flow**). Understanding this movement is essential for irrigation, drainage and erosion control.

19.2 Saturated Flow and Darcy's Law

Saturated flow occurs when the soil pores are completely filled with water (below the water table or just after heavy irrigation).

The governing law is **Darcy's law**:

- **Q = K x A x (dh/L)**

Where:

- Q = volume of water flowing per unit time (cm3/s or m3/s).
- K = **hydraulic conductivity** (cm/s or m/day).
- A = cross-sectional area of flow (cm2).
- dh/L = hydraulic gradient (change in head per unit distance).

In simple form: **flux (q) = Q/A = K x (dh/L) = K x i**, where i is the hydraulic gradient.

Key points

- Flow is driven by **gravity and pressure (hydrostatic) potential**.
- K is highest when the soil is saturated.
- Flow occurs mainly through **large (macro) pores**.

19.3 Unsaturated Flow

Unsaturated flow occurs when the pores are only partly filled with water (the common field condition above the water table).

- Flow is driven mainly by **matric potential gradients** (suction), not by gravity alone.
- As the soil dries, the **unsaturated hydraulic conductivity decreases rapidly** (fewer water-filled pores).
- Darcy's law is used in modified form with unsaturated K.
- Water moves from wetter (higher potential) to drier (lower potential) zones.

Difference: In saturated flow, K is constant for a given soil; in unsaturated flow, K changes with water content.

19.4 Infiltration

Definition: Infiltration is the **process by which water enters the soil surface** (downward entry of water into the soil).

- **Infiltration rate** = volume of water entering per unit area per unit time (cm/h or mm/h).
- **Initial infiltration rate** is high (dry soil) and decreases with time to a **steady (basic) infiltration rate**.
- Factors affecting infiltration: soil texture, structure, initial moisture, surface crusting, organic matter, vegetation cover, slope, compaction.

Typical infiltration rates (standard ranges):

Soil type	Infiltration rate (mm/h)	Class
Coarse sand	> 30	Very rapid
Sandy loam	15-30	Rapid
Loam	8-15	Moderate
Silt loam	2.5-8	Moderate to slow
Clay loam	0.5-2.5	Slow
Clay (wet)	< 0.5	Very slow

19.5 Hydraulic Conductivity

Definition: Hydraulic conductivity (K) is the **ability of a soil to transmit water** under a unit hydraulic gradient. It is the proportionality constant in Darcy's law.

- Units: cm/s, cm/h or m/day.
- **Saturated hydraulic conductivity (Ks)** is the maximum (all pores full).
- Depends on **pore size and continuity** (macro-pores) more than on total porosity.
- Sandy soils: high K (fast); clayey soils: low K (slow).

19.6 Percolation

Definition: Percolation is the **downward movement of water through the soil profile** after infiltration, i.e., the movement of water through the soil body toward the water table.

- Infiltration is entry at the surface; percolation is movement within the profile.
- Percolation rate is also expressed as cm/h or mm/h.

19.7 Permeability

Definition: Permeability is the **property of a soil that allows water (or air) to pass through it**; it is a qualitative expression of hydraulic conductivity.

- **Permeability classes:** very slow, slow, moderate, rapid, very rapid.
- It depends on **pore size, pore continuity, texture, structure and compaction**.
- Note: **hydraulic conductivity** is the quantitative measure; **permeability** is the general property (sometimes used interchangeably, though strictly permeability is a property of the medium independent of fluid).

19.8 Drainage

Definition: Drainage is the **removal of excess water from the soil** (both surface and sub-surface).

- **Surface drainage** - removal of excess water from the surface (open drains, land shaping).
- **Sub-surface (internal) drainage** - removal of excess water from within the profile (tile drains, open ditches).
- **Natural drainage** - by gravity through the profile to the water table.
- **Artificial drainage** - by installing drains to lower the water table.

Need for drainage: waterlogging reduces aeration, restricts root growth, causes denitrification and nutrient loss, and reduces yields.

Benefits: improves aeration, root growth, microbial activity, and allows timely tillage and traffic.

19.9 Relationship of the Terms

```
RAIN / IRRIGATION
     |
     v
[INFILTRATION]  <-- entry at the surface
     |
     v
[PERCOLATION]   <-- movement down the profile
     |
     v
[HYDRAULIC CONDUCTIVITY / PERMEABILITY]  <-- the soil's ability to transmit
     |
     v
[DRAINAGE]      <-- removal of excess water
```

19.10 Worked Example 1: Darcy's Law - Discharge

Problem: A soil column has a cross-sectional area of 100 cm2 and a hydraulic conductivity of 2 cm/h. The hydraulic head difference is 10 cm over a length of 20 cm. Calculate the discharge (Q).

Solution

- Hydraulic gradient i = dh/L = 10/20 = 0.5
- Q = K x A x i = 2 x 100 x 0.5 = **100 cm3/h**

Answer: The discharge is 100 cm3 per hour.

19.11 Worked Example 2: Flux (Darcy Velocity)

Problem: Using the same soil, calculate the flux (q).

Solution

- q = Q/A = 100/100 = **1 cm/h**
- (Also q = K x i = 2 x 0.5 = 1 cm/h.)

Answer: The flux is 1 cm/h.

19.12 Worked Example 3: Infiltration and Time

Problem: A soil has a steady infiltration rate of 10 mm/h. How long will it take to infiltrate 50 mm of water (ignoring initial high rate)?

Solution

- Time = depth / rate = 50 / 10 = **5 hours**

Answer: It takes about 5 hours to infiltrate 50 mm of water.

19.13 Worked Example 4: Infiltration Volume

Problem: A rainfall of 40 mm falls on one hectare. If the infiltration rate is 12 mm/h and the storm lasts 2 hours, how much water infiltrates and how much runs off?

Solution

- Rainfall = 40 mm
- Water that can infiltrate = 12 x 2 = 24 mm
- Runoff = 40 - 24 = **16 mm**
- Volume infiltrated = 24 mm x 10,000 m2 = 0.024 m x 10,000 = 240 m3
- Volume runoff = 16 mm x 10,000 m2 = 160 m3

Answer: 24 mm (240 m3) infiltrates and 16 mm (160 m3) runs off.

19.14 Worked Example 5: Drainage / Water Table Drop

Problem: A drainage system removes 12 mm of water per day from a waterlogged field. How long will it take to remove 60 mm of excess water?

Solution

- Time = 60 / 12 = **5 days**

Answer: About 5 days are needed to remove 60 mm of excess water.

19.15 Factors Affecting Water Movement

Factor	Effect on movement
Texture	Sand - fast; clay - slow
Structure	Good aggregation - fast; massive - slow
Compaction	Reduces conductivity
Organic matter	Improves structure, increases infiltration
Initial moisture	Wet soil - lower infiltration
Surface crust	Reduces infiltration
Slope	Steeper slope - more runoff
Vegetation	Improves infiltration, reduces runoff

19.16 Tamil Nadu Context

- **Coastal sandy soils** have very high infiltration and hydraulic conductivity but low water-holding capacity; need frequent, light irrigation.
- **Black clay soils (vertisols)** have low infiltration when wet (swelling) but crack when dry (high initial infiltration); need careful water management.
- **Cauvery delta alluvial soils** are moderately permeable; paddy culture suits them.
- **Waterlogging** in parts of the delta and in low-lying areas requires drainage.
- TNAU recommends **raised beds and furrow irrigation** in heavy soils and **drip irrigation** in sandy and water-scarce areas.

19.17 Exam Focus

Key points

- Saturated flow follows Darcy's law: Q = K x A x (dh/L).
- Unsaturated flow is driven by matric potential; K decreases as soil dries.
- Infiltration = entry of water at the surface; percolation = movement down the profile.
- Hydraulic conductivity is the quantitative measure; permeability is the general property.
- Drainage removes excess water (surface and sub-surface).
- Sandy soils have high conductivity and infiltration; clay soils low.
- Infiltration rate decreases with time to a steady rate.
- Waterlogging harms aeration and yields; drainage corrects it.

Short-answer questions

1. *State Darcy's law and explain its terms.*

Model answer: Darcy's law is Q = K x A x (dh/L), where Q is the discharge, K is the hydraulic conductivity, A is the cross-sectional area and dh/L is the hydraulic gradient. It describes saturated water flow in soil.

2. *Distinguish infiltration and percolation.*

Model answer: Infiltration is the entry of water into the soil at the surface; percolation is the downward movement of water through the soil profile after infiltration.

3. *What is hydraulic conductivity?*

Model answer: Hydraulic conductivity is the ability of a soil to transmit water under a unit hydraulic gradient. It is the proportionality constant in Darcy's law and depends mainly on pore size and continuity.

Long-answer questions

1. *Explain saturated and unsaturated flow of soil water and Darcy's law.*

Outline: Introduction - saturated flow, Darcy's law and terms - unsaturated flow and matric potential - difference between the two - hydraulic conductivity - worked examples - conclusion.

2. *Discuss infiltration, hydraulic conductivity, percolation, permeability and drainage.*

Outline: Definitions - infiltration and factors - hydraulic conductivity - percolation - permeability - drainage (types, need, benefits) - relationship diagram - Tamil Nadu examples - conclusion.

Lecture 20: Soil Air - Composition, Gaseous Exchange - Effect on Crop Growth

20.1 Introduction

Soil air is the gaseous phase of the soil, occupying the pore spaces not filled with water. It is essential for the respiration of plant roots and soil microorganisms. The composition of soil air differs from that of atmospheric air because of biological activity in the soil.

20.2 Composition of Soil Air

Gas	Atmospheric air (%)	Soil air (%)
Nitrogen (N2)	78.0	78.0 - 80.0
Oxygen (O2)	20.9	10 - 20 (varies)
Carbon dioxide (CO2)	0.04	0.2 - 3.0 (can be higher)
Water vapour	variable	Usually near saturation
Other gases	traces	traces (CH4, N2O, H2S)

Key differences from atmospheric air

- **Less oxygen** (used by roots and microbes).
- **More carbon dioxide** (produced by respiration and decomposition).
- **Nearly saturated with water vapour** (humidity near 100%).
- **More variable** in composition - changes with depth, moisture and biological activity.

20.3 Soil Air and Soil Water - Inverse Relationship

- Air and water together fill the pore space.
- As water content **increases**, air content **decreases**, and vice versa.
- At saturation, the soil has **no air** (anaerobic) - harmful to most crops.
- Ideal soil has about **25% water and 25% air** by volume.
- **Air-filled porosity** is the fraction of the total volume occupied by air.

20.4 Gaseous Exchange (Soil Respiration and Aeration)

Gaseous exchange is the process by which soil air is renewed - O2 enters and CO2 leaves the soil.

Mechanisms of gaseous exchange

1. **Mass flow (convection)** - bulk movement of air due to pressure differences (caused by temperature, wind, barometric changes, water movement).
2. **Diffusion** - movement of individual gas molecules from higher to lower concentration; the **main mechanism** for O2 and CO2 exchange.
3. **Movement with water** - dissolved gases carried by water moving into or out of the soil.

Driving force: the **concentration gradient** of O2 and CO2 between the atmosphere and the soil air.

20.5 Diagram: Soil Aeration and Exchange

```
ATMOSPHERE            SOIL SURFACE           SOIL
O2 high   --------->  (O2 enters)   --------->  O2 used by roots
CO2 low   <---------  (CO2 leaves)  <---------  CO2 produced
=======================================================
DIFFUSION is the main mechanism (O2 in, CO2 out)
Mass flow also occurs (pressure/temperature changes)
```

```
==================================================
Good aeration  = pores well connected to the atmosphere
Poor aeration  = pores filled with water (waterlogging)
```

20.6 Oxygen and CO2 Requirements

- Most crops need soil air with at least **10% oxygen**; below this, root growth is restricted.
- **Oxygen diffusion rate (ODR)** is a measure of aeration; crops generally need ODR above about **20-30 x 10^-8 g cm^-2 min^-1** (standard range).
- **CO2** above about **10%** is harmful; high CO2 reduces root growth and nutrient uptake.
- **Critical values vary with crop;** wetland crops (paddy) can tolerate low oxygen because they have air spaces (aerenchyma) in their roots.

20.7 Effect of Poor Aeration (Waterlogging) on Crops

- **Root respiration** is reduced; roots die.
- **Nutrient uptake** falls (water and nutrients cannot be absorbed).
- **Denitrification** - nitrate is converted to N2 gas and lost.
- **Toxic substances** - reduced compounds form: **methane (CH4)**, **hydrogen sulphide (H2S)**, **ferrous iron (Fe2+)**, **manganese (Mn2+)** and organic acids - all harmful to roots.
- **Growth and yield** decline; plants become stunted and yellow (chlorosis).
- **Soil structure** deteriorates; puddling occurs.

20.8 Effect of Soil Air on Crop Growth (Summary)

Air condition	Effect on crops
Good aeration (O2 > 10%)	Healthy roots, good nutrient uptake, high yield
Moderate aeration	Some reduction in growth
Poor aeration (waterlogged)	Root death, denitrification, toxic compounds, low yield
Excess CO2 (> 10%)	Inhibits root growth, reduces uptake
Wetland (paddy)	Tolerated by aerenchyma in adapted crops

20.9 Factors Affecting Soil Aeration

- **Soil texture:** sandy soils are well aerated; clay soils poorly aerated when wet.
- **Soil structure:** well-aggregated soils have good pore continuity.
- **Moisture content:** high moisture reduces air-filled porosity.
- **Compaction:** reduces pore space, reduces aeration.
- **Organic matter:** high OM increases respiration (more O2 demand) but improves structure.
- **Temperature:** higher temperature increases microbial activity and O2 demand.
- **Depth:** aeration decreases with depth.

20.10 Management of Soil Aeration

- **Drainage** of waterlogged soils (surface and sub-surface drains).
- **Adding organic matter** to improve structure and porosity.
- **Avoiding compaction** (controlled traffic, minimum tillage).
- **Raised beds and ridges** for heavy, poorly drained soils.
- **Aerating tillage** (sub-soiling) for compacted layers.
- **Crop selection:** grow tolerant crops (paddy, certain grasses) in wet soils.
- **Irrigation management:** avoid over-irrigation that fills all pores.

20.11 Worked Example: Air-Filled Porosity

Problem: A soil has a total porosity of 50% and a volumetric water content of 35%. Calculate the air-filled porosity.

Solution

- Air-filled porosity = total porosity - volumetric water content
- = 50 - 35 = **15%**

Answer: The air-filled porosity is 15%. If the volumetric water content rises above 45-50% (near saturation), aeration becomes poor and roots may suffer.

20.12 Worked Example: Water and Air Fractions

Problem: A soil of total volume 200 cm3 has 90 cm3 of water. If the porosity is 50%, find the air content.

Solution

- Pore volume = 50% of 200 = 100 cm3
- Water volume = 90 cm3
- Air volume = 100 - 90 = **10 cm3** (5% of total volume)
- Air-filled porosity = 10/200 x 100 = **5%** (very poor aeration)

Answer: Only 5% of the soil volume is air - the soil is nearly waterlogged and aeration is poor.

20.13 Exam Focus

Key points

- Soil air occupies the pore spaces not filled with water.
- Soil air has less O2, more CO2 and is nearly saturated with water vapour compared with atmospheric air.
- Air and water are inversely related in the pore space.
- Gaseous exchange occurs mainly by diffusion (O2 in, CO2 out); mass flow also occurs.
- Most crops need at least 10% O2 in soil air.
- Poor aeration causes root death, denitrification, toxic compounds (CH4, H2S, Fe2+) and yield loss.
- Paddy tolerates low oxygen (aerenchyma).
- Management: drainage, organic matter, avoiding compaction, raised beds.

Short-answer questions

1. *Compare the composition of soil air with atmospheric air.*

Model answer: Soil air has less oxygen (10-20% vs 20.9%), more carbon dioxide (0.2-3% vs 0.04%) and is nearly saturated with water vapour, compared with atmospheric air. Nitrogen is similar in both.

2. *How does gaseous exchange occur in soil?*

Model answer: Mainly by diffusion, driven by concentration gradients of O2 and CO2 between the atmosphere and soil air. Mass flow (convection) due to pressure and temperature changes also occurs.

3. *Why is waterlogging harmful to crops?*

Model answer: Waterlogging fills the pores with water, reducing oxygen. Root respiration stops, nutrient uptake falls, denitrification occurs and toxic substances (CH4, H2S, Fe2+) form, so roots die and yield falls.

Long-answer questions

1. *Describe the composition of soil air, gaseous exchange and the effect of aeration on crop growth.*

Outline: Introduction - composition table - air-water relationship - gaseous exchange (diffusion, mass flow) - O2 and CO2 requirements - effect of poor aeration - diagram - management - conclusion.

2. *Explain how soil aeration affects crop growth and how it can be managed.*

Outline: Introduction - aeration and root respiration - nutrient uptake - denitrification - toxic compounds - factors affecting aeration - management (drainage, organic matter, raised beds, avoiding compaction) - Tamil Nadu examples - conclusion.

Lecture 21: Soil Temperature - Source, Amount and Flow of Heat in Soil and Effect on Crop Growth

21.1 Introduction

Soil temperature is the degree of hotness or coldness of the soil. It is a physical property that strongly influences seed germination, root growth, microbial activity, nutrient availability and the rate of chemical reactions in the soil.

21.2 Sources of Heat in Soil

- **Solar radiation (the sun)** - the **main (primary) source** of soil heat.
- **Heat from the Earth's interior** - geothermal heat; very small for surface soils.
- **Chemical and biological reactions** - decomposition of organic matter and respiration release small amounts of heat.
- **Warm rain and irrigation water** - add heat in some conditions.
- **Latent heat** - released when water vapour condenses in the soil.

21.3 Heat Balance and Flow of Heat in Soil

Soil gains heat mainly from solar radiation and loses it by radiation, convection, evaporation and conduction.

Heat flow in soil occurs mainly by **conduction** (transfer of heat from warmer to cooler regions through the soil solids and water). The governing law is **Fourier's law**:

- **Q = -K x A x (dT/dx)**

Where:

- Q = rate of heat flow.
- K = **thermal conductivity** of the soil.
- A = cross-sectional area.
- dT/dx = temperature gradient.

Thermal properties of soil

- **Thermal conductivity (K):** the ability to conduct heat; highest for quartz (sand), water and moist soils; lowest for dry, organic soils (air is a poor conductor).
- **Specific heat:** the heat needed to raise the temperature of unit mass by 1 degree C; water has a high specific heat (so wet soils warm slowly).
- **Thermal diffusivity:** thermal conductivity divided by volumetric heat capacity; measures how fast temperature changes spread.

21.4 Daily and Seasonal Variation of Soil Temperature

- **Daily (diurnal) variation:** soil temperature rises during the day (afternoon maximum) and falls at night (early morning minimum). The amplitude decreases with depth; at about 30-50 cm the daily variation is small.
- **Seasonal variation:** temperature is highest in summer and lowest in winter; the amplitude decreases with depth and lags with depth.
- **At great depth** (below about 10-15 m), temperature is nearly constant (affected only by geothermal heat).

Temperature wave: both daily and seasonal temperature waves show a **time lag** and **decreasing amplitude** with depth.

21.5 Factors Affecting Soil Temperature

Factor	Effect
Solar radiation	Main source; more radiation - higher temperature
Latitude	Lower latitude (near equator) - warmer soil
Altitude	Higher altitude - cooler soil (Nilgiris cooler than plains)
Season	Summer warm; winter cool
Time of day	Afternoon warmest; early morning coolest

Factor	Effect
Soil colour	Dark soils absorb more heat and warm faster
Moisture	Wet soils warm slowly (high specific heat, evaporation cools)
Texture	Sandy soils warm faster; clay soils slowly
Organic matter	High OM - lower temperature (insulation, poor conductor)
Vegetation/mulch	Shade and insulation keep soil cooler
Slope/aspect	Sun-facing slope warmer
Ground cover	Bare soil warmer than covered soil

21.6 Effect of Soil Temperature on Crop Growth

A. Seed germination

- Each crop has an **optimum temperature range** for germination.
- Too cold: slow or no germination; too hot: poor germination.
- Typical optimum for many tropical crops: **20-30 degrees C**.

B. Root growth

- Roots grow best at moderate temperatures; extremes reduce growth.
- Cold soils reduce water and nutrient uptake (roots less active).

C. Nutrient availability and uptake

- Warm soils increase microbial mineralisation, releasing N, P, S.
- Low temperature reduces mineralisation and uptake (especially P).

D. Microbial activity

- Warm, moist soils have high microbial activity; cold soils slow it.
- Nitrogen fixation and decomposition are temperature-sensitive.

E. Chemical reactions

- Higher temperature speeds up chemical weathering and reactions.

F. Water and air movement

- Temperature gradients cause water vapour movement and affect viscosity.

G. Overall crop growth and yield

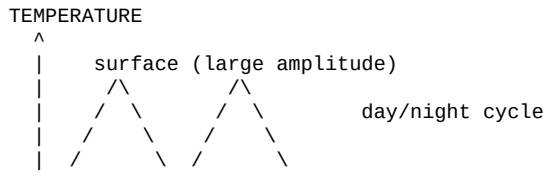
- Optimum temperature gives good growth and yield; extremes (heat or cold stress) reduce them.

21.7 Optimum Soil Temperature Ranges (Standard)

Crop / process	Optimum soil temperature (degrees C)
Germination (most tropical crops)	20 - 30
Germination (cool-season crops)	10 - 20
Root growth (most crops)	15 - 25
Microbial activity (general)	25 - 35
Nitrification	25 - 35
Paddy (wetland)	25 - 30

(Values are standard ranges; exact optima vary with crop and variety.)

21.8 Diagram: Daily Soil Temperature Variation with Depth



Mounish Raj · MouRa · Agri AI

```
  +------------------------> time
     (afternoon peak, early morning minimum)

With depth:
- amplitude decreases
- temperature wave lags (delay)
At ~30-50 cm: daily variation almost nil
```

21.9 Management of Soil Temperature

- **Mulching** - keeps soil cool in summer and warm in winter; conserves moisture.
- **Soil colour and surface** - dark surfaces warm faster (useful in cold regions).
- **Irrigation** - cools soil (evaporation); timing affects temperature.
- **Vegetation/cover crops** - shade and insulate.
- **Raised beds and ridges** - warm faster in spring.
- **Greenhouses/polythene mulch** - raise soil temperature for early crops.
- **Drainage** - removes excess water, allows faster warming.

21.10 Worked Example: Temperature Gradient and Heat Flow (Concept)

Problem: The soil surface is at 30 degrees C and the soil at 50 cm depth is at 25 degrees C. Calculate the temperature gradient.

Solution

- Temperature difference = 30 - 25 = 5 degrees C
- Distance = 50 cm = 0.5 m
- Temperature gradient = 5 / 0.5 = **10 degrees C per metre**

Answer: The temperature gradient is 10 degrees C/m. Heat flows from the warmer surface downward according to Fourier's law.

21.11 Worked Example: Amplitude with Depth (Concept)

Problem: The daily temperature amplitude at the surface is 12 degrees C. If the amplitude halves for every 10 cm depth, what is the amplitude at 30 cm?

Solution

- Surface amplitude = 12 degrees C
- 10 cm: 6 degrees C
- 20 cm: 3 degrees C
- 30 cm: 1.5 degrees C

Answer: The amplitude at 30 cm is about 1.5 degrees C, showing that daily temperature variation is much reduced at depth.

21.12 Tamil Nadu Context

- **Summer soil temperatures** in the plains (e.g., Coimbatore, Madurai) can exceed **35-40 degrees C** at the surface, affecting germination and requiring mulching.
- **The Nilgiris (higher altitude)** have cooler soils, suited to temperate crops (tea, potato).
- **Mulching** is recommended by TNAU to moderate soil temperature and conserve moisture.
- **Paddy soils** maintain moderate temperatures under standing water, buffering extremes.

21.13 Exam Focus

Key points

- Solar radiation is the main source of soil heat.
- Heat flows mainly by conduction; Fourier's law applies.
- Thermal conductivity is highest in moist, sandy soils; lowest in dry, organic soils.
- Daily and seasonal temperature variations decrease in amplitude and lag with depth.

- Dark, sandy, dry, bare soils warm fastest.
- Soil temperature affects germination, root growth, microbial activity and nutrient availability.
- Optimum germination temperature for most tropical crops: 20-30 degrees C.
- Management: mulching, irrigation, drainage, raised beds, polythene mulch.

Short-answer questions

1. *What is the main source of heat in soil and how does heat flow?*

Model answer: The main source is solar radiation. Heat flows mainly by conduction from warmer to cooler regions, governed by Fourier's law.

2. *How does soil temperature vary with depth?*

Model answer: Both daily and seasonal temperature variations decrease in amplitude and show a time lag with increasing depth. At about 30-50 cm the daily variation is almost nil, and below 10-15 m temperature is nearly constant.

3. *List four factors affecting soil temperature.*

Model answer: Solar radiation, soil colour, moisture content and texture (also latitude, altitude, season, organic matter and vegetation).

Long-answer questions

1. *Explain the source, amount and flow of heat in soil and the factors affecting soil temperature.*

Outline: Introduction - sources of heat - heat balance - heat flow by conduction (Fourier's law) - thermal properties (conductivity, specific heat, diffusivity) - daily and seasonal variation - factors table - management - conclusion.

2. *Discuss the effect of soil temperature on crop growth.*

Outline: Introduction - germination - root growth - nutrient availability and uptake - microbial activity - chemical reactions - optimum ranges table - management of temperature - Tamil Nadu examples - conclusion.

Lecture 22: Soil Reaction (pH) - Definition, pH Scale, Classification - Soil Acidity and Alkalinity, Buffering, Factors Affecting Soil pH, Effect on Nutrient Availability

22.1 Definition of Soil pH

Soil pH is the **negative logarithm (base 10) of the hydrogen ion activity (concentration)** in the soil solution.

- **pH = -log10 [H+]**

It is a measure of the **acidity or alkalinity** of the soil solution. Because it is a logarithmic scale, a change of one pH unit means a **ten-fold change** in H+ concentration.

22.2 The pH Scale

- The pH scale runs from **0 to 14** (practically 1-14 for soils).
- **pH 7 = neutral** (pure water at 25 degrees C).
- **pH < 7 = acidic** (more H+).
- **pH > 7 = alkaline (basic)** (more OH-).
- Most **agricultural soils** lie between **pH 4.5 and 9.0**.

22.3 Classification of Soils by pH

pH range	Class	Common description
< 4.5	Extremely acid	Acid sulphate soils
4.5 - 5.0	Very strongly acid	Strongly leached
5.1 - 5.5	Strongly acid	Acidic soils
5.6 - 6.0	Moderately acid	Slightly acidic

pH range	Class	Common description
6.1 - 6.5	Slightly acid	Near neutral
6.6 - 7.3	Neutral	Ideal for most crops
7.4 - 7.8	Slightly alkaline	Calcareous
7.9 - 8.4	Moderately alkaline	Calcareous
8.5 - 9.0	Strongly alkaline	Sodic soils
> 9.0	Very strongly alkaline	Highly sodic

(The optimum pH for most crops is **6.0-7.5**.)

22.4 Soil Acidity

Soil acidity is the condition where the soil solution has excess H+ (and Al3+) ions, giving pH below 7.

Types of soil acidity

1. **Active acidity** - the H+ present in the soil solution; measured directly by pH.
2. **Exchangeable acidity** - H+ and Al3+ held on the exchange complex; released into solution as needed.
3. **Residual acidity** - acidity not neutralised by liming to a target pH; requires further lime.

Sources of acidity

- **Leaching of bases** (Ca, Mg, K, Na) by rainfall, leaving H+ on the complex.
- **Organic acids** from organic matter decomposition.
- **Nitrogen fertilizers** (ammonium fertilizers produce H+ on nitrification).
- **Acid rain** (H2SO4, HNO3).
- **Oxidation of pyrite** (acid sulphate soils).
- **Uptake of cations by crops** (releases H+).

Effect: reduces nutrient availability (P, Mo), increases Al and Mn toxicity, reduces microbial activity and legume nodulation.

22.5 Soil Alkalinity

Soil alkalinity is the condition where the soil has excess hydroxyl (OH-) or carbonate/bicarbonate ions, giving pH above 7.

Sources of alkalinity

- **Calcium carbonate (lime)** in calcareous soils.
- **Sodium carbonate and bicarbonate** in sodic soils.
- **Weathering of basic rocks.**
- **Irrigation with alkaline water** in arid regions.

Types

- **Calcareous soils** - pH 7.5-8.5; due to CaCO3; moderate problem.
- **Sodic (alkali) soils** - pH 8.5-10.5; due to exchangeable Na and Na2CO3; severe problem (poor structure, dispersion).

Effect: reduces availability of P, Fe, Zn, Mn, B; causes nutrient deficiencies (e.g., Fe chlorosis in calcareous soils).

22.6 Buffering in Soil

Definition: Buffering is the **resistance of a soil to change in pH** when an acid or base is added.

- A soil with **high buffering capacity** resists pH change; one with low buffering changes easily.
- **Buffering is due to:**

- **Clay minerals** (high CEC).

- **Organic matter** (humus with many exchange sites).

- **Carbonate/bicarbonate systems** (in calcareous soils).

- **Exchange reactions** that release or absorb H+.

Significance

- High-buffered soils (heavy clay, high OM) need **more lime** to raise pH than light sandy soils.
- Buffering protects crops from sudden pH changes.
- It affects fertilizer efficiency and lime recommendations.

22.7 Factors Affecting Soil pH

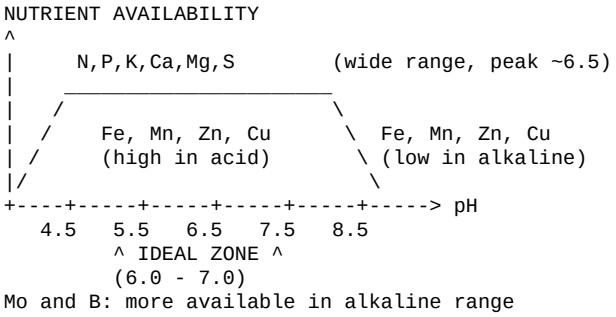
Factor	Effect on pH
Parent material	Basic rocks -> alkaline; acidic rocks -> acidic
Climate/rainfall	High rainfall -> leaching -> acidic
Organic matter	Decomposition -> organic acids -> acidic
Fertilizers	Ammonium fertilizers -> acidic; nitrate -> less acidic
Liming	Raises pH
Gypsum/acidifying amendments	Lower pH
Irrigation water	Alkaline water raises pH; acidic water lowers
Cropping	Cation uptake lowers pH; anion uptake raises
Flooding (paddy)	Tends toward neutral
Salts	Sodic salts raise pH

22.8 Effect of pH on Nutrient Availability

pH range	Nutrient availability
4.5-5.5	N, P, K, S, Ca, Mg low; Fe, Mn, Al high (toxicity); Mo very low
6.0-7.0	**Maximum availability of N, P, K, S, Ca, Mg and micronutrients**
6.5-7.5	**Ideal for most crops**
7.5-8.5	P, Fe, Zn, Mn, B, Cu low; Ca, Mg, K OK
8.5-10.5	P, Fe, Zn, Mn low; Na high (toxicity); poor structure

General principle: The **availability of most plant nutrients is maximum between pH 6.0 and 7.0.** Phosphorus is most available at pH 6.5; micronutrients (Fe, Mn, Zn, Cu) are more available in acid soils; Mo is more available in alkaline soils.

22.9 Diagram: Nutrient Availability vs pH



22.10 Management of Soil pH

- **Acidic soils:** apply **lime** (CaCO3 or dolomite) to raise pH; grow acid-tolerant crops.
- **Alkaline/calcareous soils:** apply **gypsum** (for sodic soils), elemental sulphur, organic matter; grow alkaline-tolerant crops.
- **Sodic soils:** gypsum + leaching + drainage.
- **Acid sulphate soils:** lime + drainage (careful - drainage can worsen acidification).

- **Fertilizer choice:** use appropriate fertilizers to avoid further acidification.

22.11 Worked Example 1: pH from H+ Concentration

Problem: A soil solution has a hydrogen ion concentration of 1 x 10^-5 mol/L. Calculate the pH.

Solution

- pH = -log10 [H+]
- pH = -log10 (1 x 10^-5) = **5.0**

Answer: The soil is **strongly acidic** (pH 5.0).

22.12 Worked Example 2: H+ Change with pH

Problem: How many times more acidic is a soil of pH 4 than a soil of pH 6?

Solution

- pH difference = 6 - 4 = 2 units
- Each unit = 10-fold change
- Ratio = 10^2 = **100 times**

Answer: The pH 4 soil is **100 times more acidic** than the pH 6 soil.

22.13 Worked Example 3: Classifying a Soil

Problem: A soil has pH 8.7 and ESP of 18%. Classify it.

Solution

- pH 8.7 -> strongly alkaline.
- ESP 18% (>15%) -> sodic.
- Therefore the soil is **sodic (alkali)**.

Answer: A strongly alkaline sodic soil; it needs gypsum and leaching for reclamation.

22.14 Tamil Nadu Context

- **Laterite and red soils** (Nilgiris, parts of Salem) are often **acidic** (pH 5.0-6.0); TNAU recommends **liming**.
- **Black soils** are neutral to slightly alkaline (pH 7.0-8.5), often calcareous.
- **Coastal and sodic soils** (parts of Ramanathapuram, delta) may be **alkaline to strongly alkaline**; need gypsum.
- TNAU recommends **lime application based on soil test** for acidic soils in the western and hill zones.

22.15 Exam Focus

Key points

- pH = -log10 [H+]; scale 0-14; 7 = neutral.
- pH < 7 acidic; pH > 7 alkaline; most soils 4.5-9.0.
- Active, exchangeable and residual acidity.
- Acidity from leaching, organic acids, ammonium fertilizers, acid rain, pyrite oxidation.
- Alkalinity from carbonates, sodium salts; calcareous and sodic soils.
- Buffering is the resistance to pH change; due to clay, humus and carbonates.
- Nutrient availability is maximum at pH 6.0-7.0.
- Acidic soils: lime; sodic soils: gypsum; each pH unit is a 10-fold change.

Short-answer questions

1. *Define soil pH and classify soils by pH.*

Model answer: Soil pH is the negative logarithm of the hydrogen ion concentration (pH = -log [H+]). Soils are classified as extremely acid (<4.5), strongly acid (5.1-5.5), moderately acid (5.6-6.0), neutral (6.6-7.3), moderately alkaline (7.9-8.4) and strongly alkaline (8.5-9.0).

2. *What is soil buffering?*

Model answer: Soil buffering is the resistance of a soil to change in pH when an acid or base is added. It is due to clay minerals, organic matter and carbonate systems with high exchange capacity.

3. *How does pH affect nutrient availability?*

Model answer: Most nutrients (N, P, K, S, Ca, Mg) are maximally available at pH 6.0-7.0. In acid soils Fe, Mn and Al become excessive (toxicity) and Mo is deficient; in alkaline soils P, Fe, Zn, Mn and B become deficient.

Long-answer questions

1. *Explain soil reaction (pH), the pH scale, soil acidity and alkalinity, and buffering.*

Outline: Definition of pH - scale and classification - soil acidity (types, sources, effects) - soil alkalinity (sources, types) - buffering and its causes - factors affecting pH - conclusion.

2. *Discuss the effect of soil pH on nutrient availability and crop growth, with management.*

Outline: Introduction - nutrient availability vs pH diagram - acid soil problems - alkaline soil problems - management (lime, gypsum, organic matter, fertilizer choice) - Tamil Nadu examples - conclusion.

Lecture 23: Soil Electrical Conductivity - Definition, Factors Affecting EC, Salinity Limits

23.1 Introduction

Soil salinity refers to the amount of soluble salts in the soil. It is measured by the **electrical conductivity (EC)** of the soil solution or extract, because salt solutions conduct electricity. EC is the standard measure of soil salinity and is used to classify soils as normal, saline or saline-sodic.

23.2 Definition of Electrical Conductivity

Electrical conductivity (EC) is the **ability of a solution (or soil extract) to conduct an electric current**, which is proportional to the concentration of dissolved ions (salts).

- Unit: **deciSiemens per metre (dS/m)** (also written as mmhos/cm; 1 dS/m = 1 mmho/cm).
- Sometimes expressed as **microSiemens/cm (microS/cm)**; 1 dS/m = 1000 microS/cm.
- EC is measured with a **conductivity meter** on a **soil-water extract**.

23.3 Methods of Measuring EC

- **Saturation extract (ECe):** the EC of the extract from a saturated soil paste - the **standard** measure of soil salinity.
- **Soil-water suspension (1:1, 1:2, 1:5):** EC of a soil-water suspension - simpler but needs conversion to ECe.
- **Field salinity sensors** and **EM (electromagnetic) induction meters** for rapid field mapping.

The **saturated paste extract (ECe)** is the reference because it best represents the soil solution as plants experience it.

23.4 Relationship between EC and Salt Content

- Higher salt concentration -> higher EC.
- ECe of 1 dS/m corresponds to roughly **640 mg/L (ppm)** of total dissolved salts (a commonly used conversion).
- Rule of thumb: **EC (dS/m) x 640 = approximate total dissolved salts (mg/L or ppm).**

23.5 Classification of Soils by Salinity (Salinity Limits)

The standard classification (based on ECe of the saturation extract):

Class	ECe (dS/m)	Effect on crops
Non-saline (normal)	0 - 2	No harmful effect
Slightly saline	2 - 4	Sensitive crops affected
Moderately saline	4 - 8	Many crops affected

Class	ECe (dS/m)	Effect on crops
Strongly saline	8 - 16	Only tolerant crops grow
Very strongly saline	> 16	Only very tolerant crops grow

Standard salinity threshold: Soils with **ECe > 4 dS/m** are called **saline** (at 25 degrees C). This is the widely used limit.

23.6 Saline, Sodic and Saline-Sodic Soils (EC, ESP, pH)

Property	Normal	Saline	Sodic (alkali)	Saline-sodic
ECe (dS/m)	< 4	> 4	< 4 (usually)	> 4
ESP (%)	< 15	< 15	> 15	> 15
pH	6.5-7.5	< 8.5	8.5-10.5	< 8.5 (usually)
Appearance	normal	white salt crust	black, dispersed	salt crust + poor structure
Reclamation	-	leaching + drainage	gypsum + leaching	gypsum + leaching + drainage

23.7 Factors Affecting Soil EC

- **Salt content / fertilizer application** - more salts, higher EC.
- **Irrigation water quality** - saline water raises EC.
- **Evaporation (climate)** - arid climate concentrates salts (higher EC).
- **Rainfall and leaching** - high rainfall leaches salts (lower EC).
- **Drainage** - poor drainage retains salts (higher EC).
- **Water table depth** - shallow, salty water table raises EC by capillary rise.
- **Soil texture** - clay soils retain more salts than sandy soils.
- **Soil moisture** - EC of a suspension depends on the soil:water ratio (dilution lowers EC).
- **Organic matter and fertilizers** - additions change EC.
- **Temperature** - EC increases with temperature (about 2% per degree C).

23.8 Effect of Salinity on Crops

- **Osmotic effect** - high salts lower osmotic potential, reducing water uptake ("physiological drought").
- **Specific ion toxicity** - excess Na, Cl or B harms plants.
- **Nutrient imbalance** - salts interfere with nutrient uptake.
- **Reduced germination and growth** - seedlings are most sensitive.
- **Yield reduction** - yield falls as ECe rises above the crop's threshold.

Crop salt tolerance (threshold ECe and slope):

Crop	Threshold ECe (dS/m)	Tolerance
Barley	8.0	Tolerant
Cotton	7.7	Tolerant
Wheat	6.0	Moderately tolerant
Rice (paddy)	3.0	Moderately sensitive
Maize	1.7	Moderately sensitive
Groundnut	3.2	Moderately sensitive
Beans, clover	1.5-2.0	Sensitive

(Values are standard published thresholds; yield declines above the threshold.)

23.9 Worked Example 1: Estimating Total Dissolved Salts

Problem: A soil has an ECe of 6 dS/m. Estimate the total dissolved salts.

Solution

- Total dissolved salts (mg/L) = ECe x 640
- = 6 x 640 = **3,840 mg/L (ppm)**

Answer: The soil solution contains about 3,840 mg/L of dissolved salts - a moderately saline soil.

23.10 Worked Example 2: Classifying a Soil

Problem: A soil has ECe = 5 dS/m, ESP = 8% and pH 8.2. Classify the soil.

Solution

- ECe 5 > 4 -> saline.
- ESP 8 < 15 -> not sodic.
- pH 8.2 < 8.5.
- Therefore the soil is **saline** (not sodic).

Answer: A saline soil; reclamation by leaching with good water and drainage.

23.11 Worked Example 3: Conversion of Units

Problem: A conductivity meter reads 2,400 microS/cm. Convert to dS/m.

Solution

- 1 dS/m = 1000 microS/cm
- 2400 microS/cm = 2400 / 1000 = **2.4 dS/m**

Answer: The EC is 2.4 dS/m - a slightly saline soil.

23.12 Worked Example 4: Yield Loss Estimate (Concept)

Problem: For a crop with a threshold ECe of 3 dS/m and a yield decline of 12% per dS/m above the threshold, estimate the yield loss at ECe = 6 dS/m.

Solution

- Excess ECe above threshold = 6 - 3 = 3 dS/m
- Yield decline = 3 x 12 = **36%**

Answer: About 36% yield loss, so the relative yield is about 64%.

23.13 Management of Saline Soils

- **Leaching** with good-quality water to wash salts below the root zone.
- **Drainage** to remove the leached salts.
- **Grow salt-tolerant crops** (barley, cotton, sugar beet, salt-tolerant grasses).
- **Use raised beds / ridge planting** to avoid salt accumulation near seed.
- **Add organic matter** to improve structure and water movement.
- **Use gypsum** for saline-sodic soils.
- **Avoid over-irrigation and poor-quality water.**

23.14 Tamil Nadu Context

- **Coastal soils** of Nagapattinam, Ramanathapuram and Kanyakumari face **salinity** from seawater intrusion and brackish groundwater.
- **Cauvery delta** has some **saline patches** in low-lying areas; needs drainage.
- **Saline water irrigation** in coastal belts requires careful management and salt-tolerant crops.
- TNAU recommends **drainage, leaching and salt-tolerant crops** for coastal saline soils.

23.15 Exam Focus

Key points

- EC measures the ability of a soil extract to conduct electricity; proportional to salt content.

- Unit: dS/m (1 dS/m = 1 mmho/cm = 1000 microS/cm).
- ECe (saturation extract) is the standard measure of salinity.
- Soils with ECe > 4 dS/m are saline; ESP > 15% (with high pH) are sodic.
- Factors: salt content, irrigation water, climate, drainage, water table, texture, temperature.
- Salinity causes osmotic drought and ion toxicity; yield falls above crop threshold.
- Estimate total dissolved salts: EC (dS/m) x 640 = mg/L.
- Management: leaching, drainage, salt-tolerant crops, organic matter, gypsum.

Short-answer questions

1. *Define electrical conductivity of soil and give its unit.*

Model answer: Electrical conductivity is the ability of a soil extract to conduct an electric current, proportional to the concentration of dissolved salts. It is measured in deciSiemens per metre (dS/m), where 1 dS/m = 1 mmho/cm.

2. *What is the salinity limit for a saline soil?*

Model answer: A soil is classified as saline when the electrical conductivity of its saturation extract (ECe) exceeds 4 dS/m at 25 degrees C, with an ESP below 15% and pH usually below 8.5.

3. *How does salinity affect crops?*

Model answer: High salts lower the osmotic potential, reducing water uptake (physiological drought), and excess Na, Cl or B cause specific ion toxicity. Germination and growth are reduced and yield falls above the crop's salinity threshold.

Long-answer questions

1. *Explain soil electrical conductivity, the factors affecting it and the salinity limits.*

Outline: Definition - units and measurement (ECe, suspensions) - relation to salt content - salinity classification table - saline, sodic and saline-sodic comparison - factors affecting EC - worked examples - conclusion.

2. *Discuss the effect of salinity on crop growth and its management.*

Outline: Introduction - EC and salinity - osmotic and toxicity effects - crop salt tolerance table - yield loss - management (leaching, drainage, tolerant crops, organic matter, gypsum) - Tamil Nadu coastal examples - conclusion.

Lecture 24: Soil Colloids - Definition, Properties, Classification: Inorganic and Organic Colloids

24.1 Introduction

Soil colloids are the most chemically active part of the soil. They are very small particles (less than 1 micrometre, usually less than 0.2 micrometre) with a very large surface area. They are responsible for ion exchange, water retention, swelling, and many other important soil properties.

24.2 Definition of a Soil Colloid

A **soil colloid** is a **substance of very fine particle size (colloidal size, 1 nanometre to 1 micrometre) dispersed in the soil solution**, with a very large specific surface area and a surface electrical charge.

- Colloids do not settle out (they remain suspended).
- They exhibit the **Tyndall effect** (scatter light).
- They have **Brownian movement** (random motion).

24.3 Properties of Soil Colloids

- **Very small size** - less than 1 micrometre (often < 0.2 micrometre).
- **Very large specific surface area** - thousands of times more than sand.
- **Surface electrical charge** - mostly **negative** (some positive); hence they attract cations.
- **Adsorption and ion exchange** - hold and exchange ions (CEC and AEC).
- **Water adsorption** - high water-holding capacity.

- **Swelling and shrinkage** - some colloids (montmorillonite) swell when wet.
- **Coagulation (flocculation) and dispersion** - clump together or disperse depending on cations.
- **Plasticity and cohesion** - give soil these properties.
- **Brownian movement and Tyndall effect** - colloidal behaviour.
- **Hygroscopicity** - absorb moisture from the air.

24.4 Classification of Soil Colloids

Soil colloids are classified into two main groups:

1. **Inorganic colloids** (mineral colloids)

- **Silicate clays** (layer silicates) - kaolinite, montmorillonite, illite, chlorite.

- **Non-silicate (oxide) colloids** - iron oxides, aluminium oxides (sesquioxides).

- **Amorphous colloids** - allophane.

2. **Organic colloids**

- **Humus** (humic and fulvic acids, humin).

A third category sometimes added: **organo-mineral complexes** (clay + humus bound together).

```
SOIL COLLOIDS
=====================================================
INORGANIC (mineral)               ORGANIC
|                                 |
+-- Silicate clays                 +-- Humus
|     (kaolinite, montmorillonite,  |     (humic acid,
|      illite, chlorite)            |      fulvic acid, humin)
|                                 |
+-- Oxide colloids
|     (Fe2O3, Al2O3, gibbsite)
|                                 ORGANO-MINERAL
+-- Amorphous                     COMPLEXES
      (allophane)                 (clay + humus)
=====================================================
```

24.5 Inorganic Colloids

A. Silicate clays (crystalline layer silicates)

- The most important and abundant inorganic colloids.
- Made of tetrahedral (T) and octahedral (O) sheets.
- Types: **1:1 (kaolinite)**, **2:1 (montmorillonite, illite, vermiculite)**, **2:2 (chlorite)**.
- Carry **permanent negative charge** from isomorphous substitution.
- Have high **CEC**.
- (Structure and classification detailed in Lecture 25.)

B. Non-silicate (oxide) colloids

- **Iron oxides** - haematite, goethite, ferrihydrite.
- **Aluminium oxides/hydroxides** - gibbsite, boehmite.
- Common in highly weathered soils (red, laterite).
- Carry **variable (pH-dependent) charge** - positive in acid soils (anion exchange), negative in alkaline soils.
- Low CEC; high **phosphate fixation**.

C. Amorphous colloids

- **Allophane and imogolite** - from volcanic ash.
- High surface area, high pH-dependent charge, high water retention.

24.6 Organic Colloids (Humus)

- **Humus** is the organic colloidal fraction of soil.
- Amorphous, dark, complex, resistant to further decomposition.

- Components: **humic acid** (soluble in alkali, insoluble in acid), **fulvic acid** (soluble in both), **humin** (insoluble).
- Carries a very **high negative charge per unit mass** (high CEC, often 150-300 cmol/kg).
- **Very high water-holding capacity** (holds several times its weight in water).
- Improves soil structure, buffering, and nutrient retention.
- (Detailed in Lecture 28.)

24.7 Comparison of Inorganic and Organic Colloids

Feature	Inorganic (mineral) colloids	Organic colloids (humus)
Nature	Crystalline (or amorphous oxides)	Amorphous
Main types	Silicate clays, oxides	Humic/fulvic acid, humin
Size	< 1 micrometre	< 1 micrometre
Charge	Mostly negative (some positive)	Highly negative
CEC	3-150 cmol/kg (varies)	150-300 cmol/kg (high)
Surface area	High (up to 800 m2/g)	Very high
Water holding	High	Very high
Swelling	Varies (montmorillonite high)	Little
Colour	Grey to red	Dark brown to black
Effect on structure	Varies	Very favourable
Stability	Stable	Decomposes slowly

24.8 Properties Compared: Clay vs Humus

Property	Silicate clay	Humus
Charge source	Isomorphous substitution + broken edges	Carboxyl and phenolic groups (pH-dependent)
Charge type	Mostly permanent	Mostly variable
CEC (cmol/kg)	3-150	150-300
Swelling	High (2:1), low (1:1)	Low
Colour	Light (kaolinite), grey	Dark
Aggregating effect	Moderate	Strong
Buffering	Moderate	High

24.9 Importance of Soil Colloids in Agriculture

- **Nutrient holding (CEC):** colloids hold plant nutrients (Ca, Mg, K, NH4) against leaching.
- **Ion exchange:** supply nutrients to plant roots (see Lecture 27).
- **Water holding:** retain water for plant use.
- **Structure:** bind particles into stable aggregates.
- **Buffering:** resist pH change.
- **Fertilizer management:** soils with high CEC need less frequent fertilization; sandy (low colloid) soils need frequent, light doses.
- **Soil classification:** colloid type (clay mineralogy) is used in classification.

24.10 Worked Example: Surface Area Comparison

Problem: A cubic soil particle of 1 mm side is divided into cubes of 1 micrometre side. Calculate how many times the surface area increases.

Solution

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- Original particle: 1 mm = 1000 micrometres.
- Number of small cubes along one edge = 1000/1 = 1000.
- Total small cubes = 1000 x 1000 x 1000 = 10^9.
- Surface area of original cube = 6 x (1000)^2 = 6 x 10^6 square micrometres.
- Surface area of all small cubes = 10^9 x 6 x (1)^2 = 6 x 10^9 square micrometres.
- Increase factor = (6 x 10^9) / (6 x 10^6) = **1000 times**.

Answer: The surface area increases 1000-fold. This explains why colloids have enormous surface area and high reactivity.

24.11 Tamil Nadu Context

- **Red and laterite soils** contain **kaolinite and iron oxide colloids** (low CEC, high P fixation).
- **Black soils (vertisols)** contain **montmorillonite colloids** (high CEC, high swelling).
- **Alluvial soils** contain a mixture of illite, kaolinite and smectite with humus.
- **Organic matter-rich soils** (hill soils of the Nilgiris) have high humus colloids.

24.12 Exam Focus

Key points

- Soil colloids are particles < 1 micrometre with very large surface area and surface charge.
- They carry mostly negative charge, so they attract and hold cations.
- Properties: adsorption, ion exchange, water holding, swelling, flocculation, buffering.
- Classification: inorganic (silicate clays, oxides, amorphous) and organic (humus).
- Silicate clays are crystalline layer silicates with permanent charge and CEC.
- Oxide colloids have variable (pH-dependent) charge and high P fixation.
- Humus is amorphous, highly charged (CEC 150-300 cmol/kg), high water holding.
- Colloids are the storehouse of nutrients and the key to soil fertility.

Short-answer questions

1. *Define soil colloids and list any four properties.*

Model answer: Soil colloids are very fine particles (less than 1 micrometre) dispersed in the soil solution with a large specific surface area and surface charge. Properties: large surface area, surface charge (mostly negative), ion exchange (CEC), high water-holding capacity, swelling, and flocculation/dispersion.

2. *Classify soil colloids.*

Model answer: Soil colloids are classified into inorganic colloids (silicate clays, oxide colloids and amorphous colloids) and organic colloids (humus). Organo-mineral complexes form a third related group.

3. *Why is humus an important colloid?*

Model answer: Humus has a very high negative charge (CEC 150-300 cmol/kg), holds large amounts of water and nutrients, buffers pH and binds particles into stable aggregates, greatly improving soil fertility and structure.

Long-answer questions

1. *Describe soil colloids, their properties and classification.*

Outline: Definition - properties (size, surface area, charge, exchange, water, swelling, flocculation) - classification (inorganic: silicate clays, oxides, amorphous; organic: humus) - diagram - comparison of inorganic and organic colloids - importance - conclusion.

2. *Compare inorganic and organic soil colloids and discuss their importance in agriculture.*

Outline: Introduction - inorganic colloids (types) - organic colloids (humus, components) - comparison table - properties (charge, CEC, water, structure) - importance in nutrient holding, exchange, buffering, structure - Tamil Nadu examples - conclusion.

Lecture 25: Silicate Clays: Structure and Classification - 1:1, 2:1, 2:2 Clay Minerals, Amorphous Clay Minerals and Their Properties

25.1 Introduction

Silicate clays are crystalline, layered (sheet) silicates that form the most important group of soil colloids. Their structure and type determine the CEC, swelling, water retention and fertility of a soil. This lecture describes their structure and classification.

25.2 Structure of Silicate Clays

Silicate clays are built from two types of **sheets**:

A. Silica tetrahedral sheet (T sheet)

- Made of **SiO4 tetrahedra**.
- Each tetrahedron has a silicon atom at the centre and four oxygen atoms at the corners.
- Three basal oxygens are shared with neighbouring tetrahedra, forming a **hexagonal network**.
- The fourth (apical) oxygen points toward the octahedral sheet.

B. Alumina octahedral sheet (O sheet)

- Made of **octahedra** with **aluminium (or magnesium)** at the centre and oxygen/hydroxyl at the corners.
- Aluminium is surrounded by six oxygens/hydroxyls.
- Forms a sheet by sharing edges.

Combination: T and O sheets combine in different ratios to give **1:1**, **2:1** and **2:2** clay minerals.

```
SILICA TETRAHEDRAL SHEET (T)
--------------------------------------
Si4+ at centre, 4 O around; tetrahedra
share 3 oxygens -> hexagonal sheet

ALUMINA OCTAHEDRAL SHEET (O)
---------------------------------------
Al3+ (or Mg2+) at centre, 6 OH/O around
-> octahedral sheet

COMBINATIONS:
1:1  = T-O               (kaolinite)
2:1  = T-O-T             (montmorillonite, illite)
2:2  = T-O-T + brucite   (chlorite)
```

25.3 Classification of Silicate Clays

Silicate clays are classified by the **ratio of sheets** and the **interlayer**:

Type	Sheet arrangement	Example	Expanding?	CEC (cmol/kg)
1:1	T-O	Kaolinite, halloysite	No	3-15
2:1 expanding	T-O-T with water/cations	Montmorillonite (smectite), vermiculite	Yes	80-150 (mont.), 100-150 (verm.)
2:1 non-expanding	T-O-T with K in interlayer	Illite (mica-like)	No	20-40
2:2 (2:1:1)	T-O-T + brucite interlayer	Chlorite	No (limited)	10-40

25.4 The 1:1 Clay Minerals (Kaolinite Group)

Structure: One tetrahedral sheet + one octahedral sheet (T-O). The layers are held together by **hydrogen bonds** between the OH of the octahedral sheet and the O of the next tetrahedral sheet.

Properties

- **Non-expanding** - layers held tightly by H-bonds.
- **Low CEC** (3-15 cmol/kg) - little isomorphous substitution.
- **Low surface area** (external only, about 10-30 m2/g).

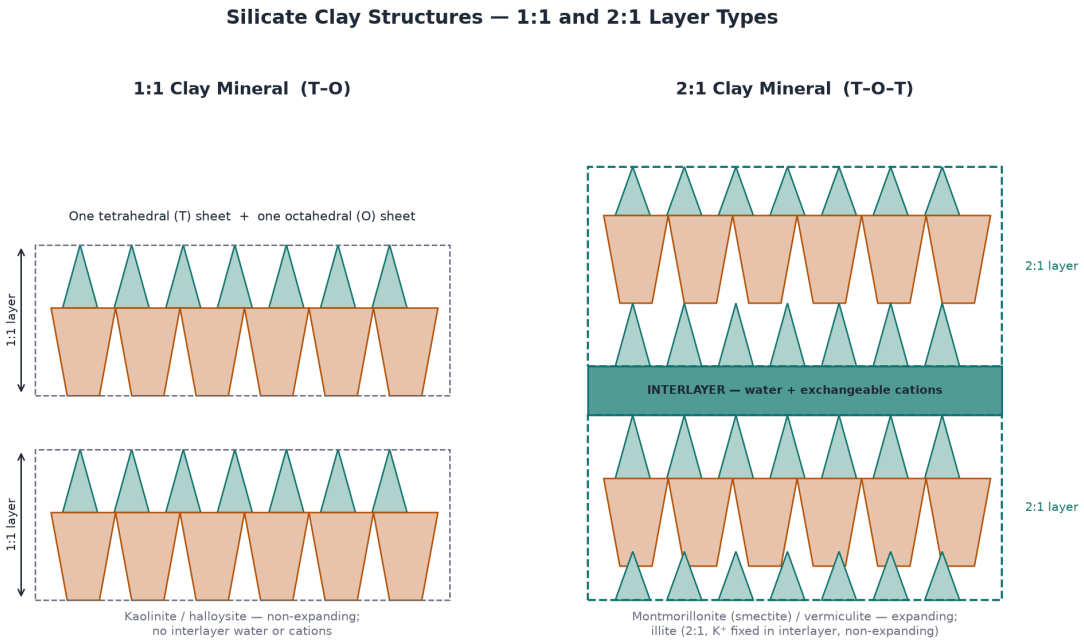


Figure 25.1 Silicate clay structures — 1:1 and 2:1

Tetrahedral–octahedral sheet arrangements of 1:1 (T–O) and 2:1 (T–O–T) layer silicates.

- **Low swelling** and shrinkage.
- **Stable structure**; good for building and pottery.
- **Low water-holding** compared with 2:1 clays.

Formation: Strong weathering and leaching (humid tropics) - kaolinite is the end product of feldspar weathering.

Occurrence: Red soils, laterite soils (Tamil Nadu).

25.5 The 2:1 Clay Minerals

Structure: One octahedral sheet sandwiched between two tetrahedral sheets (T-O-T).

Types:

A. Expanding 2:1 - Montmorillonite (Smectite group)

- Water and exchangeable cations in the **interlayer**.
- **High swelling and shrinkage** (shrink-swell clay; cracks when dry).
- **Very high CEC** (80-150 cmol/kg) - high isomorphous substitution.
- **Very high surface area** (internal + external, up to 800 m2/g).
- **High water holding**; sticky when wet, hard when dry.
- **Formation:** poor drainage, high bases, moderate weathering.
- **Occurrence:** black soils (vertisols) of Tamil Nadu.

B. Expanding 2:1 - Vermiculite

- High CEC (100-150 cmol/kg), moderate swelling.
- Interlayer holds water and cations.

C. Non-expanding 2:1 - Illite (Mica group)

- **Potassium (K+)** in the interlayer holds layers together - non-expanding.
- **Moderate CEC** (20-40 cmol/kg).
- Less swelling than montmorillonite.
- **Formation:** weathering of mica; moderate leaching.
- **Occurrence:** many soils; common in alluvium.

25.6 The 2:2 (2:1:1) Clay Minerals (Chlorite Group)

Structure: A 2:1 layer (T-O-T) with a **brucite [Mg(OH)2] interlayer sheet** - hence written as 2:2 or 2:1:1.

Properties

- **Non-expanding** (interlayer is a stable hydroxide sheet).
- **Low to moderate CEC** (10-40 cmol/kg).
- Moderate surface area.
- **Formation:** from Mg-rich parent material or alteration of other clays.
- Stable, little swelling.

25.7 Amorphous Clay Minerals

Definition: Clay-sized secondary minerals with **no ordered crystal structure**.

Examples

- **Allophane** and **imogolite** - hydrous aluminium silicates from volcanic ash.
- **Iron oxides/hydroxides** - ferrihydrite, goethite, haematite.
- **Aluminium hydroxides** - gibbsite.
- **Opaline silica.**

Properties

- **Very high surface area** and high water retention.
- **High pH-dependent (variable) charge** - positive in acid, negative in alkaline.

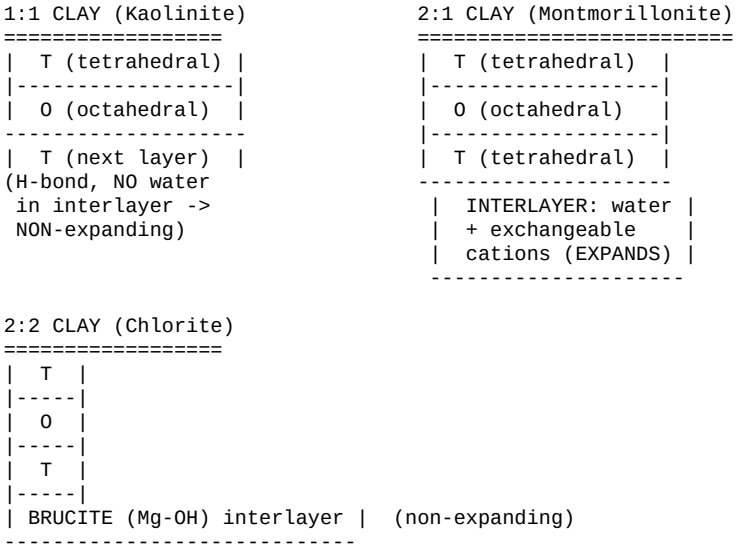
- **High phosphate fixation.**
- Weak, friable aggregates.

Occurrence: volcanic soils (Andisols), young soils, and highly weathered tropical soils (oxide-rich).

25.8 Comparison: 1:1 vs 2:1 vs 2:2 Clays

Property	1:1 (Kaolinite)	2:1 (Montmorillonite)	2:1 (Illite)	2:2 (Chlorite)
Sheet ratio	T-O	T-O-T	T-O-T	T-O-T + brucite
Expanding	No	Yes	No	No
Interlayer	None	Water + cations	K+	Brucite (Mg-OH)
CEC (cmol/kg)	3-15	80-150	20-40	10-40
Surface area (m2/g)	10-30	Up to 800	65-100	70-100
Swelling	Low	Very high	Low	Low
Charge source	Broken edges	Isomorphous substitution	Isomorphous + edges	Isomorphous
Water holding	Low	Very high	Moderate	Moderate
Cohesion/plasticity	Low	High	Moderate	Moderate
Soil example	Red soil	Black soil	Alluvium	Mg-rich soils

25.9 Diagram: Clay Structures



25.10 Effect of Clay Type on Soil Behaviour

Soil property	Kaolinite soil	Montmorillonite soil
CEC	Low	High
Water holding	Lower	Higher
Swelling/shrinkage	Low	Very high
Cracking	Little	Deep cracks when dry
Tillage	Easier	Difficult (sticky/hard)
Fertility	Lower	Higher
Erosion	Moderate	Low (but runoff)
Example	Red soil	Black soil

25.11 Worked Example: CEC of a Clay Mixture

Problem: A soil contains 30% montmorillonite (CEC 100 cmol/kg) and 20% kaolinite (CEC 10 cmol/kg) in its clay fraction. Estimate the clay CEC.

Solution

- Contribution of montmorillonite = 0.30 x 100 = 30 cmol/kg
- Contribution of kaolinite = 0.20 x 10 = 2 cmol/kg
- Total (of the whole soil, if these are percentages of whole soil) = 30 + 2 = **32 cmol/kg**
- (If these are percentages of the clay fraction, divide by the clay fraction.)

Answer: The estimated contribution is about 32 cmol/kg; the montmorillonite dominates the CEC because of its much higher charge.

25.12 Tamil Nadu Context

- **Red soils** - kaolinite dominant (1:1); low CEC, good drainage, moderate fertility.
- **Black soils (vertisols)** - montmorillonite dominant (2:1 expanding); high CEC, deep cracking, high fertility but difficult tillage.
- **Alluvial soils** - illite, kaolinite and smectite mixture.
- **Laterite soils** - kaolinite with iron and aluminium oxides (oxide-rich).
- TNAU crop recommendations consider clay type for irrigation and tillage advice.

25.13 Exam Focus

Key points

- Silicate clays are layered (sheet) silicates built from tetrahedral (T) and octahedral (O) sheets.
- 1:1 = T-O (kaolinite): non-expanding, low CEC (3-15).
- 2:1 = T-O-T: montmorillonite (expanding, high CEC 80-150) and illite (K in interlayer, non-expanding, CEC 20-40).
- 2:2 = T-O-T + brucite interlayer (chlorite): non-expanding, low-moderate CEC.
- Montmorillonite has very high surface area (up to 800 m2/g) and high swelling.
- Kaolinite has low surface area (external only) and low swelling.
- Amorphous clays (allophane) have high pH-dependent charge and high P fixation.
- Clay type controls CEC, water holding, swelling and fertility.

Short-answer questions

1. *Name the two building sheets of silicate clays.*

Model answer: The silica tetrahedral sheet (T) and the alumina octahedral sheet (O). Tetrahedra have silicon at the centre with four oxygens; octahedra have aluminium (or magnesium) at the centre with six oxygens/hydroxyls.

2. *Distinguish 1:1 and 2:1 clay minerals.*

Model answer: A 1:1 clay (kaolinite) has one tetrahedral and one octahedral sheet, is non-expanding, has low CEC (3-15 cmol/kg) and low surface area. A 2:1 clay (montmorillonite) has one octahedral sheet between two tetrahedral sheets, is expanding, has high CEC (80-150 cmol/kg) and very high surface area.

3. *What is a 2:2 clay? Give an example.*

Model answer: A 2:2 (or 2:1:1) clay has a 2:1 layer with a brucite (Mg(OH)2) interlayer sheet. Chlorite is an example; it is non-expanding with low to moderate CEC.

Long-answer questions

1. *Describe the structure and classification of silicate clays with diagrams.*

Outline: Introduction - T and O sheets - combination into 1:1, 2:1, 2:2 - kaolinite (structure, properties) - montmorillonite - illite - chlorite - amorphous clays - comparison table - diagram - conclusion.

2. *Compare 1:1, 2:1 and 2:2 clay minerals and explain how clay type affects soil properties.*

Outline: Introduction - structure of each - CEC, surface area, swelling, water holding - comparison table - effect on soil behaviour (fertility, tillage, cracking) - Tamil Nadu examples - conclusion.

Lecture 26: Sources of Charge - Positive and Negative Charge - pH Dependent Charge, Isomorphous Substitution

26.1 Introduction

Soil colloids (clay and humus) carry **electrical charges** on their surfaces. These charges are responsible for the **adsorption and exchange of ions (cations and anions)** - the basis of soil fertility and nutrient supply. Charges may be **negative** or **positive**, and may be **permanent** or **pH-dependent (variable)**.

26.2 Types of Charge on Soil Colloids

- **Negative charge** - attracts and holds **cations** (Ca2+, Mg2+, K+, Na+, NH4+, H+, Al3+). Most soil colloids carry net negative charge.
- **Positive charge** - attracts and holds **anions** (Cl-, NO3-, SO4^2-, PO4^3-). Present mainly on oxide and edge surfaces.

26.3 Sources of Negative Charge

A. Isomorphous substitution (permanent charge)

- The substitution of one ion by another of **similar size but lower charge** within the crystal lattice, without changing the crystal structure.
- Example: **Al3+ replaced by Mg2+** in the octahedral sheet, or **Si4+ replaced by Al3+** in the tetrahedral sheet.
- Each such substitution leaves a **net negative charge** on the lattice.
- This charge is **permanent** - it does not change with pH.
- Main source of charge in **2:1 clays** (montmorillonite, illite).

B. Broken edges (edge charge)

- At the broken edges of clay crystals, **exposed OH groups** and unsatisfied bonds give charge.
- This charge is **pH-dependent** - negative at high pH, positive at low pH.

C. Dissociation of acidic groups on organic matter

- **Carboxyl (-COOH)** and **phenolic (-OH)** groups on humus dissociate to give H+ and leave a **negative charge**.
- This charge is **pH-dependent** - more negative at higher pH.

D. Deprotonation of hydroxyl groups on oxide surfaces

- Fe/Al oxide surfaces develop negative charge at high pH.

26.4 Sources of Positive Charge

A. Protonation of hydroxyl groups (oxide and edge surfaces)

- At **low pH**, exposed OH groups on oxides and clay edges gain a proton (H+) and become **positively charged (-OH2+)**.
- This charge is **pH-dependent** - positive at low pH, decreases as pH rises.

B. Isomorphous substitution rarely gives positive charge; positive charge is mainly from protonation.

C. Broken edge bonds may be positive at low pH.

26.5 pH-Dependent (Variable) Charge

Definition: pH-dependent charge is the charge that **varies with the pH** of the soil solution; it arises from the **adsorption or desorption of H+ (protons)** on surface hydroxyl (-OH) and carboxyl (-COOH) groups.

- **Low pH (acidic):** more H+ in solution -> surfaces take up H+ -> **positive charge** (or less negative). Anion exchange increases.
- **High pH (alkaline):** H+ is removed from surfaces -> **negative charge** increases. Cation exchange increases.

Point of Zero Charge (PZC): the pH at which the net surface charge is **zero**. Below PZC the surface is net positive; above PZC it is net negative.

- **Kaolinite:** PZC about pH 4-5.
- **Iron/aluminium oxides:** PZC about pH 7-9 (hence positive charge in acid soils).

Example: In acid soils, iron and aluminium oxides carry positive charge and **fix phosphate** (anion adsorption) - a major problem in red and laterite soils.

26.6 Isomorphous Substitution

Definition: Isomorphous substitution is the **replacement of one ion by another of similar size but different (usually lower) charge** in the crystal lattice of a clay mineral, without changing the crystal structure.

Common examples

- **Si4+ replaced by Al3+** in the tetrahedral sheet -> one negative charge per substitution.
- **Al3+ replaced by Mg2+** in the octahedral sheet -> one negative charge per substitution.
- **Al3+ replaced by Fe2+** (or Fe3+) in the octahedral sheet.

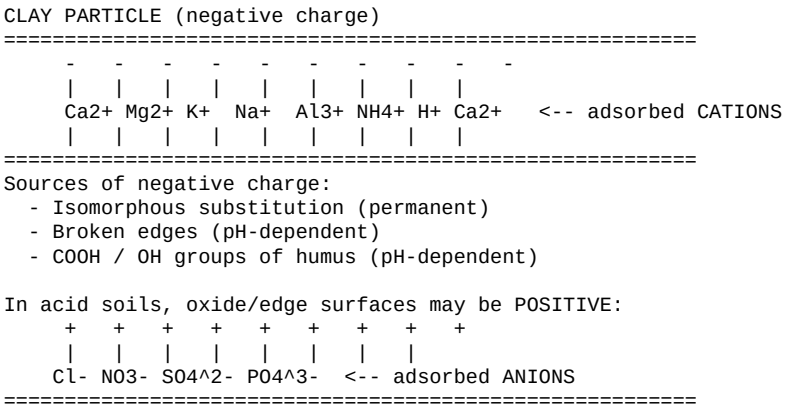
Characteristics

- Gives **permanent (pH-independent) negative charge**.
- The charge does **not** change with pH.
- **Montmorillonite** has high isomorphous substitution (high CEC); **kaolinite** has little (low CEC).
- It is the **main source of CEC** in 2:1 clays.

26.7 Comparison: Permanent vs pH-Dependent Charge

Feature	Permanent charge	pH-dependent (variable) charge
Source	Isomorphous substitution	Protonation/deprotonation of OH, COOH
Effect of pH	Does not change	Changes with pH
Sign	Negative (usually)	Negative (high pH) or positive (low pH)
Main in	2:1 clays (montmorillonite)	Oxides, edges, organic matter
CEC contribution	Constant	Increases with pH
Example	Montmorillonite, illite	Kaolinite edges, Fe/Al oxides, humus

26.8 Diagram: Charges on a Clay Surface



26.9 Significance of Surface Charge

- **Cation exchange capacity (CEC):** negative charge holds cations - the basis of nutrient retention.
- **Anion exchange capacity (AEC):** positive charge holds anions (including phosphate).
- **Nutrient availability:** controls the supply of Ca, Mg, K and NH4 to plants.
- **Flocculation/dispersion:** high charge + Ca2+ -> flocculation (good structure); Na+ -> dispersion (poor structure).

- **Phosphate fixation:** positive charge in acid soils fixes phosphate, reducing availability.
- **Lime and fertilizer management:** pH-dependent charge means liming increases CEC (more negative charge) and improves nutrient retention.
- **Buffering:** charge development contributes to pH buffering.

26.10 Worked Example: Charge from Isomorphous Substitution (Concept)

Problem: In a clay lattice, 5% of the Si4+ in the tetrahedral sheet is replaced by Al3+. If each substitution gives one negative charge, and there are 1000 silicon positions per unit, calculate the number of negative charges.

Solution

- Si positions replaced = 5% of 1000 = 50
- Each replacement gives one negative charge
- Total negative charges = **50**

Answer: 50 negative charges per 1000 Si positions, giving the clay a net negative charge (permanent, from isomorphous substitution).

26.11 Worked Example: PZC Interpretation

Problem: An iron oxide has a point of zero charge (PZC) of pH 8.0. What charge does it carry in a soil of pH 5.5 and of pH 9.0?

Solution

- Below PZC (pH 5.5 < 8.0) -> surface is **positively charged** (adsorbs anions, fixes phosphate).
- Above PZC (pH 9.0 > 8.0) -> surface is **negatively charged** (adsorbs cations).

Answer: At pH 5.5 the oxide is positive (anion adsorption, phosphate fixation); at pH 9.0 it is negative (cation adsorption).

26.12 Tamil Nadu Context

- **Red and laterite soils** contain iron and aluminium oxides with high PZC -> carry **positive charge in acid conditions** -> strong **phosphate fixation**. TNAU recommends band application of P fertilizer and organic matter to reduce fixation.
- **Black soils** contain montmorillonite with **permanent negative charge** -> high CEC and good nutrient retention.
- **Liming acidic soils** increases negative charge and CEC, improving nutrient retention.

26.13 Exam Focus

Key points

- Soil colloids carry negative (mostly) and positive charges.
- Negative charge holds cations; positive charge holds anions.
- Sources of negative charge: isomorphous substitution (permanent), broken edges, humus COOH/OH groups.
- Sources of positive charge: protonation of OH groups at low pH.
- Isomorphous substitution = replacement of an ion by a lower-charge ion of similar size; gives permanent negative charge.
- pH-dependent charge varies with pH; PZC is the pH of zero net charge.
- Oxides (high PZC) carry positive charge in acid soils and fix phosphate.
- CEC and AEC depend on these charges; liming increases negative charge.

Short-answer questions

1. *What is isomorphous substitution? Give one example.*

Model answer: Isomorphous substitution is the replacement of one ion by another of similar size but lower charge in a clay lattice, without changing the structure. Example: Al3+ replaced by Mg2+ in the octahedral sheet, leaving a permanent negative charge.

2. *Distinguish permanent and pH-dependent charge.*

Model answer: Permanent charge arises from isomorphous substitution and does not change with pH (as in montmorillonite). pH-dependent charge arises from protonation/deprotonation of hydroxyl and carboxyl groups and changes with pH (as on oxide and edge surfaces and humus).

3. *What is the point of zero charge (PZC)?*

Model answer: The PZC is the pH at which the net surface charge of a colloid is zero. Below the PZC the surface is net positive; above it the surface is net negative.

Long-answer questions

1. *Explain the sources of negative and positive charge on soil colloids.*

Outline: Introduction - negative charge (isomorphous substitution, broken edges, humus groups) - positive charge (protonation of OH) - pH-dependent charge and PZC - isomorphous substitution detail - diagram - significance - conclusion.

2. *Discuss permanent and pH-dependent charge and their significance in soil fertility.*

Outline: Introduction - permanent charge (isomorphous substitution, CEC) - pH-dependent charge (oxides, edges, humus) - PZC - effect on CEC, AEC and phosphate fixation - liming and charge - Tamil Nadu examples - conclusion.

Lecture 27: Ion Exchange - Cation and Anion Exchange Capacity and Base Saturation

27.1 Introduction

Soil colloids carry charges and adsorb ions on their surfaces. These adsorbed ions can be **replaced (exchanged)** by other ions from the soil solution. This **ion exchange** is the basis of nutrient retention and supply in soil, and of lime and gypsum recommendations.

27.2 Definition of Ion Exchange

Ion exchange is the **reversible replacement of one adsorbed ion on a soil colloid by another ion of like charge from the soil solution**.

- **Cation exchange** - exchange of cations (positive ions) on the negatively charged colloid.
- **Anion exchange** - exchange of anions (negative ions) on positively charged surfaces.

The exchange is **stoichiometric** (equal charge for equal charge) and **reversible**.

27.3 Cation Exchange

Definition: Cation exchange is the replacement of one cation on the soil colloid by another cation from the soil solution.

Example: When a Ca-saturated clay is treated with a potassium solution:

- Clay-Ca + 2 K+ <-> Clay-2K + Ca2+

The K+ replaces Ca2+ on the colloid (two K+ for one Ca2+, to balance charge), and Ca2+ goes into solution.

27.4 Cation Exchange Capacity (CEC)

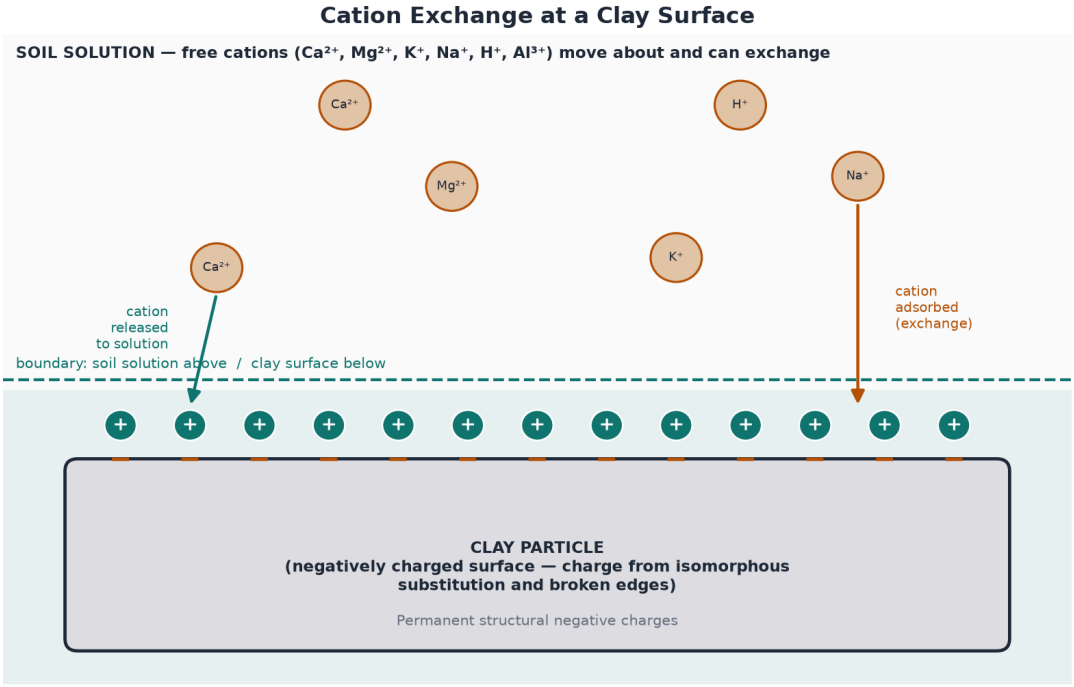
Definition: Cation exchange capacity (CEC) is the **total amount of exchangeable cations a soil can hold (adsorb) at a given pH**, expressed as the number of **centimoles of charge per kilogram (cmol/kg)** of soil (older unit: milliequivalents per 100 g, meq/100 g; 1 cmol/kg = 1 meq/100 g).

- CEC is a measure of the **negative charge** of the soil and hence of its **nutrient-holding capacity**.
- High CEC = high fertility and nutrient retention; low CEC = poor nutrient retention (leaching).

27.5 Anion Exchange Capacity (AEC)

Definition: Anion exchange capacity (AEC) is the **total amount of exchangeable anions a soil can hold** on positively charged surfaces, expressed in cmol/kg.

- AEC occurs on **positive charges** (oxide and edge surfaces in acid soils).



Cation exchange: a cation in solution displaces another from the clay surface. CEC = capacity to hold exchangeable cations (cmol(p⁺)/kg).

Figure 27.1 Cation exchange at a clay surface

Exchangeable cations held on a negatively charged clay surface, in equilibrium with the soil solution.

- Important for **phosphate, sulphate, chloride and nitrate** retention.
- AEC is generally much smaller than CEC; soils are net negatively charged.

27.6 Exchangeable Cations and Their Amounts

The main exchangeable cations are **Ca2+, Mg2+, K+, Na+** (bases) and **H+, Al3+** (acid-forming).

Typical proportions on the exchange complex (well-drained, neutral soil):

- Ca2+: 60-80%
- Mg2+: 10-20%
- K+: 2-6%
- Na+: 1-3%
- H+ and Al3+: 0-10% (more in acid soils)

27.7 Base Saturation

Definition: Base saturation is the **percentage of the CEC occupied by the basic (metallic) cations - Ca, Mg, K and Na** - i.e., the fraction of exchange sites not occupied by H+ or Al3+.

- **Base saturation (%) = (sum of exchangeable bases / CEC) x 100**
- Exchangeable bases = Ca + Mg + K + Na (all in cmol/kg).

Interpretation

- High base saturation (> 80%): fertile, neutral to alkaline soil.
- Low base saturation (< 50%): acidic, leached soil.
- Base saturation is used in soil classification (e.g., Alfisols have >35% base saturation; Ultisols <35%).

27.8 Exchangeable Sodium Percentage (ESP)

Definition: ESP is the percentage of the CEC occupied by sodium:

- **ESP = (exchangeable Na / CEC) x 100**
- ESP > 15% -> sodic (alkali) soil.
- Used with EC and pH to classify saline, sodic and saline-sodic soils.

Sodium adsorption ratio (SAR): a related measure from water analysis:

- SAR = Na+ / sqrt[(Ca2+ + Mg2+)/2] (concentrations in mmol/L).

27.9 CEC Ranges by Soil Type / Colloid

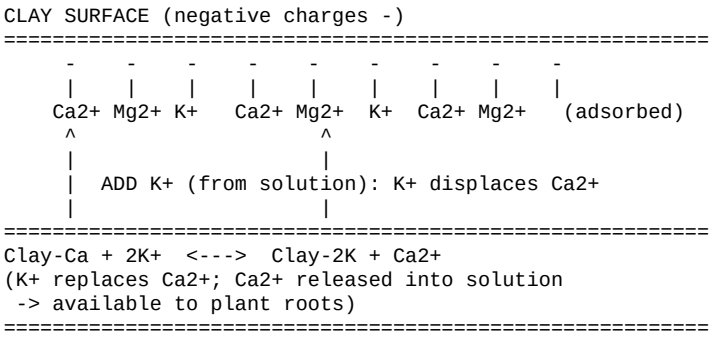
Soil / colloid	CEC (cmol/kg)
Sand	1-5
Sandy soil	2-10
Loam	10-20
Silt loam	15-25
Clay soil	25-60
Kaolinite	3-15
Illite	20-40
Chlorite	10-40
Montmorillonite	80-150
Vermiculite	100-150
Humus (organic matter)	150-300
Peat / organic soil	50-200

General: CEC increases with **clay content, organic matter and the type of clay** (montmorillonite > illite > kaolinite).

27.10 Factors Affecting CEC

- **Clay content and type:** more clay and montmorillonite -> higher CEC.
- **Organic matter:** humus has very high CEC; more OM -> higher CEC.
- **Soil pH:** CEC increases with pH (pH-dependent charge); liming raises CEC.
- **Texture:** clayey soils have higher CEC than sandy soils.

27.11 Diagram: Cation Exchange at a Clay Surface



27.12 Worked Example 1: CEC from Exchangeable Cations

Problem: A soil has exchangeable Ca = 12 cmol/kg, Mg = 4 cmol/kg, K = 1 cmol/kg, Na = 0.5 cmol/kg and H+Al = 2.5 cmol/kg. Calculate the CEC.

Solution

- CEC = sum of all exchangeable cations
- = 12 + 4 + 1 + 0.5 + 2.5 = **20 cmol/kg**

Answer: CEC = 20 cmol/kg.

27.13 Worked Example 2: Base Saturation

Problem: Using the soil above, calculate the base saturation.

Solution

- Exchangeable bases = Ca + Mg + K + Na = 12 + 4 + 1 + 0.5 = 17.5 cmol/kg
- CEC = 20 cmol/kg
- Base saturation = (17.5 / 20) x 100 = **87.5%**

Answer: Base saturation = 87.5%, indicating a fertile, near-neutral soil.

27.14 Worked Example 3: ESP

Problem: For the same soil, calculate the ESP.

Solution

- ESP = (exchangeable Na / CEC) x 100 = (0.5 / 20) x 100 = **2.5%**

Answer: ESP = 2.5%, well below 15%, so the soil is not sodic.

27.15 Worked Example 4: Gypsum Requirement (Basics)

Problem: A sodic soil has a CEC of 25 cmol/kg and an ESP of 20%. If the target ESP is 10%, how much exchangeable Na (in cmol/kg) must be replaced? (Then gypsum is applied to supply the equivalent Ca.)

Solution

- Present exchangeable Na = 20% of 25 = 5 cmol/kg
- Target exchangeable Na = 10% of 25 = 2.5 cmol/kg
- Na to be replaced = 5 - 2.5 = **2.5 cmol/kg**
- Gypsum ($CaSO_4.2H_2O$) supplies Ca^{2+} to replace Na^+; the gypsum requirement is calculated from this Na value using the soil depth and bulk density (see note).

Answer: 2.5 cmol/kg of Na must be replaced; gypsum is applied in equivalent amount (the exact field dose depends on soil depth and bulk density).

27.16 Worked Example 5: Gypsum Requirement in Field Terms (Concept)

Problem: A sodic soil has a CEC of 25 cmol/kg and a present ESP of 20%. The target ESP after reclamation is 10%. Estimate the gypsum required for the top 15 cm of one hectare, given a bulk density of 1.4 g/cm3.

Solution

- Step 1 - Na to be replaced:

- Present exchangeable Na = 20% of 25 = 5 cmol/kg

- Target exchangeable Na = 10% of 25 = 2.5 cmol/kg

- Na to be replaced = 5 - 2.5 = **2.5 cmol/kg**

- Step 2 - soil mass in the top 15 cm of 1 ha:

- Volume = 10,000 m2 x 0.15 m = 1,500 m3

- Mass = 1,500 m3 x 1.4 t/m3 = **2,100 tonnes** (from Lecture 14)

- Step 3 - gypsum needed:

- 2.5 cmol/kg = 0.025 mol of charge per kg of soil

- Per tonne (1,000 kg): 0.025 x 1,000 = 25 mol of charge

- Gypsum (CaSO4.2H2O) has a molecular mass of 172 g/mol and supplies 1 mol Ca per mol of charge

- Gypsum per tonne of soil = 25 x 172 g = 4,300 g = 4.3 kg

- For 2,100 tonnes: 4.3 x 2,100 = 9,030 kg = **about 9.0 tonnes/ha**

Answer: Approximately **9 tonnes of gypsum per hectare** are needed to lower the ESP from 20% to 10% in the top 15 cm. This is an order-of-magnitude estimate; the exact field dose must be confirmed by a laboratory gypsum requirement test.

27.17 Importance of Ion Exchange in Agriculture

- **Nutrient retention and supply:** exchangeable cations are the reserve of plant nutrients (Ca, Mg, K, NH4).
- **Fertilizer management:** high-CEC soils hold nutrients; low-CEC (sandy) soils need frequent, light doses.
- **Liming:** lime (Ca) replaces H+ and Al3+ on the complex, raising pH and base saturation.
- **Gypsum for sodic soils:** Ca replaces Na on the complex.
- **Soil fertility assessment:** CEC and base saturation are standard soil test values.
- **Soil classification:** base saturation is a differentiating criterion (Alfisols vs Ultisols).
- **Water quality and salinity:** ESP and SAR guide irrigation water use.

27.18 Tamil Nadu Context

- **Black soils** have high CEC (montmorillonite) -> good nutrient retention, high fertility.
- **Red soils** have low CEC (kaolinite) -> need frequent fertilization and organic matter.
- **Coastal sodic soils** have high ESP -> need gypsum for reclamation (TNAU recommendation).
- TNAU soil test laboratories report CEC, base saturation and ESP for fertilizer and amendment advice.

27.19 Exam Focus

Key points

- Ion exchange is the reversible replacement of an adsorbed ion by another of like charge.
- Cation exchange occurs on negatively charged colloids; anion exchange on positive surfaces.
- CEC = total exchangeable cations; unit cmol/kg (= meq/100 g).
- CEC ranges: sand 1-5, loam 10-20, clay 25-60, montmorillonite 80-150, humus 150-300.
- Base saturation = (exchangeable bases / CEC) x 100; high (>80%) = fertile.
- ESP = (exchangeable Na / CEC) x 100; ESP > 15% = sodic.

- CEC increases with clay, montmorillonite and organic matter, and with pH.
- Exchange is the basis of liming, gypsum application and nutrient supply.

Short-answer questions

1. *Define cation exchange and CEC.*

Model answer: Cation exchange is the reversible replacement of one cation on the soil colloid by another cation from the solution. CEC is the total amount of exchangeable cations a soil can hold, expressed in cmol/kg.

2. *What is base saturation?*

Model answer: Base saturation is the percentage of the CEC occupied by the basic cations Ca, Mg, K and Na. It is calculated as (sum of exchangeable bases / CEC) x 100 and indicates soil fertility and leaching status.

3. *How is ESP calculated and what does it indicate?*

Model answer: ESP = (exchangeable sodium / CEC) x 100. An ESP above 15% indicates a sodic (alkali) soil that needs gypsum for reclamation.

Long-answer questions

1. *Explain ion exchange, CEC, AEC and base saturation with worked examples.*

Outline: Definition of ion exchange - cation exchange with equation - CEC (definition, unit, ranges) - AEC - base saturation - ESP - diagram - worked examples - significance - conclusion.

2. *Discuss the importance of cation exchange capacity in soil fertility and management.*

Outline: Introduction - CEC and nutrient retention - factors affecting CEC - CEC ranges table - base saturation and soil classification - liming and gypsum - fertilizer management - Tamil Nadu examples - conclusion.

Lecture 28: Soil Organic Matter - Properties and Its Influence on Soil Health

28.1 Introduction

Soil organic matter (SOM) is the organic fraction of the soil - all the material derived from living organisms, including fresh plant and animal residues, partially decomposed material and stable humus. It is usually only 1-5% of the soil by weight in mineral soils, but it has a **disproportionately large effect** on soil properties and health.

28.2 Composition of Soil Organic Matter

SOM consists of:

- **Living organisms** (roots, microbes, soil fauna) - a small fraction.
- **Fresh and partially decomposed residues** (litter, crop residues, manure).
- **Humus** - the stable, dark, amorphous, resistant end product.
- **Non-humic substances** - carbohydrates, proteins, lipids, organic acids (readily decomposed).
- **Humic substances** - humic acid, fulvic acid and humin (resistant).

Elemental composition (approximate):

Element	% of SOM (approx.)
Carbon	52 - 58
Oxygen	34 - 40
Hydrogen	4 - 6
Nitrogen	3 - 5
Sulphur, phosphorus	small amounts

Rule of thumb: SOM contains about **58% carbon**; therefore:

- **SOM (%) = Organic carbon (%) x 1.724** (the van Bemmelen factor).

28.3 Humus and Its Fractions

Humus is the stable, dark, complex organic material formed by humification.

- **Humic acid** - soluble in alkali, insoluble in acid; dark brown to black; high molecular weight.
- **Fulvic acid** - soluble in both acid and alkali; light yellow to brown; lower molecular weight; more reactive.
- **Humin** - insoluble in both; tightly bound to clay; most resistant.

28.4 Properties of Soil Organic Matter

- **Colloidal** - very fine, high surface area.
- **High CEC** (150-300 cmol/kg) - holds nutrients (Ca, Mg, K, NH4).
- **High water-holding capacity** - holds several times its weight in water.
- **High buffering capacity** - resists pH change.
- **Dark colour** - gives soil its dark colour.
- **Binding agent** - binds particles into stable aggregates.
- **Source of nutrients** - supplies N, P, S and micronutrients on mineralisation.
- **Chelating agent** - forms complexes with metals (Fe, Zn, Cu), improving availability.
- **Energy source for microbes** - fuels soil biological activity.
- **Low bulk density** - lowers soil bulk density, increases porosity.

28.5 Decomposition of Organic Matter

Decomposition is the breakdown of organic matter by microorganisms into simpler compounds and eventually into CO2, water and nutrients.

Processes

1. **Mineralisation** - complete breakdown to inorganic nutrients (N, P, S, CO2, H2O).
2. **Humification** - formation of stable humus.

Factors affecting decomposition

- **Temperature:** optimum 25-35 degrees C; faster when warm.
- **Moisture:** adequate moisture needed; too dry or too wet slows it.
- **Aeration:** aerobic decomposition is faster (anaerobic gives peat and methane).
- **C:N ratio:** narrow C:N (< 20) decomposes fast (legumes); wide C:N (> 30) decomposes slowly (straw, sawdust).
- **Lignin content:** high lignin resists decomposition.
- **Nutrients:** adequate N and P speed decomposition.
- **Soil type:** clay soils protect organic matter (organo-mineral complexes); sandy soils lose it faster.

28.6 The Carbon-Nitrogen (C:N) Ratio

- Microbes need carbon and nitrogen in about a **24:1** ratio (a commonly cited value; often quoted as 25:1) for their own growth.
- **C:N of crop residues:** legumes ~15-20:1; cereals ~50-80:1; sawdust ~400:1.
- **Narrow C:N (low ratio)** -> rapid decomposition and net N release (mineralisation).
- **Wide C:N (high ratio)** -> slow decomposition and **net N immobilisation** (microbes take up soil N), causing temporary N deficiency in crops.
- **Management:** add N fertilizer when incorporating wide-C:N residues (e.g., straw).

28.7 Influence of Soil Organic Matter on Soil Health

SOM is central to **soil health** - the capacity of soil to function as a living ecosystem.

Physical health

- Improves **structure** and aggregation.
- **Lowers bulk density**, increases porosity.
- **Increases water-holding capacity** and infiltration.

- Reduces **erosion** and crusting.
- Improves **tilth and workability**.
- **Moderates temperature.**

Chemical health

- **High CEC** - retains nutrients.
- **Buffering** - resists pH change.
- **Chelation** - improves micronutrient availability.
- **Nutrient reservoir** - N, P, S, micronutrients.
- Reduces **aluminium toxicity** in acid soils (by complexing Al).

Biological health

- **Feeds microbes** - energy and nutrients.
- Increases **microbial biomass and diversity**.
- Supports **nitrogen fixation**, mycorrhizae and earthworms.
- Enhances **nutrient cycling** and disease suppression.

28.8 Diagram: Role of Soil Organic Matter

```
SOIL ORGANIC MATTER
================================================
PHYSICAL:                CHEMICAL:
- structure              - CEC (nutrient holding)
- porosity               - buffering
- water holding          - chelation
- erosion control        - nutrient reservoir
- temperature
================================================
BIOLOGICAL:
- microbial food
- N fixation
- nutrient cycling
- disease suppression
================================================
-> SOIL HEALTH / FERTILITY / PRODUCTIVITY
```

28.9 Sources and Maintenance of SOM

Sources

- Crop residues and stubble.
- Green manures (sunhemp, daincha - widely used in Tamil Nadu).
- Farmyard manure (FYM), compost, vermicompost.
- Crop roots and root exudates.
- Animal wastes and poultry manure.

Maintenance and improvement

- Add organic manures regularly.
- Grow green manure and cover crops.
- Retain crop residues (avoid burning).
- Practise crop rotation with legumes.
- Reduce tillage (conservation agriculture).
- Avoid excessive burning and erosion.

28.10 Worked Example 1: Organic Matter from Organic Carbon

Problem: A soil has an organic carbon content of 0.58%. Calculate the organic matter content.

Solution

- SOM (%) = Organic carbon (%) x 1.724
- = 0.58 x 1.724 = **1.0%**

Answer: The soil contains about 1.0% organic matter.

28.11 Worked Example 2: Organic Carbon from Organic Matter

Problem: A soil has 2.5% organic matter. Calculate the organic carbon.

Solution

- Organic carbon = SOM / 1.724 = 2.5 / 1.724 = **1.45%**

Answer: The organic carbon content is about 1.45%.

28.12 Worked Example 3: Nitrogen from C:N Ratio

Problem: A soil organic matter sample has 1.16% carbon and a C:N ratio of 10. Calculate the nitrogen content.

Solution

- N = C / (C:N ratio) = 1.16 / 10 = **0.116% N**

Answer: The nitrogen content is about 0.116%.

28.13 Worked Example 4: Mass of Organic Matter per Hectare

Problem: A soil has 1.0% organic matter in the top 15 cm. Bulk density is 1.4 g/cm3. Calculate the mass of organic matter per hectare.

Solution

- Soil mass (top 15 cm of 1 ha) = 2,100 tonnes (from Lecture 14).
- Organic matter = 1.0% of 2,100 = **21 tonnes/ha**

Answer: About 21 tonnes of organic matter are present in the top 15 cm of one hectare.

28.14 Worked Example 5: Carbon Stock

Problem: Using the soil above, calculate the organic carbon stock per hectare.

Solution

- Organic carbon = 0.58% of 2,100 tonnes = **12.2 tonnes/ha**
- (Check: 1.0% SOM / 1.724 = 0.58% C; 0.58% of 2,100 = 12.2 t.)

Answer: About 12.2 tonnes of organic carbon per hectare.

28.15 Tamil Nadu Context

- **Red soils** are generally **low in organic matter** (0.5-1.0%) - need regular manuring.
- **Black soils** have moderate organic matter (0.5-1.5%).
- **Hill soils of the Nilgiris** have **higher organic matter** (2-5%) due to cool, humid conditions.
- **Paddy soils of the Cauvery delta** maintain organic matter under continuous cropping with manure.
- TNAU recommends **green manuring with sunhemp/daincha**, **FYM application** and **residue recycling** to maintain SOM.

28.16 Exam Focus

Key points

- Soil organic matter = all organic material in soil; humus is its stable fraction.
- SOM is only 1-5% by weight but has large effects.
- SOM contains about 58% carbon; SOM = organic carbon x 1.724.
- Humus fractions: humic acid, fulvic acid, humin.
- Properties: colloidal, high CEC (150-300), high water holding, buffering, binding, chelation.
- Decomposition: mineralisation + humification; controlled by temperature, moisture, aeration, C:N ratio.
- C:N ratio: narrow = fast decomposition and N release; wide = immobilisation.
- SOM improves physical, chemical and biological soil health; central to soil health.

Short-answer questions

1. *Define soil organic matter and name the humus fractions.*

Model answer: Soil organic matter is the organic fraction of the soil derived from plant and animal residues, including living organisms, decomposing residues and stable humus. The humus fractions are humic acid, fulvic acid and humin.

2. *How is organic matter related to organic carbon?*

Model answer: Soil organic matter contains about 58% carbon, so organic matter (%) = organic carbon (%) x 1.724 (the van Bemmelen factor).

3. *Why is the C:N ratio important?*

Model answer: A narrow C:N ratio (below about 20) gives rapid decomposition and net nitrogen release (mineralisation). A wide C:N ratio (above about 30) causes slow decomposition and net nitrogen immobilisation, which can temporarily starve crops of nitrogen.

Long-answer questions

1. *Describe the composition, properties and decomposition of soil organic matter.*

Outline: Definition - composition (elements, fractions) - humus and its fractions - properties (colloidal, CEC, water, buffering, binding) - decomposition (mineralisation, humification) - factors affecting decomposition - C:N ratio - worked examples - conclusion.

2. *Discuss the influence of soil organic matter on soil health.*

Outline: Introduction - soil health concept - physical effects (structure, porosity, water, erosion) - chemical effects (CEC, buffering, chelation, nutrients) - biological effects (microbes, N fixation, cycling) - maintenance (green manure, FYM, residues) - Tamil Nadu examples - conclusion.

Lecture 29: Soil Taxonomy - Structure, Differentiating Characters and Limitations; Soil Orders - Keys to Soil Order; Different Soil Orders and Distribution in the World and Soils of Tamil Nadu

29.1 Introduction

Soil taxonomy is a systematic method of **classifying soils** into groups based on their measurable properties (diagnostic horizons, moisture and temperature regimes, etc.). The system used widely today is **Soil Taxonomy**, developed by the **United States Department of Agriculture (USDA)**, also called the **US Comprehensive Soil Classification System**.

29.2 Objectives of Soil Classification

- To organise knowledge about soils.
- To identify and name soils consistently.
- To predict soil behaviour for management.
- To transfer technology between similar soils.
- To map and inventory soils (soil survey).

29.3 Structure of Soil Taxonomy - The Six Categories

Soil Taxonomy is a **hierarchical system** with **six categories**, from most general to most specific:

Category	Basis	Example
1. Order	Soil-forming processes / major diagnostic features	Alfisol
2. Suborder	Genetic / moisture & temperature regime	Udalf
3. Great group	Specific horizon features, base status	Hapludalf
4. Subgroup	Central concept (Typic) or intergrade	Typic Hapludalf
5. Family	Texture, mineralogy, temperature class	Fine-loamy, mixed, thermic
6. Series	Detailed properties, similar profiles	Named after a place

Mnemonic: "Orders Suborders Great groups Subgroups Families Series" (most general to most specific).

Nomenclature: Names are built by adding syllables. The **order** is indicated by the last syllable:

- -alf (Alfisol), -ent (Entisol), -ept (Inceptisol), -oll (Mollisol), -ox (Oxisol), -ult (Ultisol), -and (Andisol), -id (Aridisol), -ist (Histosol), -od (Spodosol), -ert (Vertisol), -el (Gelisol).

29.4 Diagnostic Horizons

Soil Taxonomy uses **diagnostic horizons** - specific horizons with defined properties - to classify soils. Important ones:

- **Mollic epipedon** - thick, dark, soft, base-rich surface horizon (Mollisols).
- **Umbric epipedon** - like mollic but low base saturation (acid).
- **Ochric epipedon** - thin, light-coloured surface horizon.
- **Argillic horizon** - subsurface horizon of illuvial clay (Alfisols, Ultisols).
- **Kandic horizon** - low-activity clay horizon (Ultisols).
- **Oxic horizon** - highly weathered, oxide-rich horizon (Oxisols).
- **Spodic horizon** - illuvial Fe-Al-humus horizon (Spodosols).
- **Cambic horizon** - weakly developed subsurface horizon (Inceptisols).
- **Calcic horizon** - carbonate accumulation (Aridisols, Mollisols).
- **Salic horizon** - salt accumulation.
- **Gypsic horizon** - gypsum accumulation.
- **Duripan / Fragipan / Plinthite** - cemented or dense layers.

Diagnostic epipedons (surface) and **diagnostic subsurface horizons** together with **soil moisture and temperature regimes** are the key differentiating characters.

29.5 Soil Moisture and Temperature Regimes

Moisture regimes

- **Aquic** - saturated much of the time (reduced).
- **Udic** - moist most of the time (humid).
- **Ustic** - moist in the growing season, dry part of the year (semi-arid).
- **Aridic** - dry most of the time (arid).
- **Xeric** - moist winters, dry summers (Mediterranean).

Temperature regimes

- **Hyperthermic** (> 22 degrees C mean annual), **Thermic** (15-22), **Mesic** (8-15), **Frigid** (0-8), **Cryic** (< 0).

Tamil Nadu soils are mostly **ustic to udic** moisture regimes and **hyperthermic** temperature regime.

29.6 Differentiating Characters and Limitations

Differentiating characters

- Diagnostic horizons (epipedons and subsurface).
- Soil moisture and temperature regimes.
- Base saturation, CEC, clay activity.
- Presence of carbonates, salts, gypsum, plinthite.

Limitations of Soil Taxonomy

- **Complex nomenclature** - names are difficult to pronounce and remember.
- **Requires detailed laboratory data** (not always available).
- **Ignores some genetic (process) information** - it is largely property-based.
- **Boundaries are arbitrary** in places.
- **Does not directly indicate fertility or productivity.**
- **Not ideal for all local needs** - national systems (e.g., India's) are also used.

- **High cost and time** for detailed field and lab work.

29.7 The Twelve Soil Orders

Order	Key feature	Common environment	World distribution
Entisols	Little horizon development; recent	All climates; floodplains, dunes	Widespread (~16% of land)
Inceptisols	Weakly developed horizons	All climates; young soils	Widespread
Vertisols	Shrink-swell clays, deep cracks	Semi-arid, base-rich	~2%; India (black soils), Australia
Aridisols	Dry, salic/calcic horizons	Arid deserts	~12%; N. Africa, Middle East, W. India
Mollisols	Dark, base-rich mollic epipedon	Grasslands	~7%; prairies, steppes
Alfisols	Argillic horizon, base saturation > 35%	Humid to sub-humid forests	~10%; temperate & tropical
Ultisols	Argillic/kandic horizon, low base saturation, acid	Humid tropics/subtropics	~8%; SE USA, tropics
Oxisols	Oxic horizon, highly weathered, low CEC	Humid tropics	~8%; Amazon, Africa, S. Asia
Spodosols	Spodic horizon (Fe-Al-humus)	Cool humid conifer forests	~3%; boreal regions
Histosols	Organic soils (peat)	Wetlands, bogs	~1%; peatlands
Andisols	Volcanic ash soils	Volcanic regions	~1%; Japan, Andes, Indonesia
Gelisols	Permafrost-affected	Polar regions	~9%; tundra

(Percentages are approximate global land-area shares from standard soil survey data.)

29.8 Keys to Soil Order (Simplified)

A simplified key to identify the soil order:

1. **Organic soil** (high organic matter, wet) -> **Histosol**
2. **Permafrost** present -> **Gelisol**
3. **Volcanic ash** parent material, high P retention -> **Andisol**
4. **Spodic horizon** (Fe-Al-humus accumulation) -> **Spodosol**
5. **Oxic horizon** (highly weathered, low CEC) -> **Ox isol** (Oxisol)
6. **Shrink-swell clays**, deep cracks -> **Vertisol**
7. **Aridic moisture regime** (dry, salic/calcic) -> **Aridisol**
8. **Mollic epipedon** (dark, base-rich) -> **Mollisol**
9. **Argillic/kandic horizon** with **low base saturation** -> **Ultisol**
10. **Argillic horizon** with **high base saturation** -> **Alfisol**
11. **Weakly developed** horizons (cambic) -> **Inceptisol**
12. **Little development** (no diagnostic horizons) -> **Entisol**

```
SIMPLIFIED KEY TO SOIL ORDERS
=====================================================
Organic? --------------------------------> HISTOSOL
Permafrost? -----------------------------> GELISOL
Volcanic ash? ---------------------------> ANDISOL
Spodic horizon? -------------------------> SPODOSOL
Oxic horizon? ---------------------------> OXISOL
Shrink-swell cracks? --------------------> VERTISOL
Arid (dry)? -----------------------------> ARIDISOL
Dark base-rich surface (mollic)? --------> MOLLISOL
Clay horizon + low base sat.? -----------> ULTISOL
Clay horizon + high base sat.? ----------> ALFISOL
Weak development (cambic)? --------------> INCEPTISOL
Little development? ---------------------> ENTISOL
=====================================================
```

29.9 Diagram: Soil Orders of the World (Text Map)

```
SOIL ORDERS OF THE WORLD (schematic)
=====================================================
POLAR (N/S):        GELISOLS (permafrost)
BOREAL:             SPODOSOLS, HISTOSOLS
TEMPERATE GRASS:    MOLLISOLS (prairies, steppes)
TEMPERATE FOREST:   ALFISOLS, INCEPTISOLS
DESERTS:            ARIDISOLS
HUMID SUBTROPICS:   ULTISOLS
HUMID TROPICS:      OXISOLS, ULTISOLS
VOLCANIC AREAS:     ANDISOLS
RIVER PLAINS:       ENTISOLS, INCEPTISOLS
SHRINK-SWELL AREAS: VERTISOLS (India, Australia)
=====================================================
India: Entisols, Inceptisols, Vertisols, Alfisols,
       Aridisols, Mollisols, Ultisols (in the hills),
       Histosols (in Kerala, Sundarbans)
=====================================================
```

29.10 Soil Orders in India (General)

The major soil orders found in India are:

- **Entisols** - recent alluvial soils, coastal sands.
- **Inceptisols** - weakly developed soils over much of the country.
- **Vertisols** - black cotton soils (regur) of the Deccan and central India.
- **Alfisols** - red and lateritic soils of the peninsular region.
- **Aridisols** - desert soils of Rajasthan.
- **Mollisols** - some dark soils of the northern plains and hills.
- **Ultisols** - acid, weathered soils of the northeastern hills and Western Ghats.
- **Histosols** - organic soils of Kerala (Kuttanad) and the Sundarbans.

29.11 Soils of Tamil Nadu

Tamil Nadu has a variety of soils, classified mainly into the following groups (traditional and Soil Taxonomy equivalents):

Traditional soil group	Area / location	Soil Taxonomy (approx.)
Red soils	Coimbatore, Salem, Vellore, Dharmapuri	Alfisols / Inceptisols
Black soils (vertisols)	Coimbatore, Madurai, Tirunelveli	Vertisols / Inceptisols
Laterite soils	Nilgiris, Kanyakumari, Salem hills	Ultisols / Oxisols (acid)
Alluvial soils	Cauvery delta (Thanjavur, Nagapattinam, Tiruvarur)	Entisols / Inceptisols
Coastal soils	Coromandel and Gulf of Mannar coasts	Entisols (sandy)
Saline / sodic soils	Coastal belts, delta patches	Aridisols / Salids / Natrargids
Forest / hill soils	Western Ghats, Nilgiris	Ultisols / Mollisols (high OM)

Key features of Tamil Nadu soils

- **Red soils** dominate (about 60-65% of the cultivated area) - low organic matter, good drainage, kaolinite clay.
- **Black soils** occur in the semi-arid interior - high CEC (montmorillonite), shrink-swell, good for cotton.
- **Laterite soils** occur in the high-rainfall western and hill zones - acidic, iron-rich.
- **Alluvial soils** of the **Cauvery delta** are the most fertile - suited to paddy.
- **Coastal soils** are sandy and saline in places - need organic matter and drainage.

29.12 The Cauvery Delta Soils

- The **Cauvery delta** (Thanjavur, Nagapattinam, Tiruvarur, parts of Trichy) has **deep alluvial soils** deposited by the Cauvery river.
- They are **fine-textured, fertile, with moderate organic matter** and good water-holding capacity.

- They support intensive **paddy (rice)** cultivation - the "granary of Tamil Nadu".
- Some areas have **salinity and drainage problems**; TNAU recommends drainage, gypsum (for sodic patches) and balanced fertilization.

29.13 TNAU Recommendations Related to Soil Types

- **Red soils:** apply organic manure, lime if acidic, balanced NPK; suitable for groundnut, millets, pulses.
- **Black soils:** manage moisture carefully, avoid working wet; suitable for cotton, sorghum, pulses.
- **Laterite soils:** lime, organic matter, phosphorus management (P fixation); suitable for cashew, rubber, tea (hills).
- **Alluvial soils (delta):** paddy-based cropping, green manure, balanced fertilizer.
- **Coastal sandy soils:** organic matter, drip irrigation, salt-tolerant crops.

29.14 Worked Example: Identifying the Soil Order

Problem: A soil has a dark surface horizon more than 25 cm thick, high base saturation, a clay-rich subsoil with base saturation above 35%, and a ustic moisture regime. Identify the likely soil order.

Solution

- Dark, thick, base-rich surface horizon -> **mollic epipedon**.
- Clay-rich subsoil with base saturation > 35% -> **argillic horizon**, high base status.
- Ustic moisture regime -> sub-humid.
- This combination indicates an **Alfisol** (specifically an Ustalf). If the surface horizon is very dark and dominant, it could be a **Mollisol**; here the argillic horizon with high base saturation points to **Alfisol**.

Answer: Likely an **Alfisol** (Ustalf), typical of the red and semi-arid soils of the interior.

29.15 Worked Example: Order from Diagnostic Horizon

Problem: A soil has deep cracks that open and close with moisture, and high shrink-swell clay. Identify the order.

Solution

- Deep cracks and shrink-swell clay are the diagnostic features of **Vertisols**.
- (These are the black cotton soils of India and Tamil Nadu.)

Answer: Vertisol (black soil / regur).

29.16 Exam Focus

Key points

- Soil Taxonomy (USDA) is a hierarchical classification with six categories: order, suborder, great group, subgroup, family, series.
- Diagnostic horizons (mollic, argillic, oxic, spodic, etc.) and moisture/temperature regimes are the differentiating characters.
- There are 12 soil orders; names end in -ent, -ept, -oll, -alf, -ult, -ox, -od, -ert, -ist, -and, -id, -el.
- Major orders: Entisols, Inceptisols, Vertisols, Aridisols, Mollisols, Alfisols, Ultisols, Oxisols, Spodosols, Histosols, Andisols, Gelisols.
- Limitations: complex names, needs lab data, property-based, arbitrary boundaries.
- India: Vertisols (black), Alfisols (red), Entisols/Inceptisols (alluvial), Aridisols (desert), Ultisols (hills).
- Tamil Nadu: red soils (Alfisols), black soils (Vertisols), laterite (Ultisols/Oxisols), delta alluvium (Entisols/Inceptisols), coastal (Entisols).
- Cauvery delta soils are the most fertile - the granary of Tamil Nadu.

Short-answer questions

1. *Name the six categories of Soil Taxonomy.*

Model answer: Order, suborder, great group, subgroup, family and series (from most general to most specific).

2. *List any four soil orders with a key feature each.*

Model answer: Vertisols (shrink-swell clays and deep cracks), Aridisols (dry, salic/calcic horizons), Mollisols (dark base-rich mollic epipedon) and Spodosols (spodic Fe-Al-humus horizon).

3. *What are the limitations of Soil Taxonomy?*

Model answer: Complex nomenclature, need for detailed laboratory data, largely property-based (ignoring some genetic information), arbitrary boundaries in places, high cost and time, and no direct indication of fertility.

Long-answer questions

1. *Describe the structure of Soil Taxonomy and its differentiating characters.*

Outline: Introduction - objectives - six categories with examples - diagnostic horizons - moisture and temperature regimes - nomenclature - limitations - conclusion.

2. *Describe the different soil orders and their distribution in the world and in Tamil Nadu.*

Outline: Introduction - the 12 soil orders with key features and distribution - simplified key - world distribution diagram - India's soil orders - soils of Tamil Nadu (red, black, laterite, alluvial, coastal) with Soil Taxonomy equivalents - Cauvery delta - TNAU recommendations - conclusion.

Appendix A: Quick Revision - Formulae, Units and Key Ranges

This appendix collects, in one place, the formulae, unit conversions and standard numerical ranges used across the 29 lectures. Use it for last-minute revision.

A.1 Master List of Formulae

Quantity	Formula	Notes
Gravimetric water content	W (%) = (mass of water / mass of oven-dry soil) x 100	Weight basis
Volumetric water content	theta = W x Db	W as fraction
Depth of water	depth = volumetric water content x soil depth	cm
Bulk density	Db = mass of oven-dry soil / total volume	g/cm3
Particle density	Dp = mass of solids / volume of solids	~2.65 g/cm3
Porosity	Porosity (%) = (1 - Db/Dp) x 100	Pore space
Air-filled porosity	total porosity - volumetric water content	%
Organic matter from C	SOM (%) = organic carbon (%) x 1.724	van Bemmelen factor
Organic carbon from SOM	OC = SOM / 1.724	%
Nitrogen from C:N	N = C / (C:N ratio)	%
CEC	sum of exchangeable cations	cmol/kg
Base saturation	(sum of bases / CEC) x 100	%
ESP	(exchangeable Na / CEC) x 100	>15% = sodic
pH	pH = -log10 [H+]	-
pF	pF = log10 (cm of water suction)	FC = 2.5, PWP = 4.2
Plasticity index	PI = LL - PL	%
Shrinkage index	SI = LL - SL	%
Liquidity index	LI = (natural WC - PL) / PI	0 to 1
Darcy's law	Q = K x A x (dh/L)	Saturated flow
Flux	q = Q/A = K x i	cm/h
Infiltration time	time = depth / rate	h
Total water potential	psi_t = psi_g + psi_m + psi_o + psi_p + psi_a	bar/kPa
Available water capacity	AWC = FC - PWP	cm

Quantity	Formula	Notes
Stokes' law	v proportional to r^2	Settling
Total dissolved salts	TDS (mg/L) approx = EC (dS/m) x 640	Salinity
Gypsum requirement	from Na to be replaced and soil mass	t/ha

A.2 Unit Conversions

From	To	Multiply by
1 dS/m	mmho/cm	1
1 dS/m	microS/cm	1000
1 bar	kPa	100
1 bar	atm	0.987 (about 1)
1 bar	cm of water	about 1020
1 cmol/kg	meq/100 g	1
1 g/cm3	t/m3	1
1 ha	m2	10,000
1 cm water over 1 ha	litres	100,000
1 mm water over 1 ha	m3	10
1 acre-furrow-slice (15 cm)	mass	about 2 million lb
1 ha-furrow-slice (15 cm)	mass	about 2.24 million kg

A.3 Key Numerical Ranges at a Glance

Property	Standard range
Particle density (mineral soil)	2.6 - 2.7 g/cm3 (avg 2.65)
Bulk density (sandy)	1.5 - 1.8 g/cm3
Bulk density (clay)	1.0 - 1.3 g/cm3
Bulk density (organic)	0.2 - 0.8 g/cm3
Porosity (sand)	35 - 45%
Porosity (loam)	45 - 55%
Porosity (clay)	50 - 60%
Sand (USDA)	2.0 - 0.05 mm
Silt (USDA)	0.05 - 0.002 mm
Clay	< 0.002 mm
Field capacity tension	-33 kPa (1/3 bar), pF 2.5
Permanent wilting point	-1500 kPa (15 bar), pF 4.2
Hygroscopic coefficient	-31 bar, pF 4.5
CEC of sand	1 - 5 cmol/kg
CEC of clay soil	25 - 60 cmol/kg
CEC of montmorillonite	80 - 150 cmol/kg
CEC of humus	150 - 300 cmol/kg
Base saturation of fertile soil	> 80%
ESP for sodic soil	> 15%
ECe for saline soil	> 4 dS/m
Ideal soil pH for most crops	6.0 - 7.5
Optimum germination temperature	20 - 30 degrees C

Property	Standard range
C:N ratio of humus	about 10 - 12
C:N ratio for mineralisation	< 20
Soil formation rate	about 1 cm in 100 - 1000 years

A.4 The 29 Lectures at a Glance (Revision Map)

No.	Lecture topic	One-line takeaway
1	Soil definition; pedological vs edaphological	Soil is a 3-D natural body; pedology classifies, edaphology manages
2	Components; three phase system; crust	50% solid, 25% water, 25% air; crust is silicate-rich
3	Rocks - igneous, sedimentary, metamorphic	Origin-based classification; sedimentary gives most soils
4	Important rocks - mineralogy	Acidic rocks give sandy soils; basic rocks give fertile soils
5	Primary minerals - silicate/non-silicate	Silicates 90% of crust; Goldich stability series
6	Secondary minerals - clays, amorphous	Weathering forms 1:1, 2:1 clays and oxides
7	Weathering - physical, chemical, biological	Disintegration + decomposition in place
8	Factors of soil formation	Jenny equation; passive (PM, relief) vs active (climate, biota, time)
9	Fundamental processes (Simenson)	Additions, losses, translocations, transformations
10	Specific processes	Podzolization, laterization, salinization, alkalization, calcification
11	Soil profile and horizons	O, A, E, B, C, R; suffix symbols; lithological discontinuity
12	Texture and textural triangle	Sand/silt/clay proportions; 12 USDA classes
13	Soil structure	Granular best; columnar = sodic; OM binds
14	Bulk density, particle density, porosity	Db < Dp; porosity = (1-Db/Dp) x 100
15	Soil colour	Munsell hue/value/chroma; red = oxidised, grey = waterlogged
16	Consistency and Atterberg limits	LL, PL, SL; PI = LL - PL
17	Soil water - forms, units, pF	Gravitational/capillary/hygroscopic; pF scale
18	Soil water potential	Matric, osmotic, gravitational; water moves down gradient
19	Movement of water	Darcy's law; infiltration, conductivity, drainage
20	Soil air and aeration	Less O2, more CO2; diffusion; waterlogging harms
21	Soil temperature	Solar source; conduction; affects germination and microbes
22	Soil reaction (pH)	6.0-7.5 ideal; lime for acid, gypsum for sodic
23	Electrical conductivity	ECe > 4 dS/m = saline; EC x 640 = TDS

No.	Lecture topic	One-line takeaway
24	Soil colloids	< 1 micrometre; high surface area; negative charge
25	Silicate clays	1:1, 2:1, 2:2 structures; CEC and swelling differ
26	Sources of charge	Isomorphous substitution (permanent); pH-dependent charge
27	Ion exchange	CEC, base saturation, ESP; liming and gypsum
28	Soil organic matter	58% C; SOM = OC x 1.724; improves all soil health
29	Soil taxonomy	Six categories; 12 orders; soils of Tamil Nadu

A.5 Common Formulae to Memorise

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Porosity (%) = (1 - Db/Dp) x 100
Volumetric water content = gravimetric WC x bulk density
SOM (%) = organic carbon (%) x 1.724
Base saturation (%) = (Ca+Mg+K+Na) / CEC x 100
ESP (%) = exchangeable Na / CEC x 100
pH = -log10 [H+]
pF = log10 (cm of water suction)
PI = LL - PL
Q = K x A x (dh/L)          (Darcy's law)
TDS (mg/L) = EC (dS/m) x 640
```

Glossary of Soil Science Terms

- **Active acidity** - hydrogen ions present in the soil solution, measured directly as pH.
- **Adhesion** - attraction between soil particles and other surfaces (water, implements, roots).
- **AEC (Anion Exchange Capacity)** - total exchangeable anions a soil can hold on positive surfaces (cmol/kg).
- **Aggregate** - a group of soil particles bound together into a structural unit (ped).
- **Alkalization** - accumulation of sodium on the exchange complex, producing sodic soil.
- **Alluvium** - sediment deposited by rivers (e.g., the Cauvery delta soils).
- **Amorphous minerals** - clay-sized secondary minerals with no ordered crystal structure (e.g., allophane).
- **Andisol** - soil order formed from volcanic ash, with high phosphate retention.
- **Anion exchange** - exchange of anions on positively charged soil surfaces.
- **Argillic horizon** - a subsurface horizon of illuvial clay accumulation.
- **Aridisol** - soil order of dry regions with salic or calcic horizons.
- **Available water capacity (AWC)** - water held between field capacity and permanent wilting point.
- **Base saturation** - percentage of CEC occupied by basic cations (Ca, Mg, K, Na).
- **Bulk density (Db)** - mass of oven-dry soil per unit total volume (solids + pores).
- **Calcic horizon** - subsurface horizon of calcium carbonate accumulation.
- **Calcification** - accumulation of calcium carbonate in the soil.
- **Catena** - a sequence of related soils down a slope.
- **Cation exchange** - replacement of one adsorbed cation by another from solution.
- **CEC (Cation Exchange Capacity)** - total exchangeable cations a soil can hold (cmol/kg).
- **Chlorite** - a 2:2 clay mineral with a brucite interlayer; non-expanding.
- **Cohesion** - attraction between soil particles themselves.
- **Colloid** - a particle less than 1 micrometre with a large surface area and surface charge.
- **Decalcification** - removal (leaching) of calcium carbonate from the soil.
- **Diagnostic horizon** - a horizon with defined properties used in soil classification.

- **Eluviation** - washing out of soil material from a horizon.
- **Entisol** - soil order with little horizon development (recent soils).
- **ESP (Exchangeable Sodium Percentage)** - percentage of CEC occupied by sodium.
- **Ferro-magnesian minerals** - dark minerals rich in iron and magnesium (olivine, pyroxene, biotite).
- **Field capacity** - water content after free drainage (about -33 kPa or 1/3 bar).
- **Flocculation** - clumping of clay particles, usually by Ca2+ or Mg2+.
- **Gelisol** - soil order affected by permafrost.
- **Gleying** - grey, bluish colouring from reduced iron under waterlogged conditions.
- **Humification** - conversion of fresh organic matter into humus.
- **Humus** - the stable, dark, colloidal organic fraction of soil.
- **Hydraulic conductivity** - the ability of a soil to transmit water (Darcy's law constant).
- **Hydrolysis** - chemical weathering of minerals by water; the main clay-forming process.
- **Illite** - a non-expanding 2:1 clay with potassium in the interlayer.
- **Illuviation** - deposition of eluviated material into a lower horizon.
- **Inceptisol** - soil order with weakly developed horizons.
- **Infiltration** - entry of water into the soil at the surface.
- **Isomorphous substitution** - replacement of an ion by another of similar size but lower charge in a clay lattice; gives permanent negative charge.
- **Kaolinite** - a 1:1 clay mineral; non-expanding, low CEC.
- **Laterization** - intense tropical weathering in which silica is removed and iron/aluminium oxides accumulate.
- **Leaching** - removal of soluble material from the soil by percolating water.
- **Liquid limit** - moisture content at which soil begins to flow like a liquid.
- **Lithological discontinuity** - a change in parent material within a soil profile.
- **Matric potential** - potential due to the attraction of water by the soil matrix; negative.
- **Mineralisation** - complete decomposition of organic matter to inorganic nutrients.
- **Mollic epipedon** - thick, dark, base-rich surface horizon.
- **Mollisol** - soil order with a dark, base-rich surface horizon (grassland soils).
- **Montmorillonite (smectite)** - an expanding 2:1 clay with high CEC and high shrink-swell.
- **Munsell colour system** - describes soil colour by hue, value and chroma.
- **Osmotic potential** - potential due to dissolved salts; negative; reduces plant water uptake.
- **Oxisol** - soil order with an oxic horizon; highly weathered, low CEC.
- **Parent material** - the material from which soil forms (rock or transported deposit).
- **Ped** - a natural soil aggregate; the unit of soil structure.
- **Pedon** - the smallest three-dimensional unit of soil.
- **Pedoturbation** - mixing of soil material within the profile by physical or biological agencies.
- **Percolation** - downward movement of water through the soil profile.
- **Permanent wilting point** - water content at which plants wilt permanently (about -1500 kPa or 15 bar).
- **pF** - logarithm (base 10) of the suction expressed in cm of water.
- **pH** - negative logarithm of the hydrogen ion concentration; measures acidity/alkalinity.
- **Plasticity** - ability of soil to be moulded and retain its shape.
- **Plasticity index** - liquid limit minus plastic limit.
- **Podzolization** - eluviation of Fe, Al and organic matter under cool humid conifer forest.
- **Porosity** - percentage of soil volume occupied by pores.
- **Primary mineral** - a mineral formed when the rock solidified.
- **Salinization** - accumulation of soluble salts in the soil.
- **Secondary mineral** - a mineral formed by weathering or new formation in soil.
- **Soil profile** - vertical section of soil from the surface to the parent material.

- **Soil structure** - arrangement of soil particles into aggregates.
- **Soil texture** - relative proportions of sand, silt and clay.
- **Spodic horizon** - subsurface horizon of illuvial Fe-Al-humus (Spodosols).
- **Stokes' law** - relates particle settling velocity to particle size (basis of mechanical analysis).
- **Subsoil** - the B horizon; zone of illuvial accumulation.
- **Topsoil** - the A horizon; mineral surface layer rich in humus.
- **Translocation** - movement of material within the soil (eluviation and illuviation).
- **Ultisol** - soil order with an argillic horizon and low base saturation (acid, weathered).
- **Vertisol** - soil order with shrink-swell clays and deep cracks (black soils).
- **Weathering** - in-place disintegration and decomposition of rocks and minerals.

References and Suggested Reading

The material in this study guide follows the standard concepts and definitions used in the following widely used soil science textbooks and course resources. Students are encouraged to consult these for further depth and for figures and photographs.

Standard Textbooks

1. **Brady, N.C. and Weil, R.R.** *The Nature and Properties of Soils.* Pearson Education (latest edition). - The standard reference for soil physical, chemical and biological properties, soil water, colloids and fertility.
2. **Biswas, T.D. and Mukherjee, S.K.** *Textbook of Soil Science.* McGraw-Hill / Tata McGraw-Hill, New Delhi. - A widely used Indian textbook covering soil genesis, physics, chemistry and classification.
3. **Sehgal, J.L.** *Pedology: Concepts and Applications.* Kalyani Publishers, New Delhi. - Detailed treatment of soil genesis, soil forming processes and soil taxonomy.
4. **Joffe, J.S.** *Pedology.* Rutgers University Press. - A classic work on the pedological concept of soil and soil formation.
5. **Brady, N.C.** *The Nature and Properties of Soils* (earlier editions) - useful for the classical definitions and standard ranges.
6. **Foth, H.D.** *Fundamentals of Soil Science.* John Wiley & Sons. - Introductory coverage of soil physical and chemical properties.
7. **Tisdale, S.L., Nelson, W.L. and Beaton, J.D.** *Soil Fertility and Fertilizers.* Macmillan. - Nutrient availability, soil reaction and fertility management.
8. **Singer, M.J. and Munns, D.N.** *Soils: An Introduction.* Prentice Hall. - Clear explanations of soil water, colloids and ion exchange.
9. **Hillel, D.** *Introduction to Soil Physics.* Academic Press. - Soil water potential, movement, infiltration and hydraulic conductivity.

Indian and Course Material

10. **ICAR e-Course material** on Fundamentals of Soil Science (Soil Science and Agricultural Chemistry), Indian Council of Agricultural Research, New Delhi.
11. **TNAU e-Course and lecture notes** - Tamil Nadu Agricultural University, Coimbatore; the SAC 101 course material and crop-wise soil management recommendations.
12. **Soil Survey Staff.** *Soil Taxonomy: A Basic System of Soil Classification for Making and Interpreting Soil Surveys.* USDA Natural Resources Conservation Service, Agriculture Handbook. - The source of soil orders, diagnostic horizons and moisture/temperature regimes.
13. **IARI (Indian Agricultural Research Institute)** publications on soil survey and soil fertility in India.
14. **NBSS & LUP (National Bureau of Soil Survey and Land Use Planning), Nagpur** - soil maps and classification of Indian soils.

Notes on Figures and Values

- All numerical values such as particle density (2.65 g/cm3), bulk density ranges, CEC ranges, Atterberg limits, pF values and salinity thresholds are **standard published ranges** drawn from the above texts. Where exact values depend on the specific soil, a range has been given rather than a single figure.
- Soil colour notations follow the **Munsell Soil Colour Charts**.
- Tamil Nadu soil descriptions follow the general soil survey information for the state and TNAU recommendations.

A Note to the Student

This guide is meant to supplement classroom teaching and the prescribed textbook. Use it to revise lecture by lecture, attempt the short- and long-answer questions at the end of each lecture, and cross-check the tables and worked examples against your textbook. For examinations, always write definitions clearly, use the standard tables and formulae, and quote ranges rather than uncertain exact values.

All the best for your examinations.